

Guaranteed indirect delivery in directed networks

Extremal bounds, sparse constructions, and verified finite cases

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1 Results and scope

1.1 Introduction

A messenger has just obtained a message at a station s of a directed network. A pursuer occupies another station, and both positions are visible. They move alternately, the messenger first, using one directed connection or waiting. The messenger wants to deliver to a designated recipient t before interception. Receipt completes the task immediately, even if the pursuer is at t : the pursuer can intercept the messenger but cannot disable the recipient.

A direct connection $s \rightarrow t$ makes delivery immediate. We study the additional communication supported by the network: source–recipient pairs without a direct connection for which delivery is nevertheless guaranteed against every allowed initial pursuer position. Write $I(G)$ for their number. The three extremal quantities are

$$M_V(n) = \max_{|V(G)|=n} I(G), \quad M_E(m) = \max_{|E(G)|=m} I(G), \quad M(n, m) = \max_{|V(G)|=n, |E(G)|=m} I(G).$$

Isolated vertices are allowed. Removing them preserves I , so the arc-only maximum is finite: a graph with $m > 0$ arcs has at most $2m$ nonisolated vertices.

The original problem and a layered subset construction arose in Alexandra Ignatova’s 2023 project, mentored by Angel Raychev. A public 2023 abstract records that provenance (Ignatova 2023); the retained manuscript is dated February 2024 (Ignatova 2024). Ignatova and Raychev had already established the upper bound $I(G) \leq n(n-3)$ and investigated two-in/two-out-regular constructions. This account reconstructs that bound, distinguishes the original construction from its corrected analysis, and supplies new attaining families among the September 2026 extensions. The full extremal classification remains open.

1.1.1 Results and proof boundaries

The principal results are now an eventual classification, with separate finite exceptions and explicit proof boundaries.

1. **Vertex and even arc maxima.** For every $n \geq 12$,

$$M_V(n) = M_E(2n) = M(n, 2n) = n(n-3).$$

A six-type repeating construction with controlled vertex removals covers every $n \geq 85$; two independent exact solvers and full rank certificates supply the finite bridge $12 \leq n \leq 84$ (Theorem 40).

2. **Odd arc maxima.** For every odd $m = 2k + 1 \geq 25$,

$$M_E(m) = k(k-3) + 2.$$

Six local constructions cover all residue classes of the support size. Their strategies are independently verified and composed by a uniform ordinary argument (Theorem 45 and Corollary 45.1).

3. **The entire main interval.** For every sufficiently large n , every integer budget $2n \leq m \leq n(n-3)$ attains

$$M(n, m) = n(n-1) - m.$$

Thus Raychev’s full-interval conjecture is proved. Theorem 44 gives the explicit sufficient threshold $n \geq 115$. Antichain composition connects the sparse and dense constructions by an explicit polynomial algorithm (Theorem 59). The earlier exact conditional-counting algorithm remains a separate routing option (Theorem 54).

4. **Explicit smaller-order intervals.** A protected-difference construction attains every $3n \leq m \leq n(\lceil n/7 \rceil - 1)$ for $n \geq 25$. A strengthened six-type certificate attains every $2n \leq m \leq 3n$ for $n \geq 115$, with at most $9\lceil n/6 \rceil + 21$ messenger moves. Other arithmetic families give additional ranges at smaller orders (Theorems 34–39 and 52).
5. **The complete final dense band.** For every $n \geq 20$,

$$M(n, n(n-3) + r) = 2n - r - \left\lceil \frac{r}{2} \right\rceil - 1, \quad 1 \leq r \leq n.$$

At $r = 0$, the value is $2n$ for every $n \geq 10$; above $r = n$, it is zero. One direct two-type missing-graph construction and its functional extensions give all the dense assertions with two-move strategies (Theorem 58).

6. **A combined formula.** For $n \geq 115$, these results determine every $M(n, m)$ with $m \geq 24$: the sparse arc maximum, the missing-pair bound in the main interval, the final staircase, and then zero (Corollary 47). The threshold is deliberately conservative and is not asserted to be the smallest possible.
7. **Small cases.** Complete joint maxima are known through $n = 6$, with $M_V(1), \dots, M_V(6) = 0, 0, 0, 2, 4, 7$. Two independent exhaustive checks additionally prove $M_V(7) = 21$ and $M_V(8) = 32$ (Theorem 46). The arc-only maxima through eleven arcs are $0, 0, 0, 0, 1, 1, 2, 2, 4, 5, 6, 8$ (Theorem 49). The remaining odd endpoint satisfies $M_E(23) = 83$ (Theorem 53). Other finite witnesses and nonattainment results are stated separately; no unsolved small value is filled by the eventual formulas.

The new unifying results strengthen the construction and structural conclusions:

- **Protected routing (Theorem 54).** For a specified routing set of size b , put $Q = b(2n-b-1)$ and $p = a/Q$. If $(b-3)p^2(1-p) > 4\log n + \log(8/3)$, exactly a incident arrows can be chosen deterministically so that every assignment of exterior arrows preserves universal two-move delivery. This gives every exact budget $a \leq m \leq a + (n-b)(n-b-1)$ and subsumes the former full-vertex conditioning theorem, including its random-sampling guarantee. The minimum universal two-move budget satisfies $q_2(n) = \Theta(n \log n)$ (Corollary 55).
- **Sparse support (Theorems 56–57).** With $2k$ arrows, $k \geq 6$, a graph outside the k -vertex two-regular shape has $I \leq (k-1)(k-3)$. With $2k+1$ arrows, $k \geq 3$, a graph with $I > k^2 - 4k + 6$ reduces to a k -vertex induced core with at least the same score, or to three specified $k+1$ -vertex degree families. These quantitative statements recover the equality cases and replace the bespoke 23-arrow support argument.
- **Direct dense construction (Theorem 58).** Its two-move endpoint works at every $n \geq 10$. Its arbitrary-subset bridge covers $2n \leq h \leq 3n$ at every $n \geq 16$. Two explicit residue constructions enlarge the dense envelope to $n^2/4 - O(n)$ missing arrows, preserving two-move delivery and every optional subset. The first staircase step is exact for every $n \geq 16$, and the entire staircase for every $n \geq 20$.
- **Antichain composition (Theorem 59).** Incomparable hub subsets attach arbitrarily many outside vertices to a fixed two-part core. Every choice of outside arrows preserves two-move delivery. The exact budget formulas close the numerical main-interval overlap and sharpen the constructive bound to $q_2(n) \leq (5.3555763082\dots + o(1))n \log_2 n$.

- **Support concentration (Theorem 60).** With $\eta = m - 2\sqrt{I(G)}$, the nonisolated support has at most $m/2 + 4\eta$ vertices; all but 12η arrows lie in a core of size between $m/2 - 3\eta$ and $m/2$. This controls near-extremizers below the earlier exact-support cutoffs without asserting that deleting the exceptional arrows preserves delivery.

The general arguments are ordinary mathematical proofs. The accompanying Lean development formalizes the actual alternating game, a certificate-to-strategy theorem, safe branching, camping obstructions, complete universally winning 12-vertex sparse and 11-vertex dense examples, robust two-move envelopes (including six explicit examples at orders sixteen through twenty-one), complementary-block two-move delivery, protected-witness preservation and delivery under exterior-arrow changes, and the numerical staircase deduction from explicit counting hypotheses. The graph-to-counting bridge and existence of attaining blocks remain ordinary arguments. It does not claim to formalize all the general extremal theorems.

1.2 The combined joint maximum

The sparse constructions can be placed inside a larger network by adding isolated vertices. This observation joins the arc-budget extrema to the two dense regimes and gives a single eventual formula.

Corollary 47. Let $n \geq 115$ and $24 \leq m \leq n(n-1)$. Then the following cases determine $M(n, m)$ exactly:

$$M(n, m) = \begin{cases} k(k-3), & m = 2k \leq 2n, \\ k(k-3) + 2, & m = 2k + 1 < 2n, \\ n(n-1) - m, & 2n \leq m \leq n(n-3), \\ 2n - r - \lceil r/2 \rceil - 1, & m = n(n-3) + r, \quad 1 \leq r \leq n, \\ 0, & n(n-2) < m \leq n(n-1). \end{cases}$$

The two formulas at $m = 2n$ agree. For even m , the first formula also holds for every $24 \leq m \leq 2n$ on every host order $n \geq m/2$.

Proof. For even $m = 2k \geq 24$, Theorem 40 constructs an attaining k -vertex graph. If $k \leq n$, add $n - k$ isolated vertices. This does not change I : pairs between components have no delivery path; a pursuer starting at an added isolate stays there and cannot obstruct an existing delivery strategy. The arc-only upper bound supplies the matching upper bound for the joint problem.

For odd $m = 2k + 1 \geq 25$, Corollary 45.1 constructs an attaining graph on $k + 1 = (m + 1)/2$ vertices. When $m < 2n$, its support has at most n vertices. Add isolates and apply the odd arc-only upper bound. This gives $k(k-3) + 2$.

Theorem 44 supplies the third case. Corollary 33 supplies the fourth, since its threshold 20 is far below the present hypothesis. Finally Theorem 13 gives zero above $n(n-2)$. The ranges cover every stated integer budget. $\square$

Theorem 53 additionally gives $M(n, 23) = 83$ for every $n \geq 12$.

This is a complete joint classification in the displayed eventual range, with deterministic attaining constructions. It does not assert the same onset for smaller host orders, or fill the remaining finite

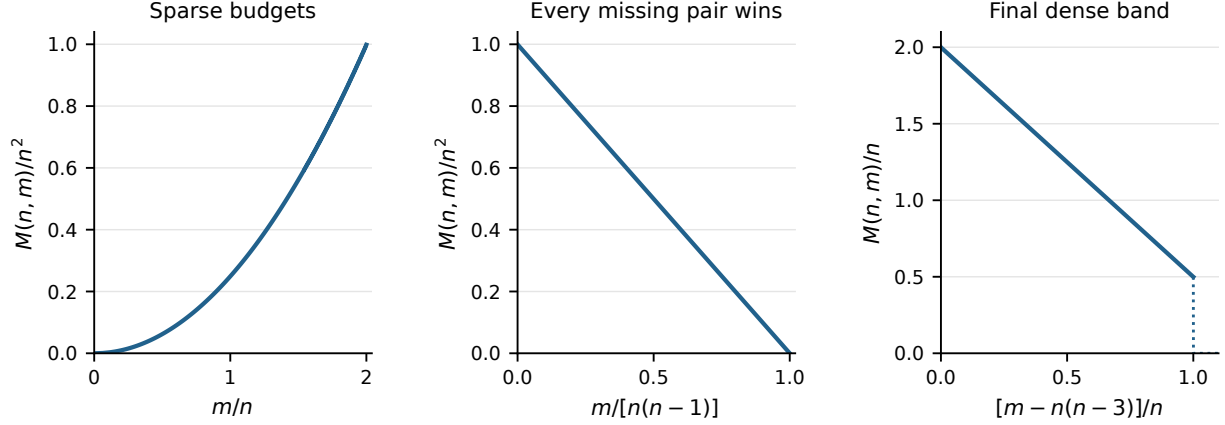


Figure 1: The three exact regimes at $n = 1800$: sparse growth, the main interval, and the final dense band. Separate axis scales show the two narrow endpoint regimes. The dotted continuation in the last panel leads to zero above $n(n-2)$.

odd and very small even arc budgets. In particular, the assertion that the full extremal problem is settled for every finite pair (n, m) would be stronger than what has been proved here.

2 Game and structural bounds

2.1 The game and its elementary structure

Let $G = (V, E)$ be a finite loopless digraph. Opposite arcs are allowed. A legal move from x goes to x or to an outgoing neighbor; write

$$C(x) = \{x\} \cup N^+(x).$$

Fix $s, t \in V$ and $c \neq s$. If $s = t$, receipt has already occurred. Otherwise the messenger moves first. On its turn it chooses $v \in C(s)$. If $v = t$, delivery wins immediately. If $v \neq t$ and $v = c$, it is captured. Otherwise the pursuer chooses $c' \in C(c)$; equality $c' = v$ is capture. The process repeats. Infinite play without delivery is a messenger loss.

Let $W_G(s, t)$ mean that, for every initial $c \neq s$, there exists a messenger strategy guaranteeing delivery against every pursuer strategy. The messenger observes c ; its strategy may depend on it. Then

$$I(G) = \#\{(s, t) : s \neq t, (s, t) \notin E, W_G(s, t)\}.$$

If $w(G)$ counts all guaranteed pairs including the diagonal, then exactly

$$w(G) = n + m + I(G).$$

The terminal rule matters. In the directed diamond $s \rightarrow a \rightarrow t, s \rightarrow b \rightarrow t$, the messenger chooses the intermediate vertex not occupied by the pursuer and delivers on its next turn. But reaching a vertex that would be safe as a *recipient* does not justify using it as an intermediate stop. The pursuer gets an intervening move before the message can be forwarded again.

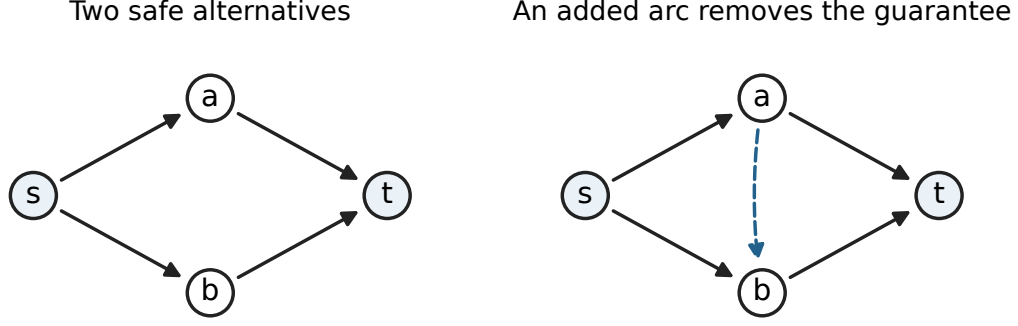


Figure 2: Two routes allow a guaranteed indirect delivery. Adding the arc from a to b lets a pursuer at a block both first moves.

2.1.1 Camping obstructions

Lemma 1 (source domination). If $c \neq s$ and $N^+(s) \subseteq C(c)$, there is no guaranteed indirect pair with source s .

Proof. The pursuer starts at c and waits while the messenger waits. Every nonterminal departure from s is either an immediate collision or is captured on the following pursuer turn. Waiting forever is not delivery. $\square$

Lemma 2 (predecessor domination). Put $D_t = N^-(t)$. If $D_t \subseteq C(c)$, an indirect guaranteed source for t must equal c .

Proof. For any other source the pursuer may start at c . Every indirect delivery must first enter D_t , where the pursuer can capture before the next messenger turn. $\square$

In particular, a source counted by I has outdegree at least two, and a recipient counted by I has indegree at least two. An indegree-one recipient is blocked by its unique predecessor. In an undirected network, interpreted as two opposite arcs per edge, the pursuer can start at t and catch the messenger upon its first arrival at a neighbor of t . Thus $I = 0$ for every such network.

The two degree conditions are necessary, not sufficient. Neither the absence of a one-vertex directed separator nor the existence of two internally disjoint routes characterizes this game. Adding $a \rightarrow b$ to the diamond destroys its only indirect guarantee, although both routes remain. Consequently I is not monotone under adding arcs.

Lemma 3 (two-move criterion). For an indirect pair (s, t) , put $B = N^+(s) \cap N^-(t)$. Delivery is guaranteed within two messenger moves exactly when

$$\text{for every } c \neq s, \quad B \not\subseteq C(c).$$

Proof. Choose an intermediate vertex of $B \setminus C(c)$, survive every pursuer response, and deliver. Conversely, any successful two-move indirect delivery must use such an intermediate. Waiting on the first move cannot reach a nonadjacent recipient on the second. $\square$

2.1.2 Components, zero minima, and the vertex bound

Removing isolated vertices preserves I . More generally,

$$I(G \sqcup H) = I(G) + I(H).$$

There are no cross-component routes; a pursuer in the other component cannot interfere with a route in the messenger's component.

Proposition 4. At every admissible (n, m) the minimum of I is zero.

Proof. For $n \geq 2$, list all nonloop arcs row by row and retain the first m . There are full outgoing rows, at most one partial row, and empty rows. Full rows have only direct recipients; empty rows cannot move. If a full row exists, its vertex blocks every indirect attempt from the partial row by Lemma 1. Otherwise successors of the partial row have no outgoing arcs, so no indirect route exists. The empty and one-vertex cases are immediate. $\square$

Theorem 5 (original upper bound). For every $n \geq 3$,

$$I(G) \leq n(n-3).$$

Proof. A recipient of indegree zero or one has no indirect guaranteed sources. A recipient of indegree $d \geq 2$ has at most $n-1-d \leq n-3$, after excluding itself and its direct predecessors. Sum over recipients. $\square$

For $n \geq 4$, equality forces every recipient to have indegree two and every missing pair to be guaranteed. Every source must then have outdegree at least two; the degree sum forces outdegree exactly two. Conversely, a two-in/two-out-regular graph with every pair guaranteed attains equality. Thus attaining the vertex bound is exactly a sparse universal-delivery construction problem.

2.2 Sharp arc-budget upper bounds

Let A be the vertices of outdegree at least two and B those of indegree at least two. Write $q = |A|$, $r = |B|$, $b = |A \cap B|$, and let $e(A, B)$ count arcs from A to B . Every counted pair belongs to $A \times B$, and direct and diagonal pairs are excluded. Hence

$$I(G) \leq qr - b - e(A, B), \quad e(A, B) \geq \max\{0, 2q + 2r - m\}. \quad (1)$$

The second inequality follows by adding the total outgoing degree in A and incoming degree in B : each arc is counted at most once except those in $A \times B$.

We also need two elementary obstructions. If x has indegree zero and $x \rightarrow t$ with t of indegree two, then t has no indirect guaranteed sources. Its other predecessor blocks every indirect route: a route from elsewhere cannot reach x , while $x \rightarrow t$ itself is direct. Dually, if t is a sink, $x \rightarrow t$, and x has outdegree two, then x has no indirect guaranteed recipients. Its other successor blocks every potentially successful indirect departure. This dual argument is proved directly and does not assume invariance under reversing the graph.

2.2.1 Even numbers of arcs

Theorem 6. If $m = 2k$ and $k \geq 5$, then

$$I(G) \leq k(k-3).$$

Proof. For $k \geq 6$, apply Theorem 56. Its two-regular core has $k(k-3)$ missing nonloop pairs, while every other support shape has $I \leq (k-1)(k-3) < k(k-3)$.

We retain a direct ordinary argument for the remaining case $k = 5$, so no finite computation is needed for this theorem. Then $q, r \leq 5$. Outside $(q, r) = (5, 5), (4, 5), (5, 4)$, formula (1) gives $I \leq 10$: for $2 \leq q, r \leq 4$ its bound is at most $(q-2)(r-2) + 6 \leq 10$; if one is at most one the result is immediate; and $(q, r) = (5, r)$ with $r \leq 3$ gives $I \leq 3r \leq 9$, as does its exchanged case.

For $q = r = 5$, all ten arcs run from A to B , with the indicated degrees two. If $A = B$, there are only ten eligible missing pairs. Otherwise an indegree-zero source in $A \setminus B$ gives two zero recipient columns, leaving at most $5 \cdot 3 - 6 = 9$ pairs. For $q = 4, r = 5$, formula (1) gives $I \leq 12 - b$, sufficient if $b \geq 2$. If $b \leq 1$, an indegree-zero source outside B again gives two zero columns. The other three columns receive six arcs, at most two from outside A , so $I \leq 4 \cdot 3 - 4 = 8$. The case $q = 5, r = 4$ uses a sink outside A and two zero source rows, with the same count. Thus $I \leq 10 = 5(5-3)$ in every remaining case. $\square$

2.2.2 Odd numbers of arcs

Theorem 7. If $m = 2k + 1$ and $k \geq 5$, then

$$I(G) \leq k(k-3) + 2.$$

Proof. This is Corollary 57.1 of the ordinary sparse support reduction. Above $k^2 - 4k + 6$, that reduction leaves only an induced k -vertex core or one of three $(k+1)$ -vertex degree-defect families. Counting their eligible missing pairs gives the bound, while $k^2 - 4k + 6 < k(k-3) + 2$ for every $k \geq 5$. $\square$

2.2.3 Equality structure

Theorem 8. For $k \geq 6$, equality in Theorem 6 holds exactly when deleting isolated vertices leaves a two-in/two-out-regular graph on k vertices with every pair guaranteed.

Proof. By Theorem 56, every other support shape has $I \leq (k-1)(k-3) < k(k-3)$. In the two-regular k -vertex core there are exactly $k(k-3)$ missing nonloop pairs. Equality therefore holds exactly when every such pair is guaranteed; the diagonal and direct pairs are always guaranteed. Adding or deleting isolates preserves I . $\square$

Theorem 9. For $k \geq 6$, equality in Theorem 7 holds exactly for graphs whose nonisolated support has the following properties: it has $k+1$ vertices; one vertex u has outdegree one and indegree two; a different vertex v has indegree one and outdegree two; all other degrees are two; $u \rightarrow v$ is absent; and every missing nonloop pair with source of outdegree two and recipient of indegree two is guaranteed.

Proof. Corollary 57.1 gives exactly this equality characterization. All other reduced cases have strictly fewer eligible missing pairs, and the remaining graphs lie below the sparse support cutoff. In the stated family, equality is equivalent to guaranteeing all $k(k-3) + 2$ eligible missing pairs. $\square$

Adding the missing arc $u \rightarrow v$ in Theorem 9 gives a two-regular digraph. It is not asserted that this completed graph is automatically universally winning.

2.3 A quantitative reduction for sparse near-extremizers

The degree bounds can distinguish a genuinely competitive graph from one whose shape has already lost a linear number of guarantees. Throughout, graphs are finite, simple and loopless, opposite arcs are allowed, and the messenger–pursuer game has the timing fixed in the main paper. An induced subgraph below retains every original arc between its vertices.

Put

$$A = \{v : d^+(v) \geq 2\}, \quad B = \{v : d^-(v) \geq 2\}, \quad q = |A|, \quad r = |B|, \quad b = |A \cap B|, \quad e = e(A, B).$$

The elementary obstructions and degree count give

$$I(G) \leq qr - b - e, \quad e \geq \max\{0, 2q + 2r - m\}. \quad (56.1)$$

We call a source row or recipient column **zero** when it contains no guaranteed indirect pair. Two facts used repeatedly are that an indegree-two recipient with an indegree-zero predecessor has a zero column, and an outdegree-two source with a sink successor has a zero row. These are separate camping arguments in the original game, not an assumption that reversing every arc preserves the game.

2.3.1 The even budget: a gap before another support shape can compete

Theorem 56 (even sparse stability). Let $k \geq 6$ and let G have exactly $2k$ arcs. At least one of the following holds:

1. Deleting isolated vertices leaves a graph on exactly k vertices, with indegree and outdegree two at every vertex.
2. $I(G) \leq (k-1)(k-3)$.

In particular, a graph with

$$I(G) > k(k-3) - (k-3)$$

must have the first shape. No universality assumption is made about that two-regular core. The statement is an ordinary proof and uses no exhaustive graph computation.

Proof. Write $C = k^2 - 4k + 3 = (k-1)(k-3)$. We have $q, r \leq k$. First dispose of the ranges away from the largest active sets. If $q, r \leq k-1$ and at least one of them is at most $k-2$, (56.1) gives $I \leq C$. Indeed, if both are at least two, then

$$I \leq qr - (2q + 2r - 2k) = (q-2)(r-2) + 2k - 4 \leq k^2 - 5k + 8 \leq C.$$

If either is at most one, $I \leq k-1 \leq C$. If $q = k$ and $r \leq k-3$, then $I \leq (k-2)r \leq (k-2)(k-3) \leq C$; the same numerical estimate applies with q, r exchanged.

Suppose $q = k, r = k-2$. If $b \geq 1$, (56.1) gives

$$I \leq (k-2)^2 - b \leq C.$$

If $b = 0$, all $2k$ arcs start in A , all its outgoing degrees are two, and every member of B is a sink. The at least $2(k-2)$ arcs entering B have at least $k-2$ distinct tails in A , each a zero row. At most two contributing rows remain, giving $I \leq 2(k-2) \leq C$. For $q = k-2, r = k$, all arcs end in B and its incoming degrees are two. When $b = 0$, every member of A has indegree zero; its at

least $2(k-2)$ outgoing arcs make at least $k-2$ recipient columns zero. This gives the same bound. Thus neither case uses reversal of the game.

Next suppose $q = r = k - 1$. Formula (56.1) suffices if $b \geq 2$, since

$$I \leq k^2 - 4k + 5 - b \leq C.$$

For $b \leq 1$, at most two arcs end outside B , so at least

$$|A \setminus B| - 2 = k - 3 - b \geq 2$$

vertices of $A \setminus B$ have indegree zero. Choose two such vertices. Let

$$x = \sum_{t \in B} (d^-(t) - 2).$$

Exactly $2 - x$ arcs end outside B , and $x \in \{0, 1, 2\}$. If one of the chosen vertices points to an indegree-two member of B , there is a zero column. Assume this does not happen for either chosen vertex. If $x = 0$, their at least four outgoing arcs would all end outside B , where only two arcs are available. If $x = 1$, there is just one higher-indegree member of B ; each chosen source needs another, distinct successor outside B , where only one arc is available. Both cases are impossible. If $x = 2$, no arc ends outside B . A single indegree-four exceptional recipient cannot contain the two distinct successors of either source. The only remaining possibility is two indegree-three recipients, and both chosen sources have exactly that same pair of outgoing neighbors. Each then has a zero row by source domination.

We have therefore obtained either an indegree-two zero column or an outdegree-two zero row. Delete it from the eligible rectangle. At most two of its at least $2k - 4$ direct pairs are deleted, leaving

$$I \leq (k-1)(k-2) - (2k-6) = k^2 - 5k + 8 \leq C.$$

Discarding diagonal exclusions here only weakens the bound.

Suppose now $q = k - 1, r = k$. If $A \subseteq B$, then $b = k - 1$ and $e \geq 2k - 2$, giving $I \leq k^2 - 4k + 3 = C$. Otherwise choose $z \in A \setminus B$. Every arc ends in B , all of whose incoming degrees are two, so z has indegree zero. Its two distinct successors give two zero columns. The other $k - 2$ columns receive $2k - 4$ arcs in total; at most two of these start outside A . Consequently

$$I \leq (k-1)(k-2) - (2k-6) = k^2 - 5k + 8 \leq C.$$

For $q = k, r = k - 1$, all arcs start in A and all its outgoing degrees are two. If $B \subseteq A$, the direct bound gives C . Otherwise a sink in $B \setminus A$ has two distinct predecessors in A ; these give two zero rows, and the identical count applies.

It remains that $q = r = k$. Every arc runs from A to B , and all these outgoing and incoming degrees are two. If $A = B$, no arc has an endpoint outside this k -vertex set: we have conclusion 1. Suppose $A \neq B$. If $b \geq 3$, an indegree-zero source in $A \setminus B$ gives two zero columns. The remaining rectangle contains at least $2k - 4$ direct pairs and $b - 2 \geq 1$ diagonal pairs, so

$$I \leq k(k-2) - (2k-4) - (b-2) = (k-2)^2 - (b-2) \leq C.$$

If $b \leq 2$, the $k - b$ indegree-zero vertices in $A \setminus B$ emit $2(k - b)$ arcs, making at least $k - b$ recipient columns zero, because each recipient has indegree two. Similarly the $k - b$ sinks in $B \setminus A$ receive

$2(k-b)$ arcs, making at least $k-b$ source rows zero. At most b rows and b columns can contribute; hence $I \leq b^2 \leq 4 \leq C$. All cases are exhausted. $\square$

The same dichotomy holds for $k = 5$ by the previously established $M_E(10) = 6 \leq 8 = (k-1)(k-3)$. That extension inherits the finite computation in Theorem 49; the proof above is ordinary for every $k \geq 6$. The threshold $k \geq 5$ cannot be lowered to four: the eight-arc, six-vertex witness in Theorem 49 has $I = 4 > 3$, and is not a four-vertex two-regular graph.

Corollary 56.1. For every $k \geq 6$,

$$M_E(2k) \leq \max\{M(k, 2k), (k-1)(k-3)\}.$$

In conjunction with already proved finite bounds, this gives

$$M_E(12) \leq 15, \quad M_E(14) \leq 24, \quad M_E(16) \leq 35, \quad M_E(22) \leq 81.$$

Proof. In conclusion 1 of Theorem 56, removing isolates preserves I and leaves an admissible graph for $M(k, 2k)$. In conclusion 2 the other entry in the maximum applies. Use $M(6, 12) = 7$, $M(7, 14) \leq M_V(7) = 21$, $M(8, 16) \leq M_V(8) = 32$, and the eleven-vertex core bound $M(11, 22) \leq 81$. These numerical consequences inherit only the finite dependencies of those existing small-order bounds; no new enumeration is needed. They are upper bounds, not asserted exact values. $\square$

2.3.2 The odd budget: three exception families and small induced cores

Theorem 57 (odd sparse support reduction). Let $k \geq 3$, let G have exactly $2k+1$ arcs, and suppose

$$I(G) > k^2 - 4k + 6.$$

Then at least one of the following holds:

1. G has an induced subgraph H on k vertices with $I(H) \geq I(G)$. Either H has $2k$ arcs and both degrees two everywhere, or H has $2k+1$ arcs, one vertex of outdegree three, one vertex of indegree three, and all other corresponding degrees two. The two degree-three vertices may coincide.
2. Removing isolates from G leaves $k+1$ vertices, with one vertex u of outdegree one and indegree two, a different vertex v of indegree one and outdegree two, and all other degrees two. Either $u \rightarrow v$ is absent or it is present; these are two separate families.
3. Removing isolates leaves $k+1$ vertices, with one vertex of indegree and outdegree one and every other degree two.

The theorem supplies a reduction of the actual delivery game, not just degree sequences. In case 1 the induced core is identified explicitly in the proof. The proof is ordinary and contains no exhaustive graph computation.

Proof. Put $T = k^2 - 4k + 6$. The count (56.1) gives $I \leq T$ outside the three corners

$$(q, r) = (k, k), \quad (k-1, k), \quad (k, k-1).$$

Here is the numerical justification. If $q, r \leq k-1$ and both are at least two, then

$$I \leq (q-2)(r-2) + 2k - 3 \leq (k-3)^2 + 2k - 3 = T.$$

If either is at most one, $I \leq k-1 \leq T$. If $q = k, r \leq k-2$, then $I \leq (k-2)r + 1 \leq (k-2)^2 + 1 < T$; the exchanged case has the same numerical bound.

Suppose $q = k-1, r = k$. At most one arc ends outside B . If $b \geq k-2$, (56.1) already gives

$$I \leq k(k-3) + 3 - b \leq k^2 - 4k + 5 < T.$$

If $b \leq k-3$, at least two members of A lie outside B , and at least one has indegree zero. Such a source has an indegree-two successor in B : either every member of B has indegree two and at most one arc ends elsewhere, or one member of B has indegree three and no arc ends elsewhere. Its two distinct successors cannot both avoid the indegree-two members of B . Delete the resulting zero column. At least $e-2 \geq 2k-5$ direct pairs remain, giving

$$I \leq (k-1)^2 - (2k-5) = T.$$

The case $q = k, r = k-1$ is proved separately using sinks in $B \setminus A$: at most one vertex outside A has positive outdegree, and a sink of indegree at least two has an outdegree-two predecessor, creating a zero row. The same count applies. Thus $q = r = k$ and $e \geq 2k-1$.

First suppose $b \leq k-4$. At least three members of $A \setminus B$ have indegree zero and emit at least six arcs. At most three of these arcs can avoid indegree-two recipients in B : the only alternatives are one indegree-three recipient and no outside arc, or no higher-indegree recipient and at most one outside arc. At least three arcs therefore enter indegree-two recipients, making at least two columns zero. Removing them gives

$$I \leq k(k-2) - (2k-5) = k^2 - 4k + 5 < T.$$

If $b = k-2$, there is an indegree-zero member of $A \setminus B$, and hence an indegree-two zero column as above. After deleting it, at least $2k-3$ direct pairs and $b-1 = k-3$ diagonals remain. Thus

$$I \leq k(k-1) - (2k-3) - (k-3) = T.$$

If $b = k-3$, there are at least two indegree-zero vertices in $A \setminus B$ and at least two sinks in $B \setminus A$. The preceding arguments give an indegree-two zero column and an outdegree-two zero row. Deleting both removes at most four direct pairs and at most two diagonal pairs. Consequently

$$I \leq (k-1)^2 - (2k-5) - \max\{0, k-5\} \leq T.$$

For $k = 3, 4$ the last expression is respectively 3, 6, exactly T ; for $k \geq 5$ it is $k^2 - 5k + 11 \leq T$. We conclude that $b \geq k-1$.

Suppose $b = k-1$, and write $A \setminus B = \{v\}$ and $B \setminus A = \{u\}$. If v has indegree zero, an indegree-two zero column gives

$$I \leq k(k-1) - (2k-3) - (k-2) = k^2 - 4k + 5 < T.$$

An outdegree-zero u similarly gives a zero row and the same bound. Therefore both exceptional degrees are positive. Because at most one arc ends outside B and at most one starts outside A , they equal one. They exhaust those two allowances; all degrees in A and B are exactly two, and every vertex outside $A \cup B$ is isolated. Thus G itself has the shape in case 2, including either choice of the arc $u \rightarrow v$.

Finally let $A = B = C$, where $|C| = k$. At most one arc starts outside C and at most one ends outside C . If all $2k+1$ arcs lie in C , the induced graph $H = G[C]$ has the degree pattern in case 1

and $I(H) = I(G)$. An arc having both endpoints outside C consumes both allowances and leaves a two-regular $2k$ -arc core H with $I(H) = I(G)$; its outside component is a single arc and has no indirect guarantees.

Assume there is exactly one crossing arc, first $C \rightarrow x$. The vertex x is a sink. The $2k$ internal arcs have indegree two at every vertex, while exactly one vertex of C has total outdegree three. If that vertex emits the crossing arc, deleting the sink leaves a two-regular core H . All counted pairs of G have both endpoints in C . A successful delivery between such endpoints can never enter the sink; against a pursuer staying in C it follows only the arcs of H . Therefore every indirect guarantee counted in G is also a guarantee in H , giving $I(H) \geq I(G)$ as required. If instead an outdegree-two vertex emits the crossing arc, its source row is zero. Of the $k^2 - 3k$ eligible missing pairs, $k - 2$ lie in that row, yielding $I \leq k^2 - 4k + 2 < T$.

For a sole crossing arc $x \rightarrow C$, its outside tail has indegree zero. If the head is the unique indegree-three vertex, removing x leaves a two-regular core. A messenger starting in C cannot reach x , so all its successful delivery play stays in that core, and again $I(H) \geq I(G)$. If the head instead has indegree two, its inaccessible predecessor makes its recipient column zero and gives the same bound $k^2 - 4k + 2 < T$. These are direct arguments with the game's original directions.

If both crossing arcs exist, they are $C \rightarrow x$ and $y \rightarrow C$. All vertices of C have total indegree and outdegree two and there are $2k - 1$ internal arcs. If $x \neq y$, then x is a sink. The source of $C \rightarrow x$ has a zero row containing $k - 2$ of the $k^2 - 3k + 1$ eligible missing pairs. Thus $I \leq k^2 - 4k + 3 < T$. If $x = y$, this one outside vertex has both degrees one, every other outside vertex is isolated, and case 3 holds. This exhausts all allowed outside arcs and proves the theorem. $\square$

For $k = 0, 1, 2$ the previously elementary values $M_E(1) = M_E(3) = 0$, $M_E(5) = 1$ are already at most $k^2 - 4k + 6$. Thus the conditional statement of Theorem 57 extends vacuously to those k as well; $k \geq 3$ is the range treated by its uniform degree proof.

Corollary 57.1 (recovering the odd bound and its equality shape). For every $k \geq 5$,

$$M_E(2k + 1) \leq k(k - 3) + 2.$$

Equality requires the distinct-exception family with $u \rightarrow v$ absent and every eligible missing pair guaranteed; these conditions also suffice.

Proof. Write $U = k^2 - 3k + 2$. We have $T < U$ for $k \geq 5$. In the induced-core cases the total numbers of missing pairs are $U - 2$ and $U - 3$. For the distinct-exception family they are U if $u \rightarrow v$ is absent and $U - 1$ if it is present, after restricting to $A \times B$. For the common-exception family that rectangle has $U - 1$ missing pairs. In particular, the induced core with $I(H) \geq I(G)$ also bounds the original graph's count. Only the first distinct-exception family can reach U , and it does so exactly when all its eligible missing pairs are guaranteed. $\square$

This support theorem neither asserts that the three exceptional families are universally winning nor that degree-preserving completion or deletion preserves strategies. Its role is to reduce any possible high-scoring counterexample to explicitly defined small families before evaluating the actual game.

2.4 Quantitative concentration near the sparse optimum

The exact support alternatives in Theorems 56 and 57 apply above specific cutoffs. The following estimate continues below those cutoffs. It says that losing only a linear number of pairs from the

leading quadratic bound forces only a bounded number of exceptional vertices and arrows. It makes no claim that deleting those exceptions preserves a delivery strategy.

Theorem 60. Let G have m arrows and $I = I(G)$. Put

$$A = \{v : d^+(v) \geq 2\}, \quad B = \{v : d^-(v) \geq 2\},$$

$$q = |A|, \quad r = |B|, \quad b = |A \cap B|, \quad a = m - 2q, \quad d = m - 2r.$$

If R and C count the source rows and recipient columns containing at least one guaranteed indirect pair, then

$$R \leq \min\{q, b + 2a + \lfloor d/2 \rfloor\}, \quad C \leq \min\{r, b + \lfloor a/2 \rfloor + 2d\}, \quad I \leq RC.$$

Define $\eta = m - 2\sqrt{I} \geq 0$. The induced core $K = G[A \cap B]$ satisfies

$$m/2 - 3\eta \leq |V(K)| \leq m/2.$$

The graph has at most $m/2 + 4\eta$ nonisolated vertices, and at most 12η arrows have an endpoint outside K .

Proof. Only vertices in A can supply nonzero indirect source rows, and only vertices in B can supply nonzero recipient columns. These are the usual camping obstructions at degree zero or one. In particular $I \leq qr \leq m^2/4$.

At most d arrows end outside B , so at least $z = \max(0, q - b - d)$ vertices of $A \setminus B$ have indegree zero. They emit at least $2z$ arrows. Let x be the total excess indegree above two inside B . At most $d - x$ arrows end outside B , and vertices of indegree at least three have total indegree at most $3x$, since $e \leq 3(e - 2)$ for $e \geq 3$. Thus at most $3d$ of the $2z$ arrows can avoid indegree-two recipients. At least $z - \lfloor 3d/2 \rfloor$ distinct indegree-two recipients have an indegree-zero predecessor.

Each such recipient has a zero indirect column. An indirect source cannot reach the indegree-zero predecessor and must approach through the other predecessor, where the pursuer can wait. Therefore

$$C \leq r - z + \lfloor 3d/2 \rfloor \leq r - q + b + d + \lfloor 3d/2 \rfloor = b + \lfloor a/2 \rfloor + 2d.$$

The identity uses $r - q = (a - d)/2$ and the common parity of a, d .

For the row bound, use the original arrow directions. At least $r - b - a$ members of $B \setminus A$ are sinks. Their incoming arrows force outdegree-two predecessors except for at most $3a$ arrows, by the same surplus count. An outdegree-two source with a sink successor has zero indirect row: the pursuer camps at the other successor. Repeating the count gives $R \leq b + 2a + \lfloor d/2 \rfloor$. This is not an appeal to symmetry under reversing the pursuit game. The displayed minimum bounds and $I \leq RC$ follow.

Put $S = a + d$. Since $q + r = m - S/2$ and $I \leq qr$, arithmetic-geometric mean gives $S \leq 2m - 4\sqrt{I} = 2\eta$. The row and column estimates also imply

$$2\sqrt{I} \leq R + C \leq 2b + \frac{5}{2}S,$$

so $b \geq \sqrt{I} - 5S/4 \geq m/2 - 3\eta$. Its upper bound is immediate.

Every nonisolated vertex outside $A \cup B$ contributes at least one to its incoming plus outgoing degree. Outgoing degrees outside A total at most a , and incoming degrees outside B total at most d . There are therefore at most S such vertices. The nonisolated support is at most

$$q + r - b + S = m - b + S/2 \leq m - \sqrt{I} + \frac{7}{4}S \leq m/2 + 4\eta.$$

Finally, the number of arrows with tail outside $A \cap B$ is at most $2(q - b) + a$, and the analogous incoming count is at most $2(r - b) + d$. Counting an arrow twice can only increase this estimate. The number of arrows outside the induced core is thus at most

$$2(q + r - 2b) + S = 2m - 4b \leq 2m - 4\sqrt{I} + 5S \leq 12\eta.$$

This proves all assertions. $\square$

For example, if $m = 2k$ and $I \geq k(k - 3) - \Delta$ with $\Delta = O(k)$, then $\eta = O(1)$. Thus the support has at most $k + O(1)$ vertices and all but $O(1)$ arrows lie inside a core of at most k vertices. The constants above are explicit but are not asserted to be optimal. Theorems 56 and 57 retain their sharper conclusions in their stated ranges. In particular, this concentration estimate does not supply the game restriction $I(K) \geq I(G)$ that is proved under the more specific hypotheses of Theorem 57.

The argument was independently reviewed and checked against both game solvers on all 4,165 labelled loopless digraphs through four vertices and 181 additional graphs. The ordinary proof, finite tests, and exact integer comparisons are recorded in `research/routing-defect-phase2-support.md` and its two verification receipts. This theorem is not presently formalized in Lean.

3 Sparse attaining constructions

3.1 Sparse constructions with a linear deficit

3.1.1 A safe branching principle

Lemma 10. Suppose distinct vertices have closed move sets intersecting in at most one vertex. If a vertex r has two distinct outgoing neighbors that each guarantee delivery to t , then r guarantees delivery to t .

Proof. For every pursuer position $c \neq r$, one of the two successors lies outside $C(c)$. Move there. It survives every intervening pursuer response, after which its guaranteed strategy applies. If the chosen vertex is t , delivery is already terminal. This is a finite concatenation of actual strategies. $\square$

Theorem 11. Let $a \geq 3$, $h = 2a + 1$, $n \geq 2h + 1$, and $\gcd(n, h) = 1$. The digraph on $\mathbb{Z}/n\mathbb{Z}$ with outgoing steps $1, a, h$ has every pair guaranteed. It has $3n$ arcs and

$$I(G) = n(n - 4).$$

Proof. The positive differences in $\{0, 1, a, 2a + 1\}$ are

$$1, a - 1, a, a + 1, 2a, 2a + 1,$$

all distinct. Since $n > 2h$, no positive difference coincides modulo n with a negative one. Distinct closed move sets therefore intersect in at most one vertex.

Translate the recipient to zero. Vertices $0, -1, -a, -h$ are already delivered or have direct arcs to it. Vertex $-a - 1$ is guaranteed via its neighbors $-a, -1$. Thus all four vertices

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

are guaranteed for $z = 0$. If $Q(z)$ is guaranteed, apply Lemma 10 in the following order:

New source	Two guaranteed successors
$z - h$	$z - a - 1, z$
$z - h - 1$	$z - h, z - 1$
$z - h - a$	$z - h, z - a$
$z - h - a - 1$	$z - h - a, z - a - 1$

This proves $Q(z - h)$. Iterating at most $n - 1$ times covers every vertex because h is coprime to n . The induction is finite; it supplies terminating strategies, not merely perpetual evasion. Finally subtract the n diagonal pairs and $3n$ arcs from n^2 . $\square$

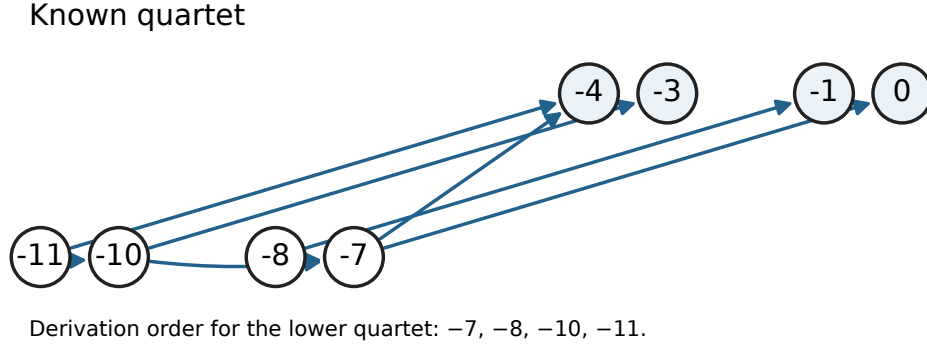


Figure 3: A guaranteed quartet produces the next quartet seven positions earlier in the step-1,3,7 construction.

The choice $a = 3$ proves the theorem for every $n \geq 15$ not divisible by seven. Adding an isolated vertex gives a linear-deficit construction for the remaining large sizes as well. The next elementary lemma removes that padding and improves the constant.

Lemma 12. For every $n \geq 25$, there is an odd integer h with $7 \leq h < n/2$ and $\gcd(n, h) = 1$.

The proof and its explicit choices are included in the construction appendix below. Taking $a = (h - 1)/2$ gives, for every $n \geq 25$,

$$n(n - 4) \leq M_V(n) \leq n(n - 3), \quad 3n \leq n^2 - M_V(n) \leq 4n.$$

These estimates motivate the sharper endpoint result: Theorem 40 subsequently proves equality in the upper bound for every $n \geq 12$.

3.2 An infinite family attaining the sparse bounds

The five-type graph used in the dense construction also attains the sparse vertex and arc-budget bounds. This requires a different strategy: its messenger may need to travel around a substantial part of the ring. The proof combines fixed local certificates with a simulation valid at every sufficiently large circumference, followed by six finite base cases.

For clarity, define the sparse graph H_k again. Its vertices are

$$(i, j) \in (\mathbb{Z}/k\mathbb{Z}) \times \{0, 1, 2, 3, 4\}.$$

The first outgoing arc follows the cycle

$$(i, 0) \rightarrow (i, 1) \rightarrow (i, 2) \rightarrow (i, 3) \rightarrow (i, 4) \rightarrow (i + 1, 0).$$

The second outgoing arc is specified by

$$\begin{aligned} (i, 0) &\rightarrow (i + 1, 0), & (i, 1) &\rightarrow (i - 1, 2), \\ (i, 2) &\rightarrow (i - 1, 4), & (i, 3) &\rightarrow (i - 1, 3), \\ (i, 4) &\rightarrow (i - 1, 1). \end{aligned}$$

Both arc maps are permutations, are loopless, and have distinct images at every vertex when $k \geq 4$. Hence H_k has $5k$ vertices and $10k$ arcs, with indegree and outdegree two everywhere. Waiting is an additional legal action, as throughout the paper.

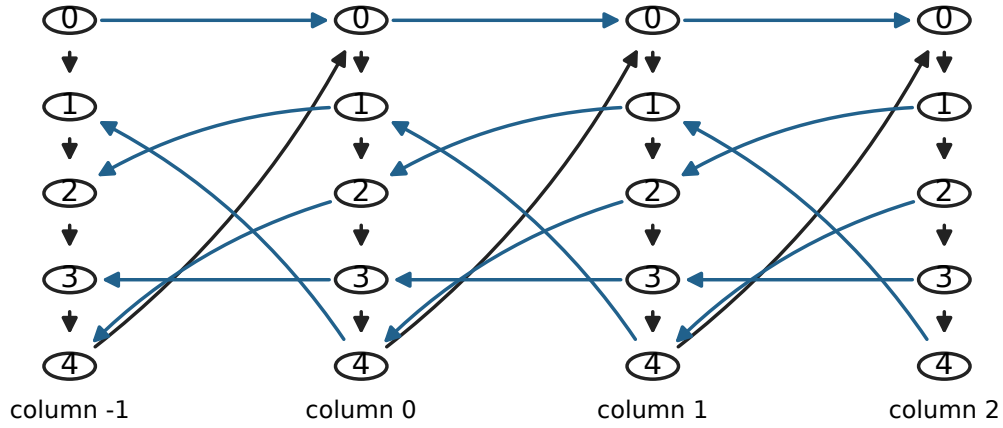


Figure 4: The five station types in consecutive columns. Black arrows follow the long cycle and blue arrows give the second permutation. Identifying columns modulo k closes the strip into the graph H_k .

Theorem 21 (uniform sparse attainment). For every integer $k \geq 4$, every source–recipient pair in H_k is guaranteed. Consequently

$$M_V(5k) = 5k(5k - 3), \quad M_E(10k) = 5k(5k - 3).$$

For $k \geq 10$, a sufficient delivery time is $8k + 16$ messenger moves. The theorem is computer-assisted: its finite ingredients are the two local objectives specified below and the six rings $4 \leq k \leq 9$.

3.2.1 The stronger local arena

Unwrap the column coordinate to the integers, retaining the same two arc rules. Confine the messenger to columns $-3, \dots, 3$, disallowing moves out of this interval. Represent the pursuer by its exact vertex in columns $-4, \dots, 4$, together with an exterior state Ω . Thus there are 35 messenger positions and 46 pursuer positions.

Within its nine-column window the pursuer has all its usual moves and its wait; a move leaving the window goes to Ω . From Ω it may wait or enter any vertex of either boundary column -4 or 4 .

This grants more freedom than a particular physical pursuer outside the window. The messenger and Ω can never coincide.

The two objectives have different stopping rules:

- A **safe transfer** ends in column zero at the start of a messenger turn, with the positions distinct. Entering column zero on a messenger move is insufficient: the intervening pursuer response must also be survived.
- A **delivery** ends immediately when the messenger enters a specified vertex $(0, t)$, before collision is checked there, exactly as in the original game.

For each objective, the certificate assigns nonnegative integer ranks to winning states and selects a legal move at every positive-rank messenger state. Every selected move decreases rank, and every legal response at a positive-rank pursuer state also decreases rank. The zero-rank states are precisely the applicable terminal successes. These conditions prove termination by induction on rank. For safe transfer, only noncollision **messenger-turn** states in column zero are seeded as successes; pursuer-turn states there still require all their responses to be safe.

The checked finite bounds are:

Local objective	Initial column	Initial noncollision states	Maximum messenger moves
Safe transfer to column zero	1	225	8
Delivery to $(0, 0)$	2	225	24
Delivery to $(0, 1)$	2	225	16
Delivery to $(0, 2)$	2	225	16
Delivery to $(0, 3)$	2	225	17
Delivery to $(0, 4)$	2	225	18

Each row covers all five source types and all 46 pursuer states other than the source itself. In particular, it includes an initially exterior pursuer. Only the bounds 8 and 24 are needed below; optimality of these local times is not claimed as a separate result.

3.2.2 Projection to a ring

Fix $k \geq 10$ and any reference column. Its nine offsets $-4, \dots, 4$ represent distinct columns of the ring. Map a physical pursuer in this window to its exact local position and every other pursuer position to Ω .

Every physical pursuer move has a permitted image. A move within the window is unchanged, one leaving it goes to Ω , and a move remaining outside is represented by waiting at Ω . A move from outside into the window can enter only a boundary column, because each arc changes the column by at most one; every such entry is permitted from Ω . At $k = 10$ the exterior consists of a single column adjacent to both boundaries, and both entries are already allowed. At $k = 9$ the two boundary columns themselves have wraparound adjacencies, so this argument would fail; those rings are deliberately covered by finite checks instead.

Every messenger move permitted by the local certificate is a genuine ring move. Its seven represented columns lie strictly inside the pursuer window. Consequently a physical collision with the

messenger is exactly a local collision; an exterior pursuer cannot coincide with it. Following a local winning strategy therefore succeeds against every physical pursuer strategy. This is a simulation of legal moves and collisions, not an assumption that winning strategies pass automatically to graph covers.

3.2.3 Composition and termination

Proof of Theorem 21. Translate the recipient’s column to zero, and let the messenger’s current column be d . Translate the safe-transfer window so that its local column zero is the physical column $d - 1$; the messenger then starts in local column one. The certificate reduces its physical column by one modulo k and finishes only after the pursuer’s response, so another transfer can begin with the correct turn order. Recenter the window between transfers; each new local certificate is valid for every resulting noncollision pursuer position.

After the least nonnegative residue of $d - 2$ modulo k such transfers, at most $k - 1$, the messenger is in column two. Apply the delivery certificate for the recipient’s type. This takes at most

$$8(k - 1) + 24 = 8k + 16$$

messenger moves. If a transfer encounters the actual recipient earlier, terminal receipt wins immediately and only shortens the play. Thus the argument guarantees finite delivery, not merely indefinite safe movement.

For $4 \leq k \leq 9$, the explicit finite-ring certificates verify every source and pursuer position for each of the five recipient types in column zero. Translating columns gives every recipient. These six checks, together with the uniform argument for $k \geq 10$, establish universality for all $k \geq 4$.

Finally, a universal graph with $n = 5k$ vertices and $2n$ arcs has

$$I(H_k) = n^2 - n - 2n = n(n - 3).$$

The vertex upper bound and the even arc-budget upper bound both equal this number. Hence both asserted extremal equalities follow. $\square$

Certificate reproduction and scope. The file `research/five-type-certificates.json` contains the fixed local arena, ranks, and chosen messenger moves, generated and directly checked by `src/five_type_certificate.py`. The independent implementation `src/five_type_local_verify.py` constructs the explicit alternating-state graph from the arc rules and checks every winning rank condition, as well as closure of the losing region. Its full ranks and selected moves are saved in `research/five-type-independent-local-certificates.json`, together with the implementation hash. Both implementations agree on the six local bounds above. The six finite bases are stored in `research/five-type-small-ring-certificates.json`, including complete arc lists, messenger-move layers, and explicit alternating-state rank certificates for the five representative recipient types. The script `src/five_type_small_certificates.py` reproduces those checks using both game solvers and verifies every certificate transition. The ordinary projection and composition proof explains why these finite arrays certify infinitely many rings. This theorem is not currently claimed as a Lean formalization.

Corollary 22 (sharp arc-budget leading term). As $m \rightarrow \infty$,

$$M_E(m) = \frac{m^2}{4} - O(m).$$

More explicitly, for every $m \geq 40$,

$$\frac{m^2}{4} - 6m \leq M_E(m) \leq \left\lfloor \frac{m}{2} \right\rfloor^2.$$

Proof. Set $k = \lfloor m/10 \rfloor \geq 4$ and $r = m - 10k$, so $0 \leq r \leq 9$. Take H_k together with r disjoint one-way arcs. Each added component has no indirect guaranteed pair, and additivity over components preserves $I(H_k)$. This graph has exactly m arcs and

$$I = 25k^2 - 15k = \frac{(m-r)^2}{4} - \frac{3(m-r)}{2} \geq \frac{m^2}{4} - 6m.$$

The upper bound follows because there are at most $\lfloor m/2 \rfloor$ possible sources and at most that many possible recipients of indirect guaranteed pairs. Thus the quadratic coefficient $1/4$ is sharp for arbitrary arc counts, not only the progression $m = 10k$. $\square$

Theorem 40 proves exact vertex maxima at every order from twelve and exact even arc maxima from 24 arcs; Corollary 45.1 proves exact odd arc maxima from 25 arcs. The padding argument above is retained as a direct proof of the leading term. The remaining small exceptions are stated in the current result summary.

3.3 Sparse endpoint attainment at every order from twelve

A second periodic construction admits controlled removal of vertices. Unlike a finite modification checked at one circumference, the removal is certified in a fixed local game that applies at every translated position. Separating the removals then provides all residue classes of the vertex count.

Theorem 40 (with finite certificates). For every integer $n \geq 12$, there is an explicit two-in/two-out-regular universally winning digraph on n vertices. Consequently

$$M_V(n) = M_E(2n) = M(n, 2n) = n(n-3).$$

Every source–recipient pair in the construction can be guaranteed within $9\lceil n/6 \rceil + 18$ messenger moves.

Proof. First construct a uniform family for $n \geq 85$. Begin with k columns, indexed modulo k , containing vertices (i, j) of types $0 \leq j \leq 5$. One permutation follows the Hamilton cycle

$$0_i \rightarrow 1_i \rightarrow 2_i \rightarrow 3_i \rightarrow 4_i \rightarrow 5_i \rightarrow 0_{i+1}.$$

The other permutation has arcs

$$\begin{aligned} 0_i &\rightarrow 5_i, & 1_i &\rightarrow 4_{i+1}, & 2_i &\rightarrow 1_{i+1}, \\ 3_i &\rightarrow 0_{i-1}, & 4_i &\rightarrow 2_{i+1}, & 5_i &\rightarrow 3_{i-1}. \end{aligned}$$

Choose columns S whose consecutive cyclic gaps are at least three. At each $h \in S$, remove 4_h and smooth its path separately in the two permutations:

$$\begin{aligned} 3_h \rightarrow 4_h \rightarrow 5_h &\longmapsto 3_h \rightarrow 5_h, \\ 1_{h-1} \rightarrow 4_h \rightarrow 2_{h+1} &\longmapsto 1_{h-1} \rightarrow 2_{h+1}. \end{aligned}$$

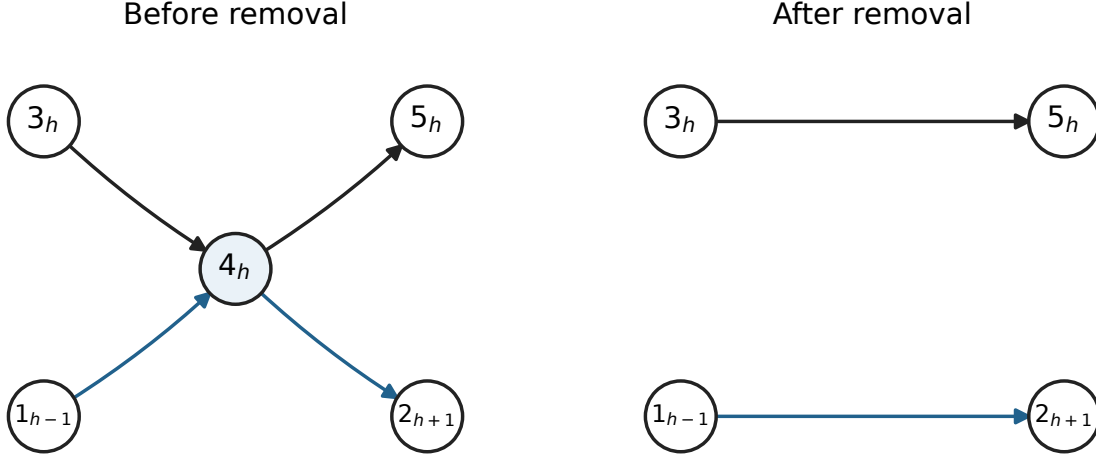


Figure 5: Remove one type-four vertex and reconnect each color separately. All undisplayed arcs stay unchanged. Spaced removals give every residue class of the total number of vertices; the local certificates prove that delivery remains possible.

All remaining arcs stay unchanged. The resulting graph has $6k - |S|$ vertices. Smoothing preserves each permutation’s bijectivity. There are no loops or coincident outgoing arcs: the modified first-permutation arc leaves type three for type five, while its other arc ends at type zero; the modified second-permutation arc leaves type one for type two two columns ahead, while its first-permutation arc ends at type two in the same column. Thus the graph remains simple and two-in/two-out regular.

We establish a uniform strategy when $k \geq 15$. Around an arbitrary reference column zero, restrict the messenger to columns $[-4, 4]$ and represent the pursuer in columns $[-6, 6]$, together with a state Ω for all outside positions. Retain both players’ waits and graph arcs. Messenger moves leaving its window are disallowed; pursuer moves leaving its window lead to Ω . From Ω allow waiting and entry at any existing vertex in columns $-6, -5, 5, 6$.

The removals inside the represented pursuer window form a subset of $\{-6, \dots, 6\}$ with consecutive gaps at least three. There are exactly 189 such subsets. For every one of these patterns, the following finite local claims have complete rank certificates:

Objective	Initial messenger column	Maximum messenger moves
Reach column zero safely at the start of a messenger turn	-1	9
Deliver to any specified existing vertex in column zero	-2	27

Each claim covers every existing initial messenger type and every noncolliding initial pursuer position, including Ω . Safe-transfer goals are seeded only at messenger turns, after the preceding pursuer response. Delivery goals instead use immediate receipt before collision. There are 1,287 certificates: seven objectives for each of the 189 patterns, with delivery to 4_0 omitted in the 36 patterns where that vertex is removed. An independent explicit alternating-state implementation

verifies all 560,885 required initial states and all 10,271,740 states of these finite games, including winning rank descent and losing-region closure.

We check that these local games represent every actual play on the ring. The thirteen pursuer columns are distinct and leave at least two exterior columns. Every graph arc changes the column by at most two, so a represented pursuer cannot jump across that exterior gap and land at the opposite represented boundary in one move. Every move from outside into the window lands in one of its two boundary layers, where Ω permits entry.

A removal outside the represented window does not change any represented-to-represented arc. The only change to an outward arc occurs when removing 4_7 redirects $1_6 \rightarrow 4_7$ to $1_6 \rightarrow 2_8$; both heads represent Ω . Outside-to-inside changes are already included among its permitted boundary entries. At the messenger boundary, any analogous changed outgoing arc still leaves the allowed messenger window. The local model therefore depends only on the removals inside the pursuer window. Their consecutive gaps are at least three, so it always has one of the 189 certified patterns.

Fix a recipient in column zero. Recenter and apply safe transfer repeatedly: each phase increases the messenger's column by one and ends on an uncaught messenger turn, ready for the next phase. After at most $k - 1$ phases it reaches column -2 , where the delivery certificate applies. The total is at most

$$9(k - 1) + 27 = 9k + 18.$$

If a transfer encounters the actual recipient earlier, play simply ends with delivery. This proves universal winning for every permitted set S .

Finally, given $n \geq 85$, put $k = \lceil n/6 \rceil$ and $r = 6k - n$. Then $k \geq 15$ and $0 \leq r \leq 5$. Choose

$$S = \{0, 3, \dots, 3(r - 1)\}$$

when $r > 0$, and take S empty when $r = 0$. The displayed gaps are three, and the wraparound gap is at least $15 - 12 = 3$. The graph has exactly n vertices and $2n$ arcs. Universal winning gives $I = n(n - 1) - 2n = n(n - 3)$, and the vertex and even arc-budget upper bounds give the three stated equalities for $n \geq 85$.

The complete primary certificates and their direct checker are `research/sparse-composition-multi-certificates.json.gz` and `src/sparse_composition_multi_certificate.py`. The independently written verifier uses direct arc formulas and an explicit alternating graph: `src/six_type_independent_spacing.py`, with full ranks in `research/six-type-independent-spacing-3-certificates.json.gz` and a summary in `research/six-type-independent-spacing-3-summary.json`. The graph definition, spacing argument and simulation are ordinary proofs; the two bounded local claims depend on the finite certificates. No Lean formalization of this theorem is asserted.

For completeness, the finite range $12 \leq n \leq 84$ is supplied by 73 explicit graphs. Each graph has $2n$ distinct nonloop arcs and both incoming and outgoing degrees two. Two independent exact game algorithms agree, and every actual-game rank obligation is checked for every recipient and every allowed initial pursuer. The finite witnesses also satisfy the displayed delivery-time bound. Their arc lists are in `research/sparse-finite-bridge-witnesses.json`, the complete ranks in `research/sparse-finite-bridge-certificates.jsonl.gz`, and the verification receipt in `research/sparse-finite-bridge-verification.json`. The unified generator `src/sparse_finite_bridge_construct.py` returns the corresponding finite witness below 85 and the uniform construction above it. This proves the remaining finite cases of the theorem. $\square$

Together with the small-order results of Theorem 46, only orders nine through eleven remain unclassified in the vertex problem. The separate Theorem 45 addresses the odd arc-budget bound. Both its finite bridge and uniform local certificates remain outside the current Lean coverage.

3.4 The odd endpoint and completion of its degree defect

The odd arc-budget equality theorem prescribes a particularly small defect: on $n = k + 1$ vertices there is one vertex u of outdegree one, a different vertex v of indegree one, all other indicated degrees are two, and the arc $u \rightarrow v$ is absent. The candidate extremal value is

$$M_E(2n - 1) = (n - 2)(n - 3).$$

The equality theorem reduces attainment to guaranteeing every missing pair whose source differs from u and whose recipient differs from v . It does not make arbitrary deletion from, or completion to, a two-regular graph preserve the relevant guarantees.

3.4.1 Two necessary structural properties

Lemma. Every odd-extremal graph with the preceding degree pattern is strongly connected. Moreover, for each vertex c , the graph with c deleted has a directed path from every $s \notin \{u, c\}$ to every $t \notin \{v, c\}$.

Proof. Each such ordered pair with $s \neq t$ is either a direct arc or a guaranteed missing pair. Against a pursuer initially at c and waiting there, any successful delivery path avoids c internally. Since its endpoints differ from c , it gives the asserted path in the deleted graph.

To prove strong connectivity without deleting a vertex, all vertices different from u already reach all vertices different from v . The unique predecessor b of v is different from u , because $u \rightarrow v$ is absent. Thus all vertices other than u reach v by first reaching b . The unique successor a of u differs from u , and u reaches every vertex by first going to a . $\square$

These are necessary conditions, not substitutes for the full game: the pursuer can move, so ordinary connectivity and stationary separators do not establish an attaining strategy.

3.4.2 Completing an odd extremizer can fail

Proposition (finite certified example). There is a graph G on 19 vertices with 37 arcs attaining $I(G) = 272 = M_E(37)$ such that adding the unique degree-completing arc yields a two-in/two-out graph which is not universally winning. Its completed indirect count is 276, whereas a universally winning two-regular graph on 19 vertices would have 304.

The example has vertex set $\{0, \dots, 18\}$ and the following outgoing neighbors:

Vertex	Outgoing neighbors	Vertex	Outgoing neighbors
0	5, 12	10	2, 17
1	15, 10	11	1, 6
2	7, 0	12	3, 8
3	1, 18	13	4, 5
4	15	14	4, 12
5	11, 9	15	0, 16
6	8, 13	16	7, 9
7	13, 17	17	14, 11
8	10, 16	18	2, 6
9	3, 14		

Its exceptional vertices are $u = 4$ and $v = 18$. The finite actual-game certificate verifies every guarantee required by odd equality and the count 272. The upper bound follows from Theorem 7 with $k = 18$.

There is also a short ordinary obstruction after adding $4 \rightarrow 18$. Fix recipient zero and start the messenger at 1, the pursuer at 4. Define

$$S_4 = \{1, 3, 5, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17\}, \quad S_{13} = \{6, 11, 12, 14\}.$$

The following four classes of messenger-turn states form a pursuer trap:

$$(s, 4) \ (s \in S_4), \quad (10, 18), \quad (17, 6), \quad (s, 13) \ (s \in S_{13}).$$

They contain the initial state and exclude every state where the messenger has reached zero. The pursuer acts as follows, in terms of the messenger's new vertex x :

Current pursuer	Response
4	If $x = 10$, move to 18. If $x = 15$ or 18, capture at x . Otherwise wait at 4.
18	If $x = 2$, capture there. If $x = 17$, move to 6. If $x = 10$, wait.
6	If $x = 11$ or 14, move to 13. If $x = 17$, wait.
13	If $x \in \{1, 3, 4, 8\}$, move to 4, capturing if $x = 4$. Otherwise wait at 13.

An immediate collision with the current pursuer already loses. Direct inspection of the outgoing-neighbor table shows that every other possible move either permits the stated capture or returns to one of the four trap classes. Waiting is included. Since infinite play without receipt loses, the messenger cannot deliver to zero. This explicitly identifies how the new arc helps the pursuer: the transition from cop position 4 to 18 after the messenger moves from 1 to 10 was unavailable before completion.

`src/odd_endpoint_completion.py` contains the fixed graph and runs both independent game solvers, checking every winning rank and every losing closure obligation. `research/odd-endpoint-completion-certificate.json` stores the complete finite certificates for the original and completed graphs. The ordinary trap proves nonuniversality without computation; the original graph's

attainment and the completed count 276 use the verified finite certificates. This example is not presently embedded in Lean. The subsequent construction settles the odd endpoint at every sufficiently large budget.

3.4.3 A useful degree-preserving search operation

Given a two-in/two-out graph, split one vertex w into vertices v, u . Assign one old incoming and one old outgoing arc to each new vertex, then add $v \rightarrow u$. This adds one vertex and one arc, giving exactly the odd equality degree pattern: v has indegree one and outdegree two, while u has indegree two and outdegree one. The forbidden opposite arc $u \rightarrow v$ is absent because the original graph was loopless.

This operation supplies a structured family to investigate, but it is not a preservation theorem. A split of the 20-vertex five-type graph attains $M_E(41) = 342$, while all twenty canonical splits of its 30-vertex member fail to attain the corresponding odd bound. Likewise, changing the two horizontal lane speeds to three produces attaining single-arc deletions at 40 and 50 vertices but failures at neighboring and larger orders. These counterchecks prevent promoting isolated success to a uniform construction. A different six-type family supplies the following uniform positive result.

3.4.4 A proved infinite family attaining the odd bound

Theorem (with finite local certificates). For every integer $q \geq 15$,

$$M_E(12q - 1) = (6q - 2)(6q - 3).$$

An explicit attaining graph has $n = 6q$ vertices and $2n - 1$ arcs. Every eligible pair can be guaranteed within $14q + 12$ messenger moves.

Proof. Use vertices (i, j) , with i modulo q and $0 \leq j \leq 5$. The first permutation follows the six types in order and then advances one column:

$$\alpha(i, j) = (i, j + 1) \ (j < 5), \quad \alpha(i, 5) = (i + 1, 0).$$

The second permutation is

$$\begin{aligned} \beta(i, 0) &= (i - 1, 0), & \beta(i, 1) &= (i - 1, 2), \\ \beta(i, 2) &= (i + 1, 5), & \beta(i, 3) &= (i + 1, 1), \\ \beta(i, 4) &= (i - 1, 4), & \beta(i, 5) &= (i + 1, 3). \end{aligned}$$

These permutations have distinct nonloop images at every vertex. Delete the single arc $u \rightarrow v$, where $u = (0, 0)$ and $v = (0, 1)$. The graph now has the exact odd equality degree pattern. We must guarantee every pair whose source differs from u and whose recipient differs from v .

Construct a local arena with messenger columns $[-4, 4]$, pursuer columns $[-6, 6]$, and a pursuer outside state Ω . The messenger may use only moves remaining in its represented columns. The pursuer may leave to Ω , wait there, and re-enter at any vertex in columns $-6, -5, 5, 6$. Include both the unchanged arena and, for each $h \in \{-6, \dots, 6\}$, the arena with the translated arc $(h, 0) \rightarrow (h, 1)$ deleted.

In every such arena the certificates prove two kinds of statement:

- From each vertex of column -1 other than the degree-one source, reach some vertex of column zero other than that source safely, at the beginning of a messenger turn, in at most 14 moves.

- From each vertex of column -2 other than the degree-one source, deliver to each recipient in column zero other than the degree-one recipient, in at most 26 moves.

All legal initial pursuer positions, including Ω , are quantified. Safe transfer goals exclude collision and are seeded only on messenger turns; delivery goals retain immediate terminal receipt. Two independent implementations check the full 97 objectives, all 44,850 required initial states, and every winning-rank and losing-closure condition. These are finite proof certificates for the displayed local statements.

For $q \geq 15$, the thirteen pursuer columns are distinct and at least two columns remain outside. Every actual move changes its column by at most one, so an outside pursuer can first re-enter only at a permitted boundary column. No represented move can jump across the outside gap. The actual game therefore projects into this stronger local arena. A deletion in the represented columns gives one of the listed translated modules; otherwise the module is unchanged.

Fix an eligible recipient. Repeated safe transfers advance the messenger one column at a time until it reaches the column two places before the recipient. At most $q - 1$ transfers are needed. Each transfer avoids the exceptional source even at its final state, so the next module's initial requirement continues to hold. The delivery module then applies. This proves finite delivery in at most $14(q - 1) + 26 = 14q + 12$ moves.

Finally, there are $(n - 1)^2$ pairs with an eligible source and recipient. Exactly $n - 2$ are diagonal. The unique arc leaving u and the unique arc entering v are distinct because $u \rightarrow v$ is absent, so precisely $(2n - 1) - 2 = 2n - 3$ arcs belong to this rectangle. Hence the number of eligible missing pairs is

$$(n - 1)^2 - (n - 2) - (2n - 3) = (n - 2)(n - 3).$$

All are guaranteed, and the odd arc-budget upper bound proves equality. $\square$

The primary proof data are generated by `src/odd_endpoint_certificates.py` and stored in `research/odd-endpoint-uniform-certificates.json.gz`. The independent checker `src/six_type_independent_odd.py` constructs its own graph and explicit alternating arena; its full ranks are in `research/six-type-independent-odd-certificates.json.gz`, with a summary and independent review in the adjacent files. The projection and finite composition argument are ordinary proofs. This result is not currently formalized in Lean. The next theorem removes the congruence restriction.

3.4.5 Every sufficiently large odd budget

Theorem 45 (the odd endpoint). For every integer $n \geq 145$,

$$M_E(2n - 1) = (n - 2)(n - 3).$$

Equivalently, for every odd $m \geq 289$,

$$M_E(m) = \frac{(m - 3)(m - 5)}{4}.$$

An explicit attaining graph has n vertices. Every pair required for attainment has a delivery strategy using at most $34\lceil n/6 \rceil + 27$ messenger moves. The construction and its composition proof are ordinary finite arguments; the bounded local strategy lemmas are supplied by independently verified finite certificates.

Proof. Put $q = \lceil n/6 \rceil$ and $r = 6q - n$, so $q \geq 25$ and $0 \leq r \leq 5$. We first describe a graph on the surviving vertices of $(\mathbb{Z}/q\mathbb{Z}) \times \{0, \dots, 5\}$. Its two initial permutations are

$$\alpha(i, j) = \begin{cases} (i, j+1) & j < 5, \\ (i+1, 0) & j = 5, \end{cases} \quad \beta(i, j) = (i + e_j, \sigma(j)).$$

Use the following four choices of (σ, e) , whose entries are in type order $0, 1, \dots, 5$:

Pattern	σ	e
A	$(0, 2, 5, 1, 4, 3)$	$(-1, -1, 1, 1, -1, 1)$
B	$(0, 5, 1, 4, 3, 2)$	$(-1, 1, 1, -1, -1, 1)$
C	$(3, 5, 0, 2, 4, 1)$	$(-1, 1, -1, -1, -1, 0)$
D	$(4, 1, 5, 2, 0, 3)$	$(-1, -1, 1, 1, -1, 1)$

For a set H of deleted vertices, smooth each permutation separately: from a surviving vertex, follow that permutation until first reaching another surviving vertex, and use the resulting arc. Finally delete one specified arc of the smoothed graph. The complete choices are:

r	Pattern	Tail u of deleted arc	Which arc	Direction d
0	A	$(0, 0)$	α	1
1	B	$(1, 4)$	β	1
2	C	$(0, 5)$	β	-1
3	D	$(0, 1)$	α	1
4	D	$(0, 1)$	α	1
5	C	$(0, 5)$	β	-1

The corresponding deleted vertex sets are:

- $r = 0$: $H = \emptyset$.
- $r = 1$: $H = \{(0, 3)\}$.
- $r = 2$: $H = \{(0, 0), (0, 1)\}$.
- $r = 3$: $H = \{(0, 2), (0, 3), (0, 4)\}$.
- $r = 4$: $H = \{(-1, 4), (0, 2), (0, 3), (0, 4)\}$.
- $r = 5$: $H = \{(0, 0), (0, 1), (1, 0), (1, 1), (1, 5)\}$.

Write v for the head of the final deleted arc. Every type cycle of each displayed σ has a nonzero total displacement e with $|e|$ no greater than its length ℓ . Its cyclic lift has length $\ell q / \gcd(q, e) \geq q$, whereas α is a cycle of length $6q$. At most five vertices are removed, so no permutation cycle disappears. The smoothing operation therefore preserves the permutation property on surviving vertices. Inspection of these six fixed patterns shows that the two smoothed images at each surviving vertex are distinct, are not the vertex itself, and change the column by at most two. The changes to outgoing arcs are confined to tails in columns $[-2, 2]$; removed vertices lie in columns $[-1, 1]$. These are finite checks on the displayed tables, independent of q . Because $q \geq 25$, reducing these local coordinates modulo q introduces neither a coincidence nor a loop. Thus the final graph has $n = 6q - r$ vertices, $2n - 1$ arcs, precisely one outdegree-one vertex u , precisely one indegree-one vertex v , and all other corresponding degrees two. Moreover $u \neq v$ and $u \rightarrow v$ is absent.

We now specify the finite strategy lemmas and explain exactly how they apply to arbitrary q . Work first in the infinite column strip. In a local arena the messenger is restricted to columns $[-5, 5]$; the pursuer has columns $[-7, 7]$ together with an outside state Ω . Inside these windows both players have their actual graph moves and may wait. A pursuer move leaving its window goes to Ω , from which the pursuer may wait or re-enter at any surviving vertex in columns $-7, -6, 6, 7$. Include the unchanged arena and every translation of the finite defect whose anchor belongs to $\{-10, \dots, 10\}$.

For $r \leq 4$, the certificates prove the following statements for every such arena, every admissible messenger vertex in the stated source column, and every initial pursuer position different from that vertex, including Ω :

- **Safe transfer:** from column $-d$, reach a surviving vertex of column zero different from u , at the start of a messenger turn.
- **Delivery:** from column $-2d$, deliver to any surviving recipient in column zero different from v .

In both statements the starting messenger vertex must differ from u . For $r = 5$, where $d = -1$, safe transfer instead permits either column zero or column -1 as its final column, and delivery is guaranteed from either column 2 or column 3. The exclusion of u at the starting and final transfer states remains unchanged. Safe transfer goals exclude collision and are terminal only at the start of a messenger turn. Delivery goals give receipt priority over collision, exactly as in the original game.

The maximum certified move counts are:

r	Safe transfer	Delivery
0	14	26
1	9	22
2	17	26
3	34	23
4	28	27
5	9	26

The primary checker verifies all 903 local objectives with the same windows for all six rows. A separately written checker verifies the five nonempty deletion patterns in 750 objectives, covering 425,176 required initial states and 7,590,092 explicit alternating states. The unchanged-vertex pattern has its earlier independent 97-objective certificate. Every certificate includes the full attractor ranks and checks each existential messenger move and every universal pursuer response. Thus the displayed local statements are finite lemmas, not inferences from testing increasing ring lengths.

For the projection to a cyclic graph, represent the fifteen pursuer columns by their consecutive integer lifts. All actual moves have column jump at most two. An outside pursuer can therefore first enter only one of the four boundary columns allowed from Ω . Since $q \geq 25$, the outside gap has at least ten columns, so no move can cross that gap between the two represented ends. Messenger positions remain in their eleven distinct represented columns, and Ω cannot conceal a collision there. Consequently each actual pursuer move has a permitted projected response in the stronger local arena.

The only defect anchors which can affect a represented outgoing move are within nine columns of the arena center, since changed tails lie within two columns of their anchor. Vertex deletions

and the two degree exceptions are covered by the same bound. Testing anchors from -10 to 10 therefore covers every relevant configuration. This interval has length 21 , less than q , so at most one periodic copy of the defect can occur in it. If no copy occurs, the unchanged arena applies. These observations prove the projection for every cyclic position of the local window.

Fix a source other than u and a recipient other than v . For $r \leq 4$, repeatedly apply safe transfer in direction d until the messenger is two columns before the recipient. At most $q - 1$ transfers are needed, after which the delivery module applies. Each transfer ends at a messenger turn away from u , so it satisfies its next module's initial requirements regardless of the pursuer's history.

For $r = 5$, measure the messenger's column distance to the recipient in the leftward direction by an integer $a \in \{0, \dots, q - 1\}$. Delivery applies as soon as $a \in \{2, 3\}$. Otherwise a transfer decreases this distance by one or two modulo q . If $a \geq 4$, this cannot pass below 2 without first reaching 2 or 3 . If initially $a = 0$ or 1 , at most two transfers either reach those distances or wrap to $q - 1$ or $q - 2$, after which the preceding decreasing argument applies. In every case at most q transfers suffice. This proves finite delivery, uniformly bounded by $34q + 27$ moves.

Every missing pair with source different from u and recipient different from v is therefore guaranteed. Their number is $(n - 1)^2 - (n - 2) - (2n - 3) = (n - 2)(n - 3)$, as in the preceding theorem. The odd arc-budget upper bound proves equality. $\square$

The generator `src/odd_endpoint_all_residues.py` supplies the six graph families and the complete primary proof data in `research/odd-endpoint-all-residues-certificates.json.gz`. `src/six_type_independent_odd_residues.py` independently constructs the smoothed graphs and explicit alternating games; its full proof data and summary are `research/six-type-independent-odd-residues-certificates.json.gz` and `research/six-type-independent-odd-residues-summary.json`. The original six-type pattern is additionally checked by `src/six_type_independent_odd.py`. These certificates and the ordinary projection argument establish an explicit all-order result; this new endpoint theorem has not yet been formalized in Lean.

Corollary 45.1 (the odd endpoint from 25 arcs). For every $n \geq 13$,

$$M(n, 2n - 1) = M_E(2n - 1) = (n - 2)(n - 3).$$

Consequently the exact odd arc-only extremum is

$$M_E(m) = \frac{(m - 3)(m - 5)}{4} \quad (m \geq 25 \text{ odd}).$$

Proof. Theorem 45 covers $n \geq 145$. For every one of the remaining 132 integers $13 \leq n \leq 144$, the supplementary data give a fixed simple graph with $2n - 1$ arcs, one outdegree-one vertex u , one different indegree-one vertex v , all other corresponding degrees two, and $u \rightarrow v$ absent. The displayed six-type residue families supply most of these graphs. Deleting one arc from independently verified finite two-regular examples supplies the remaining orders from 19 upward. Six further fixed witnesses cover orders 13 through 18.

For every recipient other than v , the finite certificates specify an integer rank for every explicit messenger-turn and pursuer-turn state. Terminal receipt has rank zero, nonterminal collision has rank -1 , and every other nonnegative messenger rank has a legal successor of smaller nonnegative rank. Every legal successor of a nonnegative pursuer rank has smaller nonnegative rank. The direct checker verifies these conditions against the original graph, together with the entire complementary losing closure. Every initial messenger state whose source differs from u and whose pursuer position

differs from the source has nonnegative rank. Strict rank descent therefore forces finite delivery from each required state against every pursuer policy. This proves all required guarantees for each listed graph.

The full check contains 10,236 recipient objectives and 215,967,810 explicit alternating states. The same degree count as in Theorem 45 then gives $(n - 2)(n - 3)$ guaranteed missing pairs. The arc-only upper bound applies to all support sizes, and hence also to the fixed support size n . This proves both equalities. $\square$

The 126 graphs at orders 19 through 144 are in `research/odd-endpoint-finite-bridge-graphs.json`. Their complete ranks are in the streamable archive `research/odd-endpoint-finite-bridge-certificates.jsonl.gz`, with source hashes and a full numerical summary in `research/odd-endpoint-finite-bridge-verification.json`. The producer `src/odd_endpoint_finite_certify.cpp` uses an explicit alternating-game attractor. The separate direct verifier `src/odd_endpoint_finite_verify.py` checks the original legal moves; it does not rerun or trust the producer's queue logic. Its accelerated array checks have an equivalent standard-library scalar mode. Scalar replay of all objectives at orders 19 through 24 also passes, and the earlier two independent game algorithms agree with selected saved certificates at orders 19, 28 and 144. The complete direct rank checks, rather than these additional sample comparisons, prove the finite bridge. The six smaller witnesses are fully checked by both original game algorithms and the direct transition verifier in `src/odd_endpoint_small_verify.py`; their complete proof data and summary are `research/odd-endpoint-small-certificates.json.gz` and `research/odd-endpoint-small-verification.json`. These finite certificates have not yet been imported into Lean.

Proposition (small odd exceptions; exhaustive finite check). For $6 \leq k \leq 11$, the general odd upper bound is not attained:

$$M_E(2k + 1) \leq k(k - 3) + 1.$$

In particular, $M_E(23) \leq 89$.

Proof. Equality in the upper bound $k(k - 3) + 2$ would, by the equality characterization, have precisely $n = k + 1$ nonisolated vertices, one outdegree-one vertex u , one distinct indegree-one vertex v , all other corresponding degrees two, and $u \rightarrow v$ absent. Add this missing arc. The resulting two-in/two-out digraph is a bipartite two-regular graph when its sources and recipients are represented in separate parts. Each of its bipartite cycles has even length, so alternating edge colors decompose it into two permutation factors A, B . Choose A to contain $u \rightarrow v$ and label $u = 0, v = 1$.

Up to relabeling fixing these two vertices, the A -cycle through zero is $(0, 1, \dots, \ell - 1)$ for one $\ell \geq 2$. Its other cycles are represented by an integer partition of $n - \ell$ into parts at least two. For each such choice, enumerating all permutations B with $B(i) \neq i, A(i)$ covers every candidate graph; remove $0 \rightarrow 1$ before applying any game test.

Two necessary obstructions safely prune this enumeration. Once the outgoing neighborhood of an eligible source s is complete, if some $c \neq s$ has $N^+(s) \subseteq N^+[c]$ and there is an eligible missing recipient for s , the pursuer can wait at c and capture the first nonwaiting messenger move. Once the incoming neighborhood of an eligible recipient t is complete, if $N^-(t) \subseteq N^+[c]$ and there is an eligible source outside $N^-(t) \cup \{t, c\}$, waiting at c until the messenger first enters $N^-(t)$ likewise permits capture. In a partial assignment the cop's outgoing row may be incomplete: coverage already present persists under later additions. The source or recipient neighborhood used in the

inclusion is always complete. The code checks explicitly that the required eligible missing partner exists.

All remaining completed graphs are evaluated by the exact finite reachability game, with receipt priority and finite termination. Enumeration finishes at every $7 \leq n \leq 12$ and finds no attaining graph. Its counts are:

n	Distinguished cycle partitions	Partial nodes	Completed graphs surviving local obstructions
7	7	756	0
8	11	6,338	21
9	15	59,375	2,092
10	22	800,507	64,992
11	30	10,079,292	767,210
12	42	164,276,910	16,902,091

These finite exclusions and the equality characterization prove the claimed strict bounds. $\square$

The complete enumerator is `src/odd_endpoint_exhaustive.cpp`; its finished receipts are `research/odd-endpoint-small-exhaustive.json` and `research/odd-endpoint-n12-exhaustive.json`. A known attaining 13-vertex graph passes every partial pruning test as a positive control. The independent review `research/odd-endpoint-twelve-independent-audit.md` checks the permutation coverage, both pruning arguments, and the final game semantics. It is an independent source and proof audit, not a separately implemented repetition of the enumeration. Neither this result nor the preceding odd endpoint certificates currently has Lean coverage. The proposition does not determine the exact small odd maxima.

The general support reduction in Theorem 57 and the exact finite exclusions in Theorem 53 supersede the earlier bespoke 23-arc reductions and intermediate bounds. Their original arguments and receipts are preserved in `research/history/manuscript-superseded-arguments-2026-09-13.md`. The following eleven-vertex core lemma remains a dependency of Theorem 53.

Lemma (eleven-vertex cores). Every graph on 11 vertices with 22 or 23 arcs satisfies $I(G) \leq 81$.

Proof. A vertex of outdegree at most one leaves at most ten contributing source rows, each of size at most eight, giving $I \leq 80$. A low-indegree vertex gives the same column bound. We may therefore assume minimum indegree and outdegree two. At 22 arcs all degrees equal two. At 23 arcs exactly one vertex has outdegree three and exactly one has indegree three.

If a source camping obstruction occurs, at least eight missing pairs fail at 22 arcs, and at least seven fail at 23 arcs. If a recipient camping obstruction occurs at t with pursuer position c , every source outside $N^-(t) \cup \{t, c\}$ fails. This gives at least seven failed pairs at 22 arcs and at least six at 23 arcs. The total numbers of missing pairs are 88 and 87, respectively, so every graph with one of these obstructions has $I \leq 81$.

It remains to enumerate the graphs with neither obstruction. The 22-arc case uses the two-permutation decomposition already described. For 23 arcs, the bipartite graph still has a perfect matching: a set S of source vertices emits at least $2|S|$ edges, whereas a set of possible heads T can

receive at most $2|T| + 1$. Thus $|N(S)| \geq |S|$, which implies a perfect matching by the augmenting-path criterion. Its removal leaves residual degree one at every source except the unique degree-three source, where two choices remain; recipient capacities are analogous.

Fix the matching permutation to each cycle partition with no singleton. Its cycle rotations and permutations of equal-length cycles allow the degree-three source to be placed at one representative vertex for each distinct cycle length. Enumerate all positions of the degree-three recipient, then every residual row subject to the prescribed column capacities. This covers every graph in question. Only complete source rows or recipient columns are used in camping prunes; a partial cop row suffices because any coverage already present persists under later additions.

The complete 22-arc enumeration has 120,302 graphs surviving these prunes, with maximum $I = 66$. The complete 23-arc enumeration has 792,080 such graphs, with maximum $I = 74$. Both use the exact finite game recurrence and were freshly recompiled and replayed with all counts and witnesses matching. Thus the graphs surviving the prunes also have $I \leq 81$, proving the lemma. $\square$

The source and mathematical audit, including the matching argument, symmetry reduction and rollback checks, is `research/odd-endpoint-eleven-enumeration-audit.md`. The enumerators are `src/small_exceptional_regular_max.cpp` and `src/small_exceptional_one_excess.cpp`; matching replay receipts and hashes are indexed by `research/odd-endpoint-eleven-enumeration-audit.json`. The saved maximizing witnesses additionally pass both original game solvers and all direct rank checks. This is independent review and re-execution of the same enumeration sources, rather than a second independently implemented enumeration.

Theorem 53 combines this eleven-vertex bound with Theorem 57 and the three finite exceptional-family exclusions to prove $M_E(23) = 83$. Its verified twelve-vertex witness also proves $M(n, 23) = 83$ for every $n \geq 12$ after adding isolates.

3.5 A packed sparse construction with optional arcs

Deleting a short, specified cluster of vertices from the six-type family allows an explicit sparse construction at every order at least 115. Near each deletion, only seven optional arrows must be suppressed. The local strategies allow both of two adjacent columns as entries for final delivery.

Theorem 52 (with finite local certificates). Let $n \geq 115$, put $k = \lceil n/6 \rceil$ and $r = 6k - n \in \{0, \dots, 5\}$. For every integer

$$2n \leq m \leq 20k - 11r,$$

there is an explicit universally winning digraph with n vertices and m arcs. Consequently

$$M(n, m) = n(n - 1) - m.$$

The construction has $2n$ compulsory arcs and exactly $8k - 9r$ optional arcs; **every subset** of the optional arcs works. Its range includes every $2n \leq m \leq 3n$, and every delivery takes at most

$$9\lceil n/6 \rceil + 21$$

messenger moves.

Proof. We specify the graph, prove the exact arc count, describe the finite strategy certificates and their complete scope, and lift those strategies to every order in the theorem.

3.5.1 The base graph and the seven exclusions

Begin with the six-type graph on $(\mathbb{Z}/k\mathbb{Z}) \times \{0, \dots, 5\}$. Write j_i for (i, j) . The first permutation is the Hamilton cycle

$$0_i \rightarrow 1_i \rightarrow 2_i \rightarrow 3_i \rightarrow 4_i \rightarrow 5_i \rightarrow 0_{i+1},$$

and the second permutation has arrows

$$0_i \rightarrow 5_i, \quad 1_i \rightarrow 4_{i+1}, \quad 2_i \rightarrow 1_{i+1}, \quad 3_i \rightarrow 0_{i-1}, \quad 4_i \rightarrow 2_{i+1}, \quad 5_i \rightarrow 3_{i-1}.$$

Delete 4_h in the canonical columns

$$S = \{0, 3, \dots, 3(r-1)\},$$

with $S = \emptyset$ when $r = 0$. Reconnect each permutation through each deleted vertex. The changed arrows are

$$3_h \rightarrow 5_h, \quad 1_{h-1} \rightarrow 2_{h+1}.$$

There are exactly $n = 6k - r$ vertices and $2n$ base arcs. Both indegree and outdegree remain two. The displayed formulas, $k \geq 20$, and the gaps of at least three between deleted columns show that the two outgoing arrows at each surviving vertex are distinct and nonloop. In particular the smoothed type 1 arrow has head of type 2 two columns ahead, distinct from its type 2 head in the same column.

The available optional arrow types $(i, j) \rightarrow (i + d, t)$ are

$$(j, d, t) \in \{(1, -3, 5), (3, -3, 5), (4, -3, 0), (4, 3, 1), (2, 3, 0), (3, 3, 0), (3, -3, 0), (5, -2, 0)\}.$$

Omit arrows whose source was deleted. In addition, for each $h \in S$, omit precisely the following seven arrows, where a quadruple (δ, j, d, t) denotes $(h + \delta, j) \rightarrow (h + \delta + d, t)$:

$$\begin{aligned} \mathcal{B} = \{ & (0, 1, -3, 5), (0, 3, -3, 5), \\ & (1, 1, -3, 5), (1, 3, -3, 5), (1, 3, 3, 0), \\ & (2, 3, -3, 5), (2, 4, -3, 0) \}. \end{aligned}$$

All column indices in this definition are modulo k . These optional arcs are distinct nonloop arcs outside the base graph, and their target types are 0, 1, or 5, so their targets always survive.

Originally there are $8k$ optional arrows. Each deletion removes the two arrows with its absent type 4 source and the seven further arrows in $\mathcal{B}$. None of the seven has the deleted source. The source-column sets $\{h, h+1, h+2\}$ for different holes are disjoint, including across the cyclic gap, so these losses do not overlap. Thus the optional count is exactly $8k - 9r$. Moreover

$$8k - 9r - n = 2k - 8r \geq 0$$

because $k \geq 20$ and $r \leq 5$. The full envelope has

$$2n + (8k - 9r) = 20k - 11r$$

arcs and contains every budget through $3n$.

3.5.2 The finite local strategy statements

Unwrap a window around a prospective destination column, now called zero. Retain messenger columns $[-5, 5]$ and cop columns $[-8, 8]$. Messenger exits are disallowed. A cop exit enters an outside state Ω ; from Ω the cop may wait or reenter any existing vertex in columns $-8, -7, -6, 6, 7, 8$.

For a local deletion crop $H \subseteq [-8, 8]$, construct the smoothed base arrows literally. The messenger uses only these base arrows, together with waiting. The cop receives all surviving optional arrows, except the seven relative exclusions for each hole in H , as well as its base arrows and waiting. The finite certificates prove the following statements for every crop that can arise from the canonical construction:

- From every existing vertex in column -1 , the messenger can reach column zero within nine messenger moves and hand off a messenger-turn state in which it is distinct from the cop.
- From every existing vertex in column -3 , the messenger can deliver to any specified existing vertex in column zero within 39 messenger moves.
- From every existing vertex in column -2 , the corresponding delivery bound is 38 messenger moves.

Each assertion quantifies over every cop position other than the source, including Ω . Receipt is terminal before capture; safe transfer instead requires surviving the cop reply before the handoff.

Here the finite evidence consists of complete ranks and messenger policies, not just successful simulations. There are 101 crops and 696 objectives: 101 safe-transfer objectives and 595 specified-target delivery objectives. Every delivery objective checks both entry columns. In a winning nonterminal messenger state, the supplied legal move strictly decreases the rank. At a winning cop state, every possible reply leads to a noncollision messenger state of no greater rank. Zero messenger ranks have the required terminal condition. These obligations prove all the stated local strategy bounds.

3.5.3 Why those crops cover every $k \geq 20$

Enumerate $k = 20, \dots, 29$, $r = 0, \dots, 5$, and every choice of center. Cropping the canonical hole sets to $[-8, 8]$ gives exactly 101 distinct sets. For any $k \geq 29$, the gap from the last hole back to the first is at least

$$k - 3(r - 1) \geq k - 12 \geq 17.$$

A 17-column crop has span 16, so it cannot contain holes from both sides of this gap. Its holes form a contiguous portion of one arithmetic progression of step three, containing at most five terms. Every such cropped progression already occurs in the $k = 29$ enumeration, by taking the appropriate number of holes and translating the center. No new local pattern can therefore occur at larger k .

This is a statement about the specified canonical cluster. It does not assert that arbitrary deletion sets with minimum spacing three admit these local strategies.

3.5.4 Projection of every actual cop move

Map an actual cop inside columns $[-8, 8]$ to its exact local vertex and every other cop position to Ω . All 17 represented columns are distinct modulo $k \geq 20$. Smoothed base arrows change a column by at most two; optional arrows change it by at most three. The exterior gap between the represented ends has displacement $k - 16 \geq 4$, so an arrow cannot wrap directly between those ends

while evading the local description. Equivalently, an inside-to-inside wrapped jump would require ordinary column difference at least $k - 3 \geq 17$, exceeding the represented span 16.

An outside cop can reenter only a column of absolute index at least six: every possible predecessor within three columns of $[-5, 5]$ is already represented. All possible reentries are thus among the three permitted boundary layers. Every actual cop reply has a local counterpart.

Omitting holes outside the crop is conservative. A hole immediately to the right may change a represented boundary arrow from an outside head to another outside head; both project to Ω . No other omitted hole changes a represented-to-represented base arrow. An omitted hole to the left can suppress optional arrows through its relative exclusions; the cropped arena instead permits these extra cop moves. Messenger moves inside $[-5, 5]$ are modeled literally. Consequently the projection neither grants the messenger an illegal move nor removes an actual cop response, and it preserves collisions and both terminal conditions.

The certificates grant the cop every permitted optional arrow at once, while the messenger uses only compulsory arrows. They therefore remain valid after selecting any subset of the optional arrows.

3.5.5 Global delivery and the time bound

Let the recipient's column be t . Repeated safe transfers advance the messenger one column at a time. Stop at the first of columns $t - 3, t - 2$, then use its delivery policy. If the messenger starts in either entry column, no transfer is needed.

For completeness, if its initial column is a , put $p = (t - 3 - a) \bmod k \in \{0, \dots, k - 1\}$. The forward distances to the entry columns are p and $(p + 1) \bmod k$. If $p \leq k - 2$, the first is at most $k - 2$; if $p = k - 1$, the second is zero. Thus at most $k - 2$ transfers are required. Safe transfer ends after the cop reply at a noncollision messenger-turn state, exactly the condition required for the next policy. The cop may occupy another vertex of that same column; all such states are already included. Earlier receipt terminates the game and only improves the bound.

The total number of messenger moves is at most

$$9(k - 2) + 39 = 9k + 21.$$

Hence every missing ordered pair is guaranteed. The universal missing-pair upper bound gives $M(n, m) = n(n - 1) - m$, as claimed. $\square$

Certificate boundary and reproducibility. The complete finite ranks and policies are saved in `research/dense-phase2-sparse-two-entry-certificates.json.gz`. The checker `src/dense_phase2_sparse_two_entries.py --check` reconstructs every arena, checks exact coverage of the 101 crops and 696 objectives, and directly verifies every supplied rank, policy, and required entry state. The reconstruction spells out all six smoothed source-type maps; the independent generator uses explicit messenger-turn and cop-turn states with an AND/OR priority queue, rather than the earlier collapsed-round solver. The summary is `research/dense-phase2-sparse-two-entry-check.json`. Literal global graph and exact optional-count checks are recorded in `research/dense-phase2-sparse-global-map.json`.

The smaller messenger window $[-4, 4]$ does not establish all the new column -3 entry statements. Enlarging it to $[-5, 5]$ is part of this proof, and the failed smaller-window checks are retained in the research log. The earlier broad exclusion buffers and their projection proofs remain research history;

they are unnecessary for the theorem above. Negative certificate ranks make no assertion of global impossibility or optimal delivery time. The graph construction, crop coverage, projection, and global composition are ordinary proofs using finite rank certificates; this theorem is not presently formalized in Lean.

3.6 A consecutive interval above the two-regular endpoint

The five-type construction can tolerate a controlled set of additional arcs. The proof gives the pursuer every additional move while restricting the messenger to the original moves. This verifies all intermediate graphs at once, without assuming that winning is monotone under adding arcs.

Theorem 36 (with finite local certificates). Let $n = 5k$ with $k \geq 10$. For every integer

$$2n \leq m \leq \frac{11n}{5},$$

there is an explicit graph with n vertices and m arcs satisfying

$$M(n, m) = n(n - 1) - m.$$

Every pair can be guaranteed within $8k + 18$ messenger moves.

Proof. Use the five-type graph H_k of Theorem 21 on vertices (i, j) , where i is modulo k and $0 \leq j \leq 4$. Its two outgoing arcs are

$$\begin{aligned} \alpha(i, j) &= (i, j + 1) & (j < 4), & & \alpha(i, 4) &= (i + 1, 0), \\ \beta(i, 0) &= (i + 1, 0), & \beta(i, 1) &= (i - 1, 2), \\ \beta(i, 2) &= (i - 1, 4), & \beta(i, 3) &= (i - 1, 3), & \beta(i, 4) &= (i - 1, 1). \end{aligned}$$

There are $2n$ arcs. Add any chosen subset of the k further arcs

$$(i, 0) \longrightarrow (i - 1, 4).$$

They are distinct from the original arcs and from each other, so every integer budget in the stated interval is obtained.

We verify the stronger asymmetric game in which the messenger may use only α, β and waiting, while the pursuer may also use all k additional arcs. Construct the same local arena as in Theorem 21: messenger columns $-3, \dots, 3$, pursuer columns $-4, \dots, 4$, and a pursuer state Ω representing the outside. The messenger is constrained to its local columns. The pursuer may leave into Ω , wait there, and re-enter at any vertex in either boundary column -4 or 4 . Within its columns it has the two original arcs and the additional arc above whenever its type is zero.

Two independently implemented finite attractor computations verify the following objectives against every legal initial pursuer state, including Ω :

Objective	Initial messenger column	Maximum messenger moves
Reach column zero safely at the beginning of a messenger turn	1	8
Deliver to (0, 0)	2	26
Deliver to (0, 1)	2	19
Deliver to (0, 2)	2	17
Deliver to (0, 3)	2	17
Deliver to (0, 4)	2	18

Each row covers all five initial messenger types and all 45 legal pursuer positions per type, for 225 required initial states. The transfer goal is seeded only at a messenger turn after the pursuer response, with unequal positions. Delivery goals use immediate receipt before collision, as required by the game. The rank checks verify every selected messenger move and every permitted pursuer response; these are complete finite certificates for the displayed statements.

For $k \geq 10$, the nine represented pursuer columns are distinct on the ring. All moves, including the additional ones, change the column by at most one. Therefore an actual pursuer outside those columns can first enter only a boundary column; its behavior projects into the adversarial Ω state. The actual messenger can follow every certified original move. The local certificates consequently apply at every translated column of every such ring, independently of which additional arcs the actual graph contains.

Fix a recipient in column zero. Repeatedly translate and apply the safe-transfer certificate to move the messenger one column backward until it reaches column two. This needs at most $k - 1$ transfers, each leaving a legal messenger-turn state for the next certificate. Then apply the appropriate delivery certificate. The total is at most

$$8(k - 1) + 26 = 8k + 18.$$

Thus every pair is guaranteed, and all $n(n - 1) - m$ missing pairs are counted. The missing-pair upper bound proves equality. $\square$

The primary generator and full ranks are `src/five_type_sparse_interval.py` and `research/five-type-sparse-interval-certificates.json`. The independent explicit alternating-state implementation is `src/five_type_augmented_verify.py`, with its full certificate in `research/five-type-augmented-independent-certificates.json`. The ring projection and concatenation are ordinary proofs; the bounded local facts depend on these finite certificates. This theorem is not presently formalized in Lean, and it makes no claim for the small rings $k = 4, \dots, 9$ or for budgets beyond $11n/5$.

Theorem 37 (with finite local certificates). Let $n = 5k$ with $k \geq 16$. At every integer budget $2n \leq m \leq 3n$, there is an explicit graph attaining

$$M(n, m) = n(n - 1) - m.$$

Every pair can be guaranteed within $8k + 16$ messenger moves.

Proof. Keep the original five-type graph, and add any subset of the n arcs

$$(i, j) \longrightarrow (i + 3, 0), \quad 0 \leq j \leq 4.$$

They are pairwise distinct and are not original arcs. It suffices to win with the messenger restricted to the original graph and the pursuer allowed all the additional arcs.

Use messenger columns $[-3, 3]$, pursuer columns $[-6, 6]$, and the outside state Ω . Give the represented pursuer its original and added moves; moves leaving the interval enter Ω . From Ω permit waiting and re-entry at every vertex in the six columns $-6, -5, -4, 4, 5, 6$. Independent synchronous and explicit alternating-state certificates verify safe transfer from column one to zero in at most eight moves, and delivery from column two to the five recipient types at zero in at most 24, 16, 16, 17, 18 moves. Each objective covers all 325 legal initial states: five messenger types and 65 pursuer positions per type. The transfer ends on a safe messenger turn, and delivery has immediate terminal priority, exactly as in Theorem 36.

We check the larger-jump projection explicitly. The thirteen represented pursuer columns are distinct when $k \geq 16$, and at least three consecutive columns lie outside them. No move, whose column displacement is at most three, can jump over that entire outside gap and land at the opposite represented boundary. Thus a represented move projects either to its represented destination or to Ω . An outside pursuer can enter the represented interval only in one of the six allowed boundary columns. Every actual pursuer behavior is therefore included in the finite arena. This gap condition is why the proof uses $k \geq 16$, not merely enough columns to make the represented labels distinct.

At most $k - 1$ safe backward transfers reach column two relative to the recipient. The final delivery costs at most 24 further moves, giving $8(k - 1) + 24 = 8k + 16$. The strategy is available in every intermediate graph, and its pursuer has no more moves than the certified adversary. Hence all pairs are guaranteed at each of the $n + 1$ integer budgets. Counting missing pairs proves equality. $\square$

The full primary ranks are `research/five-type-full-sparse-interval-certificates.json`, generated by `src/five_type_sparse_interval.py`. The independent 4620-state implementation and certificate are `src/five_type_long_augmented_verify.py` and `research/five-type-long-augmented-independent-certificates.json`. The statement is a uniform theorem with a finite local proof ingredient, not an extrapolation from tested circumferences. No new Lean coverage is claimed. For orders divisible by five with $n \geq 80$, Theorems 34 and 37 combine into a consecutive explicit interval starting at $2n$ and reaching order $n^{4/3}$.

3.7 Explicit attainment at every budget in a sparse interval

The circulant construction gives an exact joint extremum, not only a lower bound on the vertex maximum: for every $n \geq 25$, Theorem 11 and Lemma 12 already show that $M(n, 3n) = n(n - 4)$. The next result fills consecutive budgets above $3n$. Its proof controls the extra pursuer moves; arbitrary additions of arcs would not justify that conclusion.

Theorem 34. For every $n \geq 25$,

$$M(n, 3n) = n(n - 4).$$

Moreover, if $d \geq 4$ and

$$n > d^3 + 2d,$$

then at every integer budget $3n \leq m \leq dn$ there is an explicit graph attaining

$$M(n, m) = n(n - 1) - m.$$

In particular, this holds throughout $3n \leq m \leq 4n$ for every $n \geq 73$. More generally the explicit interval reaches arc counts of order $n^{4/3}$.

Proof. Say that a set $A \subseteq \mathbb{Z}/n\mathbb{Z}$ has distinct nonzero differences when

$$a - b = a' - b' \neq 0, \quad a, b, a', b' \in A,$$

implies $(a, b) = (a', b')$. For such a set, two distinct translates $x + A$ and $y + A$ intersect in at most one point: two intersections would give two representations of the nonzero difference $y - x$.

Choose h as in Lemma 12 and put $a = (h - 1)/2$. The set

$$A_0 = \{0, 1, a, h\}$$

has distinct nonzero differences by the proof of Theorem 11. The base graph B has steps $1, a, h$. For any larger set A with distinct nonzero differences, let H_A have the nonzero steps in A ; we will keep $0 \in A$ throughout. Every graph G satisfying

$$B \subseteq G \subseteq H_A$$

has distinct closed move sets intersecting in at most one vertex, because $C_G(x) \subseteq x + A$.

The proof of Theorem 11 therefore works inside every such G . To make this preservation explicit, fix the recipient zero. Its direct predecessors $-1, -a, -h$ and zero are guaranteed, and $-a - 1$ is guaranteed using the two successors $-a, -1$. If

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

is guaranteed, the following four applications of Lemma 10 establish $Q(z - h)$:

New source	Two guaranteed successors
$z - h$	$z - a - 1, z$
$z - h - 1$	$z - h, z - 1$
$z - h - a$	$z - h, z - a$
$z - h - a - 1$	$z - h - a, z - a - 1$

All displayed moves belong to B . Coprimality of h and n makes finitely many repetitions cover all sources. The same argument with all coordinates shifted applies to every recipient, even when G itself is not translation invariant. Thus every intermediate graph is universally winning.

It remains to enlarge A_0 . Suppose $|A| = k$ and its nonzero differences are distinct. To adjoin x , exclude the following residues:

1. The k members of A itself.
2. Residues $x = a + b - c$ with $a, b, c \in A$ and $b \neq c$. These exclude every equality between an old nonzero difference and either sign of a new difference. There are at most $k^2(k - 1)$ such residues.
3. Solutions of $2x = a + b$ with $a, b \in A$. These exclude an equality $x - a = b - x$ between the two signs of new differences. There are $k(k + 1)/2$ unordered pairs with repetition, and each congruence has at most two solutions, hence at most $k(k + 1)$ excluded residues.

Within each sign the new differences are already distinct. The three exclusions are therefore sufficient, and together rule out at most

$$k + k^2(k - 1) + k(k + 1) = k^3 + 2k$$

residues. Whenever $n > k^3 + 2k$, at least one admissible residue remains; choose the smallest representative. Under the theorem's hypothesis this process continues from size four to size $d + 1$.

The resulting H_A has dn arcs and contains the $3n$ arcs of B . For a prescribed integer m in the interval, add exactly $m - 3n$ of the remaining arcs in lexicographic order. The intermediate graph is universally winning by the preservation argument, so all $n(n - 1) - m$ missing pairs are guaranteed. This matches the elementary upper bound.

For $d = 4$ the hypothesis is $n > 72$. Taking the largest admissible d gives d of order $n^{1/3}$, and therefore an upper interval endpoint of order $n^{4/3}$. The endpoint $m = 3n$ only needs the original $n \geq 25$ construction, with no extension. $\square$

This is a deterministic construction at each stated integer budget. The underlying delivery argument may require many moves; it is not a two-move theorem. Its ordinary proof is independent of the finite tests in `src/sidon_main_interval.py`, and no new Lean coverage is asserted. It provides an initial consecutive part of Raychev's full-density interval. The stronger constructions and exact-budget argument in Theorems 38–48 subsequently cover the entire interval at every sufficiently large order.

3.8 An explicit algebraic family at intermediate density

Theorem 35. Let $p \geq 5$ be a prime. For every integer

$$4p^2 \leq m \leq p^2(p - 1),$$

there is an explicit graph attaining

$$M(p^2, m) = p^2(p^2 - 1) - m.$$

In particular, the interval extends to $m = n^{3/2} - n$ at orders $n = p^2$. For primes $p \geq 11$, Theorem 34 extends its lower endpoint to $3p^2$.

Proof. Put $A = \{(t, t^2) : t \in \mathbb{F}_p\}$, and let H_A be the directed graph with all nonzero moves in A . Let B_0 be its four-move subgraph with parameters

$$t \in \{1, -1, -2, -3\}.$$

These are distinct and nonzero when $p \geq 5$. We prove universal delivery in every graph G satisfying $B_0 \subseteq G \subseteq H_A$, including graphs without translation symmetry.

Every closed outgoing neighborhood in G is a subset of the corresponding translate of A . Distinct translates of A intersect in at most one point. Indeed, a nonzero difference

$$(t, t^2) - (s, s^2) = (t - s, (t - s)(t + s))$$

determines t, s uniquely: if the first coordinate is nonzero, it determines their difference and sum, and division by two is valid; if the first coordinate is zero, the entire difference is zero. Thus every nonzero difference has at most one ordered representation.

This intersection property gives the following safe-branching rule. Fix a recipient. If two distinct outgoing neighbors of a vertex are already universally winning positions for that recipient, the vertex is universally winning as well. Against an initial cop at another vertex, at most one of the two successors lies in the cop's closed outgoing neighborhood. Move to the other, survive the

cop's response, and use its winning strategy. The rule concerns the actual game, including the intervening pursuer move.

We first use recipient 0 and only moves of the uniform core B_0 . Put

$$u = (1, 1), \quad v = (-2, 4), \quad h = (-3, 9) = u + 2v,$$

and define

$$Q(z) = \{z, z - u, z - v, z - u - v\}.$$

The vertices $0, -u, -v$ are initially winning, by immediate receipt or a direct final move. The remaining vertex $-u - v$ has winning successors $-u, -v$, so all of $Q(0)$ is winning.

If every vertex of $Q(z)$ is winning, so is every vertex of $Q(z - h)$. Apply the safe-branching rule in the following order; each row lists the new vertex and two already winning successors:

$$\begin{array}{c|cc} z - h & z - u - v & z \\ z - h - u & z - h & z - u \\ z - h - v & z - h & z - v \\ z - h - u - v & z - h - v & z - u - v \end{array}$$

The corresponding pairs of moves are $(v, h), (u, h), (v, h), (u, h)$, respectively. They are distinct allowed moves.

Repeated propagation makes the entire line $L = \{\lambda h : \lambda \in \mathbb{F}_p\}$ winning: the nonzero vector h has additive order p , so integer multiples cover all the displayed scalars.

Use the fourth core move $w = (-1, 1)$. We have

$$w - v = (1, -3) = -h/3 \in L,$$

while $v \notin L$ because $\det(h, v) = 6 \neq 0$. If every vertex of a coset $L + z$ is winning, then every vertex of $L + z - v$ has two distinct winning successors, obtained by adding v and w . The safe-branching rule therefore fills that entire coset as well. Iterating gives all cosets $L - jv$, which are the p distinct cosets because $v \notin L$. Thus every vertex is winning.

The same proof shifted by the recipient uses only moves of B_0 and the global closed-neighborhood intersection bound. It applies to every recipient even if the extra arcs of G are not translation invariant. Hence every intermediate graph is universally winning.

The core has $4p^2$ arcs, and H_A has $p^2(p-1)$ arcs. Add the remaining arcs one at a time in any fixed order until the prescribed integer budget m is reached. All $p^2(p^2-1) - m$ missing nonloop pairs are guaranteed, attaining the elementary upper bound. Finally, when $p \geq 11$ we have $p^2 \geq 73$, so Theorem 34 covers the adjacent interval $3p^2 \leq m \leq 4p^2$. $\square$

The proof is finite and constructive: initializing a quartet takes at most two messenger moves, and each quartet propagation increases a bound by at most four. At most $p-1$ such propagations and $p-1$ further coset steps therefore give the uniform, unoptimized bound $5p-3$ messenger moves.

The prime hypothesis is intentional. In an extension field, repeated addition only covers scalars from its prime subfield, so this four-move propagation does not justify the same conclusion for arbitrary prime powers. The construction is checked at $p = 5, 7, 11$ by `src/parabola_cayley.py`; it verifies the full-parabola intersection condition, the four-move core's complete safe-branching closure, and the core/upper-graph inclusions for all 827 consecutive integer budgets in these examples. At $p = 5$,

both full game solvers and their rank certificates independently verify every recipient. These finite checks supplement the ordinary proof and are recorded in `research/parabola-cayley-checks.json`. No new Lean formalization is claimed.

4 Protected routing and the main interval

4.1 Protecting the moves used by the strategy

The distinct-difference condition in Theorem 34 is stronger than the delivery strategy needs. Its branching steps only compare pairs of the three original outgoing moves. Preserving those comparisons permits a much larger set of additional moves.

Theorem 38. For every $n \geq 25$ and every integer

$$3n \leq m \leq n \left(\left\lceil \frac{n}{7} \right\rceil - 1 \right),$$

there is an explicit graph attaining

$$M(n, m) = n(n - 1) - m.$$

The strategy uses only three outgoing moves per vertex; the remaining arcs may be added individually up to the stated budget.

Proof. Choose h using Lemma 12, put $a = (h - 1)/2$, and set

$$S = \{1, a, h\}, \quad A_0 = \{0, 1, a, h\} \subseteq \mathbb{Z}/n\mathbb{Z}.$$

The proof of Theorem 11 shows that A_0 has distinct nonzero differences. In particular, each of the six ordered differences

$$D = \{u - v : u, v \in S, u \neq v\} = \{\pm(a - 1), \pm(a + 1), \pm 2a\}$$

has just its original ordered representation in A_0 . We only preserve uniqueness for these six differences.

Starting from $A = A_0$, adjoin any residue outside

$$A \cup (A + D), \quad A + D = \{x + d : x \in A, d \in D\}.$$

No new ordered pair involving the adjoined residue has a difference in D , because $D = -D$. Thus the six protected differences retain their original unique representations. When $|A| = k$, at most $7k$ residues are excluded. If $k < \lceil n/7 \rceil$, then $7k < n$, so the extension remains possible. Choosing the smallest available representative constructs a set of size at least $\lceil n/7 \rceil$. For $25 \leq n \leq 28$, the initial set already has that size.

Let B be the uniform three-move graph with steps in S , and let H_A have all nonzero steps in A . Consider any graph

$$B \subseteq G \subseteq H_A.$$

For a source vertex r and distinct core moves $u, v \in S$, no different cop vertex c can have both $r + u$ and $r + v$ in its closed outgoing neighborhood. Indeed, that neighborhood is contained in $c + A$. If both successors belonged to it, their corresponding elements $x, y \in A$ would satisfy $x - y = u - v \in D$. Protected uniqueness forces $(x, y) = (u, v)$ and therefore $c = r$, a contradiction.

Consequently, whenever two successors obtained using core moves are universally winning for a fixed recipient, their source is universally winning: at least one of those successors is inaccessible to the initial cop and survives its response. This argument does not assert that arbitrary pairs of actual outgoing successors have the same property.

The entire proof of Theorem 11 uses exactly these protected pairs. For recipient zero, the direct predecessors $-1, -a, -h$ are winning, and $-a - 1$ uses the core moves $1, a$. If

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

is winning, the four steps establishing $Q(z - h)$ use the move pairs

$$(a, h), \quad (1, h), \quad (a, h), \quad (1, h),$$

respectively, in the order given in Theorem 34. Hence that quartet propagates in G . Since $\gcd(h, n) = 1$, its successive translates cover every source. Shifting the proof by the recipient covers every target, regardless of whether the added arcs are translation invariant.

Thus every graph between B and H_A is universally winning. The first has $3n$ arcs and the latter has at least $n(\lceil n/7 \rceil - 1)$. Add the extra arcs in lexicographic order until the prescribed integer budget is reached. All $n(n - 1) - m$ missing pairs are guaranteed, meeting the elementary upper bound. $\square$

The improvement from order $n^{4/3}$ to order n^2 comes from preserving only the distinctions actually used by the strategy. Other differences may repeat many times. The graph may also have larger closed-neighborhood intersections; the proof protects its particular pairs of core successors instead of imposing a global intersection bound.

For example, at $n = 29$ the construction gives core moves $1, 3, 7$ and the envelope $A = \{0, 1, 3, 7, 8\}$. Its protected differences are $\pm 2, \pm 4, \pm 6$, each still uniquely represented. The unprotected difference -1 occurs twice, as $0 - 1 = 7 - 8$. Thus this envelope genuinely fails the full distinct-difference condition, yet every intermediate graph with $87 \leq m \leq 116$ is universally winning.

Corollary 39. There is an absolute N such that, for every $n \geq N$ and every integer

$$3n \leq m \leq n(n - 3),$$

we have

$$M(n, m) = n(n - 1) - m.$$

For every sufficiently large multiple of five n , the same formula holds at every integer budget $2n \leq m \leq n(n - 3)$.

Proof. The upper endpoint in Theorem 38 satisfies

$$n(\lceil n/7 \rceil - 1) \geq n^2/7 - n.$$

Since $\sqrt{n/\log n} \rightarrow \infty$, this endpoint eventually exceeds $\lceil 10n^{3/2}\sqrt{\log n} \rceil$, the lower endpoint in Corollary 26. The two integer intervals overlap. Theorem 38 covers the lower one and Corollary 26 covers all remaining budgets through $n(n - 3)$, including its dense end via Theorem 20, now a consequence of the direct dense envelopes in Theorems 48 and 58. Taking a common sufficiently large threshold establishes the first statement.

For multiples of five, Theorem 37 supplies every budget from $2n$ through $3n$ once $n \geq 80$. Combining the intervals proves the second statement. $\square$

The later Theorems 44 and 59 replace this eventual overlap by a fully explicit construction for every $n \geq 115$, including the sparse range. Theorem 38 retains its smaller-order domain, arbitrary-subset guarantee, and protected-difference strategy. The probabilistic overlap in Corollary 39 is included only as a consequence of its underlying constructions.

The generator `src/main_interval_resume.py` checks protected differences at eight orders from 25 through 1000. It checks every protected source-pair/cop triple at the seven listed orders through 200, and verifies 24 selected recipient games with both independent solvers and the explicit rank-certificate checker. Those tests include intermediate graphs lacking translation symmetry. The receipt `research/main-interval-resume-checks.json` records their exact scope; the parameterized proof above is independent of these finite checks.

4.2 Protected routing sets at every exact budget

The random construction can be strengthened by reserving a small set of routing vertices. Only arrows incident with that set are used to certify delivery. All arrows outside the set can then be chosen freely, including adversarially, without invalidating any certified route.

Theorem 54 (protected routing). Let $n \geq 4$, let H be a specified set of $b \leq n$ vertices, and put

$$L = n(n-1), \quad Q = b(2n-b-1).$$

Choose an integer $0 < a < Q$ and write $p = a/Q$. If

$$(b-3)p^2(1-p) > 4\log n + \log(8/3), \tag{R}$$

there is a deterministic algorithm, polynomial in n , choosing exactly a arrows with at least one endpoint in H such that **every assignment of the arrows between vertices outside H** guarantees every source-recipient pair within two messenger moves. In particular, at every integer budget

$$a \leq m \leq a + (n-b)(n-b-1),$$

an explicit algorithm attains $M(n, m) = L - m$.

Separately, for every real $0 < p < 1$, without condition (R) or an integer exact-budget requirement, independently including each of the Q incident arrows with probability p fails to give such a robust routing set with probability at most

$$n(n-1)^2(1-p^2(1-p))^{b-3}.$$

Thus the theorem includes both the deterministic exact-budget and the random-construction conclusions.

Proof. Fix $s \neq t$ and an initial pursuer position $c \neq s$. Each $u \in H \setminus \{s, t, c\}$ certifies a safe two-move route when

$$s \rightarrow u, \quad u \rightarrow t \quad \text{are present,} \quad c \rightarrow u \quad \text{is absent.}$$

The three variables are distinct, even when $c = t$. All are incident with H . For different candidates the sets of three variables are disjoint: an equality between an incoming-to-candidate variable

and an outgoing-from-candidate variable would force a candidate to equal one of s, t, c , which is excluded.

There are at least $b - 3$ candidates, each with probability $p^2(1 - p)$ of success. There are exactly $n(n - 1)^2$ legal triples. The union bound proves the random assertion. If Z counts bad triples, it also gives

$$\mathbb{E}Z \leq n(n - 1)^2 e^{-(b-3)p^2(1-p)} < \frac{3}{8n}$$

under (R).

Let X be the number of chosen incident arrows. Then $X \sim \text{Bin}(Q, a/Q)$ and a is a mode. Since $\text{Var}(X) \leq Q/4$, Chebyshev's inequality gives $\Pr(|X - a| < \sqrt{Q}) \geq 3/4$. Here $Q \leq n(n - 1)$, so this interval contains at most $2n - 1$ integers. Modal maximality implies

$$\Pr(X = a) \geq \frac{3}{4(2n - 1)} \geq \frac{3}{8n}.$$

Therefore $\mathbb{E}[Z \mid X = a] < 1$.

For completeness, the conditional-expectation algorithm remains exact when restricted to these Q variables. After a partial assignment, let ℓ be the number of undecided incident arrows and d the number of additional present arrows required. The total number of completions is $C = \binom{\ell}{d}$. For a fixed legal triple, each candidate contributes a factor $(1 + x)^q$ if its witness pattern is already contradicted; otherwise its factor is $(1 + x)^q - x^j$, where q variables remain undecided and j of them require presence. Multiply these factors, multiply by $(1 + x)^{q_0}$ for unused variables, and extract the coefficient of x^d . This counts exactly the completions in which the triple is bad. Summing over triples gives an integer B with initial $B < C$.

For either decision on the next arrow, child counts satisfy $B = B_0 + B_1$ and $C = C_0 + C_1$. Hence some feasible child retains $B_i < C_i$. At the end $C = 1$ and $B = 0$. There are $O(n^2)$ decisions and $O(n^3)$ triples per decision, with polynomial-size polynomial products and $O(n^2 + \log n)$ -bit counts. Straightforward exact integer arithmetic is therefore polynomial in n .

For every legal triple a selected candidate differs from the pursuer and both endpoints. Moving to it is safe, and the pursuer cannot reach it on its response. The messenger then reaches the recipient; receipt precedes collision. No outside- H arrow appears in these conditions, so the same strategies work for every assignment of those arrows. There are $(n - b)(n - b - 1)$ of them. Choosing any prescribed number proves the entire displayed exact-budget interval. $\square$

Setting $H = V$ gives $Q = L$ and the former exact-budget density theorem at every original order and budget, with its polynomial algorithm and two-move guarantee. A smaller routing set adds genuine robustness and extends the low-density range; it does not merely combine old intervals.

4.2.1 The scale of the first universal two-move budget

Let $q_2(n)$ be the smallest number of arrows in a graph on n vertices for which every source-recipient pair is guaranteed within two messenger moves. This differs from minimizing delivery time among extremizers that guarantee only some of their missing pairs.

Orientation specialization. Fix a routing set H of size b , and let $Q = b(2n - b - 1)$. If

$$n(n - 1)^2 (7/8)^{b-3} < 1,$$

there is a deterministic polynomial construction using exactly $Q/2$ incident arrows that preserves universal two-move delivery under every choice of exterior arrows. Thus all budgets from $Q/2$ to $L - Q/2$ work. A sharper sufficient condition is

$$n(n-1)(n-2)(7/8)^{b-3} + n(n-1)(3/4)^{b-2} < 1.$$

To prove this, independently orient each unordered pair touching H , with its two directions equally likely. The incident count is exactly $Q/2$ for every outcome. For distinct s, t, c , a candidate $u \in H \setminus \{s, t, c\}$ is a safe intermediary when $s \rightarrow u$, $u \rightarrow t$, and $u \rightarrow c$, with probability $1/8$. The last orientation excludes $c \rightarrow u$. Different candidates use independent orientation variables. When $c = t$, only two variables are needed and the probability is $1/4$. Counting the two kinds of triples gives the sharper displayed failure bound; the coarser one also follows. Exterior arrows occur in none of these tests.

For exact derandomization, fix orientations in sequence. For a given triple and candidate with j free relevant variables, there are 2^j bad completions if a fixed variable already prevents the witness, and $2^j - 1$ otherwise. Multiply over candidates, whose variable sets are disjoint, and account for irrelevant free variables. Summing these integer counts over triples gives the total number of bad completions, with multiplicity. Initially it is smaller than the number of all completions. At each decision choose a branch preserving that strict inequality; the two child counts sum to their parent counts. At the end there is one completion and zero failures. There are polynomially many operations on integers of polynomial bit length. This specialization replaces coefficient extraction by products, while the full density and sampling assertions of Theorem 54 remain available.

Corollary 55. Set

$$b_n = \lceil 24 \log n + 4 \rceil, \quad A_n = \frac{b_n(2n - b_n - 1)}{2}.$$

Whenever $b_n \leq n$, every integer budget $A_n \leq m \leq L - A_n$ admits universal two-move delivery by a deterministic polynomial algorithm, with arbitrary exterior-arrow subsets as in Theorem 54. For every $n \geq 4096$, this extends to every integer budget

$$A_n \leq m \leq n(n-3).$$

Moreover, for every $n \geq 2$,

$$q_2(n) \geq \left\lceil \frac{n}{2} (\log_2 n - 1) \right\rceil,$$

and hence $q_2(n) = \Theta(n \log n)$.

Proof of the constructions. Use the orientation specialization. Since $(7/8)^x \leq e^{-x/8}$ for $x \geq 0$,

$$n(n-1)^2(7/8)^{b_n-3} < n^3 e^{-(b_n-3)/8} \leq e^{-1/8} < 1.$$

It gives the exact robust interval $[A_n, L - A_n]$.

For $n \geq 4096$,

$$b_n \leq 24 \log n + 5 \leq 32 \log n + 12 \leq (n-1)/13.$$

At 4096, $\log 4096 = 12 \log 2 < 9$ makes the left side of the second inequality less than 300, while its right side equals 315. The difference increases thereafter because $1/13 - 32/n > 0$. The elementary inequality $\log 2 < 3/4$ follows, for example, from $1 + 3/4 + (3/4)^2/2 > 2$.

Consequently $A_n \leq nb_n \leq n(n-1)/13$. The existing dense constructions supply every missing budget from $2n$ through $n\lceil(n-1)/13\rceil$ for $n \geq 32$, in two moves (Theorems 48 and 58). Their interval overlaps $[A_n, L - A_n]$, proving the claimed extension through $L - 2n = n(n-3)$. Since $A_n = O(n \log n)$, the upper order bound for q_2 follows.

Proof of the lower bound. For every vertex u , form the disjoint sets

$$S_u = \{u\} \cup N^-(u), \quad T_u = N^+(u) \setminus N^-(u).$$

For each unordered reciprocal pair add its two singleton endpoints as another pair of disjoint sets. These pairs separate every two vertices. A one-way arrow $s \rightarrow t$ is separated by (S_s, T_s) , and a reciprocal pair has its singleton separator. If neither arrow exists, start the pursuer at t . A two-move guarantee forces a route $s \rightarrow u \rightarrow t$ with $t \not\rightarrow u$: waiting first would leave the missing pair undelivered in the one remaining move. Thus $s \in S_u$ and $t \in T_u$.

If there are z reciprocal unordered pairs, the first n separators have total membership $n + 2m - 2z$ and the singleton separators add $2z$, giving $n + 2m$ in total.

Hansel's separating-system inequality states that this total is at least $n \log_2 n$; see (Bollobás and Scott 2006, Lemma 1). Its short proof is included here. Independently delete one uniformly chosen side of every separator. At most one vertex survives. A vertex appearing in d_v separators survives with probability 2^{-d_v} , so $\sum_v 2^{-d_v} \leq 1$. Convexity gives $n2^{-(\sum_v d_v)/n} \leq 1$, hence $\sum_v d_v \geq n \log_2 n$. Applying this inequality proves $n + 2m \geq n \log_2 n$ and the integer lower bound. $\square$

The lower-bound ingredient is classical; the conversion from the delivery game to this separating system uses the present timing convention, including a pursuer initially at the recipient. Theorem 59 improves the constructive asymptotic constant using explicit antichain codes. The order-of-growth result does not identify an optimal constant or assert that every budget from $q_2(n)$ onward works.

Verification boundary. The proofs above are ordinary mathematical arguments. The restricted exact-count implementation and its independent finite checks are recorded in `research/routing-unification-checks.json`; the independent proof review is `research/routing-independent-audit.md`. No large instance of the polynomial conditional-expectation algorithm is claimed. `ProtectedRouting.lean` verifies preservation of safe routing witnesses under arbitrary exterior-arrow changes and their implication for actual terminating delivery. The probability, counting algorithm, and asymptotic lower bound remain ordinary proofs.

4.3 Composition through incomparable hub sets

The construction below reduces a delivery problem on an arbitrarily large outside set to a family of subsets of a small core. The subsets specify which hubs an outside vertex cannot reach. Incomparability makes one outside vertex's inaccessible hubs available to another vertex. A two-part core handles the remaining configurations, including recipients inside the core. This produces an entire robust interval of exact budgets.

Theorem 59 (antichain composition). Let $k \geq 5$. Take two sets of hubs P_i, L_i , indexed modulo k . Their missing arrows are

$$P_i \rightarrow L_i, L_{i-1}, \quad L_i \rightarrow P_{i+4}, P_{i+6};$$

all other distinct hub pairs are arrows. For each of q outside vertices x , choose a set K_x of hubs subject to these conditions:

- $|K_x| \geq 3$, and K_x lies entirely in one of the two hub parts;
- a set in the P part contains no two indices differing by ± 2 ; a set in the L part contains no adjacent indices;
- the sets K_x form an antichain: no one contains another.

Include every hub-to-outside arrow, and include $x \rightarrow u$ for precisely the hubs $u \notin K_x$. Every subset of the arrows between distinct outside vertices may then be included. Every resulting graph has universal delivery within two messenger moves.

Writing $n = 2k + q$ and $W = \sum_x |K_x|$, this gives every exact budget

$$A \leq m \leq A + q(q - 1), \quad A = 2k(2k - 3) + 4kq - W.$$

At the upper endpoint the number of missing arrows is $4k + W$. The assertion holds for every optional subset, not merely a specially chosen subset of each size.

Proof. Let F denote the missing graph inside the hubs. For a missing pair (s, t) and a pursuer at $c \neq s$, we seek a hub u with arrows $s \rightarrow u \rightarrow t$, with $u \neq c$ and no arrow $c \rightarrow u$. The messenger can move to u , survive every pursuer response, and then reach t . The case $c = t$ is included.

First suppose all three vertices are hubs. For a missing hub pair, the set $D = F^+(s) \cup F^-(t)$ contains two points and two lines. If $s = P_i$, the line indices are $i, i - 1$ and the point indices are either $i, i + 1$ or $i - 1, i$. A point pursuer has its entire missing row in D only when it is s ; a line pursuer's point pair has difference two and cannot fit inside the adjacent point pair. If $s = L_i$, the point indices are $i + 4, i + 6$ and the line indices are either $i, i - 2$ or $i, i + 2$. The same argument with the two parts exchanged applies. Thus $F^+(c)$ contains a hub outside D , which is the desired intermediary.

Now consider the other nondirect configurations.

1. **Both endpoints outside.** If the pursuer is outside, choose $u \in K_c \setminus K_s$, which exists by incomparability. It is reachable from s , inaccessible from c , and reaches t . If the pursuer is a hub, its missing row consists of an adjacent line pair or a point pair of difference two. The restrictions on K_s prevent that entire row from lying in K_s . A remaining member works.
2. **Source outside, recipient a hub.** Missingness means $t \in K_s$. For an outside pursuer whose code lies in the same part, choose $u \in K_c \setminus K_s$; that part is a clique, so $u \rightarrow t$. For an outside pursuer using the other part, K_c is disjoint from K_s and has at least three members. At most two are missing predecessors of t , so one of the others works. For a hub pursuer whose missing row lies in the part containing t , the code restriction leaves a member outside K_s , and the clique supplies the last arrow. For the other hub pursuer part, both missing neighbors are reachable from s . Their adjacent or difference-two pair cannot equal the opposite kind of pair $F^-(t)$, so at least one reaches t .
3. **Both endpoints hubs, pursuer outside.** The set D above contains only two members in either part. Since K_c has at least three members in one part, choose a member outside D .

A hub source reaches an outside recipient directly. All intermediaries used in the proof are hubs, so optional arrows between outside vertices cannot invalidate a witness. Counting the compulsory and optional arrows gives the stated interval. $\square$

Equal-weight codes. For odd $k \geq 7$ and $3 \leq r \leq \lfloor k/2 \rfloor$, the number of independent r -sets in a

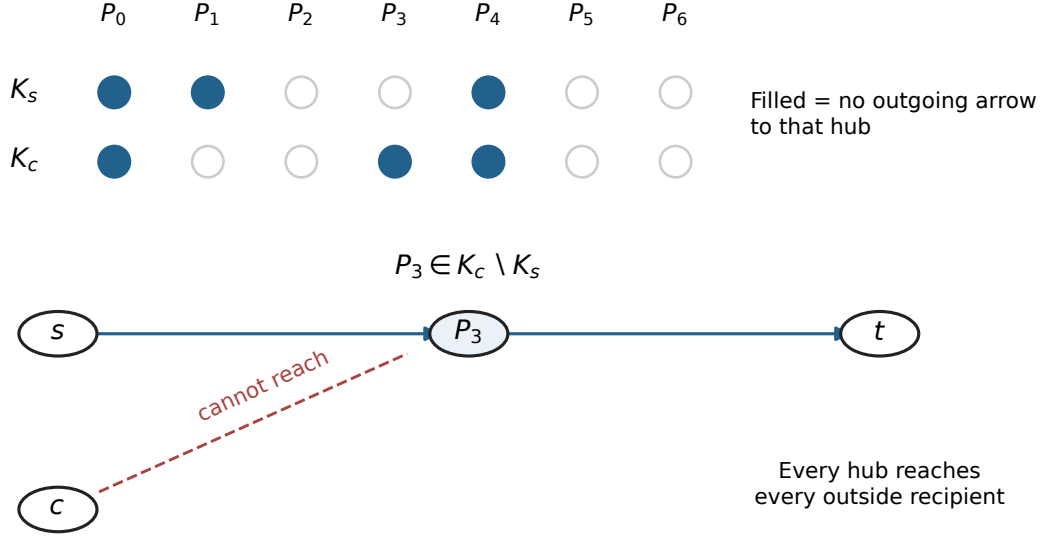


Figure 6: Two incomparable missing-hub sets leave a hub that the source can reach but the pursuer cannot. Every hub reaches the outside recipient, completing a safe two-move delivery. The theorem separately handles recipients and pursuers inside the core.

cycle of length k is

$$C(k, r) = \frac{k}{k-r} \binom{k-r}{r}.$$

To count them, distinguish a selected vertex and distribute the unselected vertices among the r nonempty gaps. This gives $\frac{k}{r} \binom{k-r-1}{r-1}$, the same expression. Multiplication by two identifies the adjacency cycle with the difference-two cycle when k is odd. The two hub parts therefore supply $2C(k, r)$ distinct codes. Equal size ensures incomparability. In particular, for any $q \leq 2C(k, r)$,

$$A = (4k - r)n - 2k(2k + 3 - r), \quad L - A - q(q - 1) = 2n + (r - 2)q.$$

Codes can be listed lexicographically; no pursuit-game search is required.

Two-move sparsity. These codes sharpen the constructive side of Corollary 55 to

$$q_2(n) \leq \left(\frac{26 \log 2}{\log(3125/108)} + o(1) \right) n \log_2 n = (5.3555763082 \dots + o(1)) n \log_2 n.$$

For an exact parameter rule at every $n \geq 30$, choose the least even $d \geq 2$ with $2C(7d + 1, 2d) \geq n$, and put $k = 7d + 1$, $r = 2d$. The core fits: when $d = 2$ it has 30 vertices; when $d \geq 4$, minimality gives

$$n > 2C(7d - 13, 2d - 4) \geq 2 \binom{5(d-2)}{2(d-2)} \geq 5(d-2)(5(d-2) - 1) \geq 14d + 2.$$

The binomial inequality uses unimodality and $2 \leq 2(d-2) \leq 5(d-2)/2$. The exact base budget is

$$A = (26d + 4)n - 2(7d + 1)(12d + 5).$$

The modal probability of a binomial distribution with $5d$ trials and success probability $2/5$ is attained at $2d$ and is at least $1/(5d+1)$. Consequently

$$C(7d+1, 2d) \geq \binom{5d}{2d} \geq \frac{(3125/108)^d}{5d+1}.$$

This proves $d \leq \log n / \log(3125/108) + O(\log \log n)$ and the displayed upper bound. Since $k = O(\log n)$, even listing all hub subsets and retaining valid codes is a polynomial algorithm. The lower bound in Corollary 55 still applies; the leading constant of $q_2(n)$ remains open.

4.4 A wider protected circle and the main-interval overlap

Here we use the protected differences of Theorem 38 without changing its strategy. Let h be an allowed odd step, $a = (h-1)/2$, and

$$A_0 = \{0, 1, a, 2a+1\}, \quad D = \{\pm(a-1), \pm(a+1), \pm 2a\}.$$

Put $p = 3a-1$ and

$$J = \{jp + t : 0 \leq j < \lfloor n/p \rfloor, 0 \leq t < a-1\}.$$

No difference between two members of J belongs to D : within a run it has magnitude at most $a-2$, while between runs, including the cyclic gap, the shortest gap is at least $2a+1$. For any translate $J+v$, remove all members of $A_0 \cup (A_0 + D)$ and then adjoin A_0 . Every protected difference still has exactly its original representation. Thus the result is an allowed envelope in Theorem 38. The excluded set has at most 19 elements, by expanding the four translates of the six differences. Averaging over v gives an envelope of size at least

$$4 + \left\lceil \frac{|J|(n-19)}{n} \right\rceil.$$

One may then apply the greedy extension from Theorem 38. In particular the old guarantee $|A| \geq \lceil n/7 \rceil$ is preserved. Searching the n translations is explicit and takes polynomial time.

Main-interval overlap certificate. This protected-circle construction, Theorem 59, and the dense constructions cover every budget from $3n$ to $n(n-3)$ for every $n \geq 115$. Combined with Theorem 52, they prove the improved order threshold in Theorem 44. Here are the complete finite and unbounded parts of the overlap argument.

For $115 \leq n \leq 390$, the finite parameter certificate supplies a valid circle step and translation and valid code parameters whose base A is at most the circle's maximum budget. It also checks that the code's terminal missing count is within the dense envelope of Theorem 48. These are integer inequalities and protected-difference checks, not extrapolations from small pursuit games. The records and an independent replay are listed below. Two orders use the following refinements:

- At $n = 116$, use $k = 11$, two allowed codes of size five, and 92 allowed codes of size four contained in neither larger code. The two larger codes exclude at most ten of the 110 available size-four codes, so there are enough. Their weight is $W = 378$, giving base budget 4176, exactly the verified protected-circle endpoint.
- At $n = 117$, use $a = 20$, $h = 41$, and $A_0 \cup [41, 59] \cup [100, 115] = \{0, 1, 20\} \cup [41, 59] \cup [100, 115]$. This has 38 members and each protected ordered difference occurs exactly once. The circle reaches 4329 arrows, above the code base 4218.

For $391 \leq n \leq 480$, choose $k = 15, r = 4$; for $481 \leq n \leq 699$, choose $k = 17, r = 5$. Their code bases are $56n - 870$ and $63n - 1088$, respectively. Each is at most $T = n(\lceil n/7 \rceil - 1)$: replace the ceiling by $n/7$, check the left endpoint, and observe that the quadratic difference increases thereafter. The code capacities are 900 and 2244, and the terminal missing counts $4n - 60$ and $5n - 102$ are within Theorem 48.

For $n \geq 700$, take $r = 3$ and an odd k with

$$(6n)^{1/3} + 5 \leq k < (6n)^{1/3} + 7.$$

Then $C(k, 3) = k(k-4)(k-5)/6 \geq n$, the core fits inside n vertices, and

$$A < 4kn < 4n((6n)^{1/3} + 7) \leq n^2/7 - n \leq T.$$

Indeed, $n/7 \geq 4(6n)^{1/3} + 29$ holds at 700, where $(6n)^{1/3} < 17$, and its difference is increasing thereafter. The terminal missing count $3n - 2k$ is between $2n$ and $3n$, so Theorem 58 completes the interval. This exhausts all orders in the assertion.

Verification and scope. The ordinary antichain proof was independently reviewed case by case. The executable constructor is `src/core_code_composition.py`; direct asymmetric witness tests are recorded in `research/core-code-composition-checks.json`. The finite overlap checks are in `research/routing-defect-phase2-code-overlap-checks.json`, with the mixed-size 116 and explicit 117 constructions in `research/routing-defect-phase2-mixed-code-checks.json`. The mixed 116 graph alone passes 1,053,860 source-recipient-pursuer witness checks. The general antichain existence, cycle counts, and numerical overlap are ordinary proofs. `RobustTwoMove.lean` formalizes the semantic implication from supplied asymmetric witnesses to actual delivery, not the combinatorial construction of those witnesses or this entire theorem.

The complete main-interval constructor is available as `src/main_interval_composed.py`. Given $n \geq 115$ and an admissible m , it selects one of the proved envelopes and outputs the exact arrow list. Its boundary checks exercise all five branches and independently replay every finite parameter certificate; see `research/main-interval-composed-checks.json`. This executable selection uses the ordinary overlap proof, not a new search for each graph.

4.5 Exact-budget consequences of protected routing

The protected-routing theorem contains the earlier full-vertex random and deterministic constructions. We retain their labels for references, but give the counting argument only once, in Theorem 54.

Theorem 43. Put $L = n(n-1)$ and $p = m/L$. Under the hypotheses of Theorem 25, there is a deterministic algorithm, polynomial in n , constructing an n -vertex, exactly m -arc graph in which every pair is guaranteed within two messenger moves.

Proof. Set $H = V$, $b = n$, and $a = m$ in Theorem 54. The former condition $(n-3)p^2(1-p) > 5 \log n$ implies its condition because $\log n > \log(8/3)$ for $n \geq 4$. The restricted completion counts become exactly the full-vertex counts. $\square$

The algorithm's polynomial bound does not imply a small practical runtime. `src/exact_budget_derandomize.py` retains the earlier full-vertex implementation and small exhaustive checks; `src/routing_unification.py` generalizes those counts to a specified routing set. Neither receipt claims execution on a large asymptotic instance.

4.5.1 An explicit threshold for the entire main interval

Theorem 44. For every integer $n \geq 115$ and every integer

$$2n \leq m \leq n(n-3),$$

a deterministic polynomial algorithm constructs a graph with exactly n vertices and m arcs in which every pair is guaranteed. Consequently $M(n, m) = n(n-1) - m$ throughout this interval.

Proof. Theorem 52 covers $2n \leq m \leq 3n$ at every order $n \geq 115$. The protected-circle construction of Theorem 38, strengthened by the run construction after Theorem 59, overlaps the antichain construction of Theorem 59. That construction in turn reaches the dense envelopes of Theorems 48 and 58. The complete overlap proof after Theorem 59 checks explicit parameter certificates for $115 \leq n \leq 390$, gives fixed parameters for $391 \leq n \leq 699$, and proves one parameter inequality for all $n \geq 700$. Thus these intervals cover every stated budget. Each construction lists its base arrows and optional arrows; include the prescribed number of optional arrows in lexicographic order. All relevant lists can be constructed in polynomial time. The missing-pair bound gives the reverse inequality. $\square$

The sufficient order threshold 115 is not asserted to be optimal. This proof is entirely by explicit combinatorial constructions and finite local strategy certificates. The sparse part retains its longer delivery strategies; there is no uniform two-move assertion for the entire interval. The protected-routing theorem remains independently useful because it allows a specified routing set and a variable incident density, with an explicit random success probability and exact-budget derandomization.

5 Dense constructions and the final staircase

5.1 The dense boundary

Theorem 13. For $n \geq 2$,

$$M(n, n(n-2)) = \left\lfloor \frac{n-2}{2} \right\rfloor,$$

and $M(n, m) = 0$ for every admissible $m > n(n-2)$.

Proof. If (s, t) is indirect guaranteed, row s misses $s \rightarrow t$. By Lemma 1, every other row c must miss an arc into $N^+(s)$. Thus every row misses at least one nonloop arc, and $m \leq n(n-2)$.

Now suppose $m = n(n-2)$. If $I = 0$, the upper bound is immediate. Otherwise the preceding argument shows that every row misses exactly one arc; write $f(v)$ for its endpoint. Necessarily $f(v) \neq v$. A possible indirect pair is $(s, f(s))$. Lemma 1 says

$$f^{-1}(s) = \emptyset, \quad f^{-1}(f(s)) = \{s\}. \quad (2)$$

These conditions suffice: for initial pursuer $c \neq s$, choose the intermediate $f(c)$. It is different from $s, f(s)$, is inaccessible on the pursuer's next move, and has an arc to $f(s)$. This is a two-move guarantee.

Pairs satisfying (2) have disjoint source and target vertices. If there are j pairs, the remaining vertices contain a cycle of the loopless functional graph of f and therefore number at least two. Hence $2j \leq n-2$. For attainment use pairs $s_i \rightarrow t_i$ in the missing-arc graph, send every t_i to a , put $a \leftrightarrow b$, and send any remaining vertex to a . Take all nonloop arcs except these missing arcs. $\square$

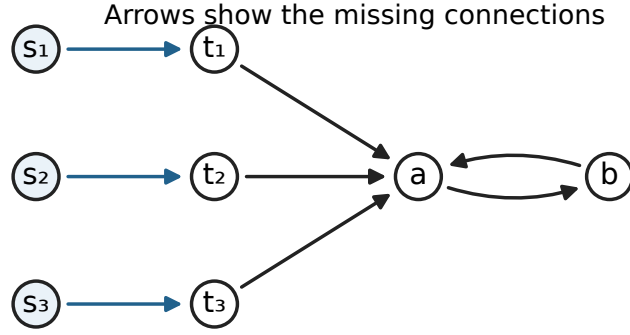


Figure 7: The missing arcs at the dense boundary form a functional graph. Each displayed two-vertex branch contributes one guaranteed indirect pair.

Theorem 14. A noncomplete graph with every pair guaranteed has at most $n(n - 3)$ arcs.

Proof. Such a graph has an indirect pair. No row can be complete: its vertex would block every indirect source other than itself and have no indirect target of its own. If a row c missed only $c \rightarrow t$, then $N^+(t) \subseteq C(c)$. Vertex t also has an indirect recipient, contradicting Lemma 1. Every row therefore misses at least two arcs. $\square$

The next section gives a uniform construction attaining this density on an infinite set of orders. Its proof is independent of the sparse-game certificate for the complementary graph proved later.

5.1.1 A periodic family with three-move delivery

Let $n = 5k$, $k \geq 4$, and write vertices as $5i + j$, with i modulo k and $0 \leq j \leq 4$. Put $\alpha(v) = v + 1$ and

$$\begin{aligned} \beta(5i) &= 5(i + 1), & \beta(5i + 1) &= 5(i - 1) + 2, \\ \beta(5i + 2) &= 5(i - 1) + 4, & \beta(5i + 3) &= 5(i - 1) + 3, \\ \beta(5i + 4) &= 5(i - 1) + 1. \end{aligned}$$

Let H have arcs $v \rightarrow \alpha(v)$, $v \rightarrow \beta(v)$, and let G be its loopless directed complement. Both maps are permutations, without fixed points or coincident images at a vertex; thus G has $n(n - 3)$ arcs.

Theorem 15. Every pair in G is guaranteed within three messenger moves. Consequently $M(5k, 5k(5k - 3)) = 10k$ for every $k \geq 4$.

Proof. For a missing pair (s, t) and initial pursuer $c \neq s$, a safe two-move intermediate is any member of

$$\{\alpha(c), \beta(c)\} \setminus F(s, t), \quad F(s, t) = \{s, t, \alpha(s), \beta(s), \alpha^{-1}(t), \beta^{-1}(t)\}.$$

Its exclusion from F ensures the two required arcs in G , while its membership in the displayed pair makes it inaccessible to the pursuer's next move.

Translate the source by a multiple of five so that $0 \leq s \leq 4$. A failure requires $c \in F - 1$, since $\alpha(c) \in F$. The complete local calculation is:

s	t	$F(s, t)$	$(c, \beta(c))$, with $c \in F - 1$, $c \neq s$
0	1	$\{0, 1, 5, 9\}$	$(-1, -9), (4, -4), (8, 3)$
0	5	$\{0, 1, 4, 5\}$	$(-1, -9), (3, -2), (4, -4)$
1	2	$\{-3, 1, 2, 6\}$	$(-4, -8), (0, 5), (5, 10)$
1	-3	$\{-4, -3, 1, 2\}$	$(-5, 0), (-4, -8), (0, 5)$
2	3	$\{-1, 2, 3, 8\}$	$(-2, -7), (1, -3), (7, 4)$
2	-1	$\{-2, -1, 2, 3\}$	$(-3, -6), (-2, -7), (1, -3)$
3	4	$\{-2, 3, 4, 7\}$	$(-3, -6), (2, -1), (6, 2)$
3	-2	$\{-3, -2, 3, 4\}$	$(-4, -8), (-3, -6), (2, -1)$
4	5	$\{-4, 0, 4, 5\}$	$(-5, 0), (-1, -9), (3, -2)$
4	-4	$\{-5, -4, 4, 5\}$	$(-6, -14), (-5, 0), (3, -2)$

The only candidate with $\beta(c) \in F$ is $(s, t, c) = (4, 5, -5)$. Every difference being tested has absolute value at most 19, so no extra equality appears modulo any $5k \geq 20$. In that exceptional state move from 4 to 0. This is legal and safe: the pursuer at -5 has no arc to 0. Afterwards the pair $(0, 5)$ is covered by the nonexceptional second row for every surviving pursuer position, and delivery takes at most two more moves. $\square$

5.2 Direct missing graphs unify the dense constructions

The following direct constructions replace the high-girth existence arguments for the numerical dense results. Its proof examines the possible safe intermediaries directly, so no girth hypothesis is needed. The parameterized families have ordinary proofs; a few small orders use explicitly listed finite constructions and independently checked witnesses. All strategies here take at most two messenger moves. Write $L = n(n - 1)$ and $h = L - m$.

For a missing-arc graph F , a missing pair (s, t) has a safe intermediary against pursuer $c \neq s$ precisely when

$$N_F^+(c) \setminus (N_F^+(s) \cup N_F^-(t)) \neq \emptyset.$$

We call the union in this expression the danger set. It contains both endpoints. A vertex outside it supplies both required messenger arcs, while being a missing outgoing neighbor of the pursuer makes it safe.

There is also a robust version. Suppose $B \subseteq F$ are missing graphs, and for every arc (s, t) of F and every $c \neq s$,

$$N_B^+(c) \not\subseteq N_F^+(s) \cup N_F^-(t). \quad (\text{R})$$

Then every intermediate missing graph $B \subseteq F' \subseteq F$ guarantees all pairs in two moves: retain the witness from B , while the danger set only shrinks. In particular, optional arcs can be chosen individually and in arbitrary order.

Theorem 58 (a unified dense construction). There are deterministic polynomial constructions with the following properties.

1. At every $n \geq 10$, a graph with $n(n - 3)$ arcs has all $2n$ missing pairs guaranteed within two moves. Thus $M(n, n(n - 3)) = 2n$.
2. At every $n \geq 16$, one compulsory set of $2n$ missing arcs and a list of n optional missing arcs satisfy (R). Hence every integer $2n \leq h \leq 3n$ attains $M(n, L - h) = h$, with two-move delivery.

3. At every $n \geq 20$, the entire final dense band is exact:

$$M(n, n(n-3) + r) = 2n - r - \left\lceil \frac{r}{2} \right\rceil - 1, \quad 1 \leq r \leq n.$$

Every counted pair in an attaining construction delivers within two moves. The first step $r = 1$ already holds at every $n \geq 16$.

4. For $n = 10q + s \geq 30$, $0 \leq s \leq 9$, put

$$H_5(n) = 10q^2 + 5q + s^2 + s.$$

A compulsory set of $2n$ missing arcs extends to an envelope with $H_5(n)$ missing arcs satisfying (R). For $n = 4q + s \geq 44$, $0 \leq s \leq 3$, the same assertion holds with

$$H_4(n) = 4q^2 - 26q + s^2 + s.$$

Thus every integer $2n \leq h \leq H(n)$ has $M(n, L-h) = h$, where $H(n) = H_5(n)$ for $30 \leq n < 44$ and $H(n) = \max\{H_5(n), H_4(n)\}$ for $n \geq 44$. Every optional subset works, and every missing pair delivers within two moves.

These statements determine every missing budget $n \leq h \leq H(n)$ at every $n \geq 30$, and every $n \leq h \leq 3n$ at every $n \geq 20$. The value is the displayed staircase for $h < 2n$ and h for $h \geq 2n$. The new cap strictly exceeds the cap of Theorem 48 throughout their common domain $n \geq 30$, preserving its time and arbitrary-subset guarantees; part 2 also covers all of its smaller orders. From $n = 52$ onward, $H(n) = H_4(n) = n^2/4 - O(n)$.

5.2.1 The two-type core

Use vertices P_i, L_i , with i modulo k , and the compulsory arcs

$$P_i \longrightarrow L_i, L_{i-1}, \quad L_i \longrightarrow P_{i+4}, P_{i+6}. \quad (\text{B})$$

There are exactly two incoming and two outgoing arcs at every vertex. The two types are disjoint, so no loop occurs. For $k \geq 5$, all pairs in the directed complement are guaranteed in two moves.

To see this, translate the source index to zero. A pursuer P_i is blocked precisely when the line part of the danger set contains both L_i, L_{i-1} . A pursuer L_i is blocked precisely when its point part contains both P_{i+4}, P_{i+6} . Thus line indices are tested for adjacency, and point indices for a difference of two. The four possible missing pairs have the following point and line danger sets:

- For $P_0 \rightarrow L_0$: point indices $\{0, 1\}$, line indices $\{0, -1\}$.
- For $P_0 \rightarrow L_{-1}$: point indices $\{-1, 0\}$, line indices $\{0, -1\}$.
- For $L_0 \rightarrow P_4$: point indices $\{4, 6\}$, line indices $\{0, -2\}$.
- For $L_0 \rightarrow P_6$: point indices $\{4, 6\}$, line indices $\{0, 2\}$.

For $k \geq 5$, the only blocked pursuer in each case is the source itself, which is not a legal initial position. This gives every even order $n = 2k \geq 10$.

For odd order $n = 2k + 1 \geq 13$, introduce a vertex w . Replace the arcs $P_0 \rightarrow L_0$ and $P_3 \rightarrow L_3$ by

$$P_0 \rightarrow w \rightarrow L_0, \quad P_3 \rightarrow w \rightarrow L_3. \quad (\text{O})$$

Each degree is still two. We verify the two-move property directly. For an old missing pair and a pursuer other than P_0, P_3, w , its old witnesses are unchanged: the danger set only gains the new

vertex w and may lose old vertices. The pursuer at w has missing neighbors L_0, L_3 , which cannot both occur in any of the four old line danger sets, whose only nonzero differences are one or two.

A pursuer at P_a , $a \in \{0, 3\}$, can use w unless the source is the other modified point or the recipient is L_0 or L_3 . In those cases L_{a-1} is safe. Explicitly, for source P_b with $a \neq b$ its remaining old recipient is L_{b-1} and the line danger set is $\{L_{b-1}\}$. For old recipient L_b , the remaining source is P_{b+1} and the line danger set is $\{L_b, L_{b+1}\}$. None contains L_{a-1} when $k \geq 6$ and $a, b \in \{0, 3\}$.

For a new pair $P_b \rightarrow w$, the danger set has point indices $\{0, 3\}$ and the single line index $b - 1$, in addition to w . Neither an adjacent line pair nor a point pair of difference two fits; the other modified point pursuer also retains its different line neighbor. For a new pair $w \rightarrow L_b$, the line indices are $\{0, 3\}$ and the point index is $b + 1$, in addition to w . Again all legal pursuers have a safe neighbor.

The remaining order $n = 11$ has the following explicit missing rows:

$$\begin{aligned} 0 : \{4, 8\}, \quad 1 : \{9, 10\}, \quad 2 : \{8, 9\}, \quad 3 : \{7, 10\}, \\ 4 : \{0, 1\}, \quad 5 : \{2, 6\}, \quad 6 : \{3, 5\}, \quad 7 : \{3, 4\}, \\ 8 : \{2, 7\}, \quad 9 : \{0, 5\}, \quad 10 : \{1, 6\}. \end{aligned}$$

Every incoming and outgoing degree is two. For each source, its two recipients in the displayed order give the following two danger sets:

s	$D(s, t_1)$	$D(s, t_2)$
0	$\{0, 4, 7, 8\}$	$\{0, 2, 4, 8\}$
1	$\{1, 2, 9, 10\}$	$\{1, 3, 9, 10\}$
2	$\{0, 2, 8, 9\}$	$\{1, 2, 8, 9\}$
3	$\{3, 7, 8, 10\}$	$\{1, 3, 7, 10\}$
4	$\{0, 1, 4, 9\}$	$\{0, 1, 4, 10\}$
5	$\{2, 5, 6, 8\}$	$\{2, 5, 6, 10\}$
6	$\{3, 5, 6, 7\}$	$\{3, 5, 6, 9\}$
7	$\{3, 4, 6, 7\}$	$\{0, 3, 4, 7\}$
8	$\{2, 5, 7, 8\}$	$\{2, 3, 7, 8\}$
9	$\{0, 4, 5, 9\}$	$\{0, 5, 6, 9\}$
10	$\{1, 4, 6, 10\}$	$\{1, 5, 6, 10\}$

Inspection of the eleven rows shows that each danger set contains only its source's row. Every legal pursuer therefore has a safe missing neighbor. This finite set-containment argument proves the eleven-vertex case and completes part 1. The accompanying certificate also lists all 220 pair/pursuer witnesses explicitly.

5.2.2 An envelope with one optional arc per vertex

For even order $n = 2k$, retain (B) as B and add the optional arcs

$$P_i \longrightarrow P_{i+6}, \quad L_i \longrightarrow L_{i+4}. \quad (\text{E})$$

For $k \geq 11$, condition (R) holds for the full envelope F . There are six types of missing pairs. Their point and line danger indices are respectively:

- $P_0 \rightarrow L_0$: $\{6, 0, 1\}$ and $\{0, -1, -4\}$;
- $P_0 \rightarrow L_{-1}$: $\{6, -1, 0\}$ and $\{0, -1, -5\}$;
- $P_0 \rightarrow P_6$: $\{6, 0\}$ and $\{0, -1, 2\}$;
- $L_0 \rightarrow P_4$: $\{4, 6, -2\}$ and $\{4, 0, -2\}$;
- $L_0 \rightarrow P_6$: $\{4, 6, 0\}$ and $\{4, 2, 0\}$;
- $L_0 \rightarrow L_4$: $\{4, 6, 5\}$ and $\{4, 0\}$.

Test line adjacency and point differences of two as before. The only blocked compulsory pair belongs to the source. All index differences tested have absolute value at most eight; $k \geq 11$ introduces no additional modular adjacency or difference of two. Thus (R) holds.

For odd order $n = 2k + 1$, with $k \geq 16$ or $k = 14$, use a different placement of the new vertex w : replace $P_0 \rightarrow L_0$ and $P_7 \rightarrow L_7$ by paths through w . Retain every optional arc in (E), and add the single optional arc $w \rightarrow P_2$. The compulsory graph has $2n$ arcs and the full envelope $3n$ arcs. We check (R).

For an old pair, every unmodified old pursuer retains its old safe witness because the only new danger vertex is w . The compulsory neighbors of the pursuer at w are L_0, L_7 . No translated line danger set in the six-case list contains both: their integer differences have absolute value at most six, while seven is not congruent to any such nonzero difference modulo the stated k .

The remaining old pursuers are P_a , $a \in \{0, 7\}$, with compulsory neighbors w, L_{a-1} . If w is in the danger set and the source differs from P_a , only the following cases arise:

- The source is the other modified point P_b . Its retained old targets are L_{b-1} and P_{b+6} . The respective line danger sets are $\{b-1, b-5\}$ and $\{b-1, b, b+2\}$.
- The recipient is L_b , $b \in \{0, 7\}$. The remaining old sources are P_{b+1} and L_{b-4} . Their line danger sets are $\{b+1, b, b-4\}$ and $\{b, b-4\}$.
- The recipient is P_2 . Its old sources are L_{-2}, L_{-4}, P_{-4} , and their line danger sets are contained in $\{2, 0, -2, -4, -5\}$.

None contains $a-1$ in its applicable case, for $k \geq 16$ or $k = 14$. Thus the remaining compulsory neighbor is safe.

It remains to check the new missing pairs. In the following lists, the danger set also contains w :

- For $P_b \rightarrow w$, $b \in \{0, 7\}$, the point indices are $\{0, 7, b+6\}$ and the only line index is $b-1$.
- For $w \rightarrow L_b$, the point indices are $\{2, b+1\}$ and the line indices are $\{0, 7, b-4\}$.
- For $w \rightarrow P_2$, the point indices are $\{2, -4\}$ and the line indices are $\{0, 7, -2, -4\}$.

The point sets contain no pair of difference two and the line sets no adjacent pair at the stated moduli. For example, the largest relevant point difference is 13, so it cannot become two when $k \geq 16$; modulo 14 it becomes -1 , which is also harmless. The line differences have absolute value at most eleven, so they introduce no modular adjacency. A modified point pursuer, when legal, retains a line neighbor L_{-1} or L_6 outside its danger set; the pursuer at w is either the source or sees only one of its two line neighbors in that set. This proves (R) at the stated odd orders.

Four explicit repairs fill the remaining odd orders. On $2k+1$ vertices, subdivide $P_0 \rightarrow L_0$ and $P_j \rightarrow L_j$ through w as above, retain the optional arcs (E), and add $w \rightarrow P_z$. Make the following substitutions in the optional list; every index is modulo k :

- At $n = 23$, use $(k, j, z) = (11, 5, 0)$, replacing $L_1 \rightarrow L_5$ by $L_1 \rightarrow L_7$ and $L_7 \rightarrow L_0$ by $L_7 \rightarrow L_2$.
- At $n = 25$, use $(k, j, z) = (12, 5, 2)$, replacing $L_1 \rightarrow L_5$ by $L_1 \rightarrow P_{10}$.

- At $n = 27$, use $(k, j, z) = (13, 6, 2)$, replacing $L_2 \rightarrow L_6$ by $L_2 \rightarrow P_{12}$.
- At $n = 31$, use $(k, j, z) = (15, 5, 0)$, replacing $L_1 \rightarrow L_5$ by $L_1 \rightarrow P_{10}$.

In each case the compulsory set has $2n$ arcs and the optional list has n distinct additional arcs. These are finite constructions: `research/dense-phase2-certificates.json` supplies, for every full-envelope pair and every legal pursuer, a compulsory neighbor outside its danger set. The direct checker `src/dense_phase2_certificates.py` verifies each listed witness using literal legal-move sets; an independent check is recorded in `research/dense-phase2-independent-root-check.json`. The certificate proves (R), so it covers every optional subset rather than just the chosen total counts. Together with the parameterized families, this establishes the bridge at every $n \geq 22$.

5.2.2.1 Six smaller envelopes For $n = 16, \dots, 21$, use the compulsory core (B), or its odd modification (O). Identify $P_i = i$, $L_i = k + i$, and, in odd order, $w = 2k$, where $k = \lfloor n/2 \rfloor$. The following ordered pairs are optional missing arcs:

$$O_{16} = \{(0, 12), (1, 13), (2, 13), (3, 15), (4, 15), (5, 9), (6, 10), (7, 10), (8, 1), (9, 2), (10, 3), (11, 4), (12, 5), (13, 6), (14, 7), (15, 0)\}$$

$$O_{17} = \{(0, 7), (0, 11), (1, 13), (2, 14), (3, 14), (4, 16), (5, 9), (6, 8), (7, 10), (8, 3), (10, 3), (12, 5), (13, 6), (14, 10), (15, 6), (16, 1), (16, 13)\}$$

$$O_{18} = \{(0, 14), (1, 14), (2, 15), (4, 16), (5, 17), (6, 9), (7, 10), (8, 11), (8, 13), (10, 1), (11, 2), (12, 3), (12, 4), (13, 5), (14, 6), (14, 9), (15, 7), (17, 8)\}$$

$$O_{19} = \{(0, 3), (0, 12), (1, 2), (2, 15), (3, 15), (4, 5), (5, 9), (6, 10), (6, 18), (7, 11), (8, 11), (8, 13), (12, 18), (13, 4), (14, 6), (15, 10), (16, 9), (17, 8), (17, 13)\}$$

$$O_{20} = \{(0, 13), (0, 16), (1, 14), (2, 15), (3, 18), (4, 10), (4, 18), (5, 10), (9, 12), (9, 15), (10, 9), (11, 17), (12, 2), (13, 3), (13, 17), (14, 5), (15, 6), (17, 11), (18, 14), (19, 8)\}$$

$$O_{21} = \{(0, 3), (1, 14), (1, 16), (2, 18), (3, 18), (6, 11), (7, 11), (8, 12), (8, 14), (9, 12), (11, 1), (11, 15), (12, 2), (13, 20), (15, 6), (15, 19), (17, 6), (17, 7), (18, 8), (18, 10), (18, 14)\}$$

Each list contains exactly n distinct arcs outside the compulsory set of $2n$ arcs. For every missing pair in the full envelope and legal initial pursuer, the archived certificate explicitly supplies $u \in N_B^+(c)$ with $(s, u), (u, t) \notin F$ and $u \notin \{s, t, c\}$. These are precisely the obligations in (R). The archives are `research/routing-defect-phase2-dense-tiny-checks.json` for orders 16, 17, 18 and `research/routing-defect-phase2-dense-small-checks.json` for 19, 20, 21; at order 18 the archived full envelope is larger, so the displayed subset inherits its witnesses.

Separate literal readback programs, importing neither the producer nor its graph helpers, check every stored witness and its complete pair/pursuer coverage. Their receipts are `research/routing-defect-phase2-dense-tiny-independent.json` and `research/routing-defect-phase2-dense-small-independent.json`. In addition, `SmallDenseEnvelopes.lean` checks all six row lists, the base and envelope arrow counts, and every required safe witness by kernel reduction. The robust-envelope semantic theorem then proves actual two-move delivery for every intermediate graph, against arbitrary legal pursuer choices. This verifies every optional subset, not just a single graph at $h = 3n$. Together with the preceding families, it completes part 2 for every $n \geq 16$.

5.2.3 Quadratic envelopes from a shared danger-set calculation

The interval can be enlarged while keeping just two compulsory missing neighbors per pursuer. In a core with indices modulo k , use

$$P_i \rightarrow L_i, L_{i-1}, \quad L_i \rightarrow P_{i+\alpha}, P_{i+\beta},$$

and optional arcs $P_i \rightarrow P_{i+d}$ for $d \in D$ and $L_i \rightarrow L_{i+e}$ for $e \in E$. Translate the source index to zero. The six full-envelope danger sets have the following point and line index parts:

pair	point indices	line indices	
$P_0 \rightarrow L_0$	$D \cup \{0, 1\}$	$\{0, -1\} \cup (-E)$	(T)
$P_0 \rightarrow L_{-1}$	$D \cup \{-1, 0\}$	$\{0, -1\} \cup (-1 - E)$	
$P_0 \rightarrow P_d$	$D \cup (d - D)$	$\{0, -1, d - \alpha, d - \beta\}$	
$L_0 \rightarrow P_\alpha$	$\{\alpha, \beta\} \cup (\alpha - D)$	$E \cup \{0, \alpha - \beta\}$	
$L_0 \rightarrow P_\beta$	$\{\alpha, \beta\} \cup (\beta - D)$	$E \cup \{0, \beta - \alpha\}$	
$L_0 \rightarrow L_e$	$\{\alpha, \beta, e, e + 1\}$	$E \cup (e - E)$	

Condition (R) asks whether the line part contains an adjacent pair, or the point part contains a compulsory pair of difference $\beta - \alpha$, belonging to a pursuer other than the source. This single calculation will establish both caps in part 4.

5.2.3.1 A residue-five family at every order from 30 Write $n = 10q + s$, where $q \geq 3$ and $0 \leq s \leq 9$, and put $k = 5q$. Take $(\alpha, \beta) = (4, 6)$ and

$$D = \{10, 15, \dots, k - 5\}, \quad E = \{5, 10, \dots, k - 5\}.$$

Add vertices $x_0, \dots, x_{s-1}$ with compulsory missing rows

$$N_B^+(x_a) = \{P_a, L_{a+3}\}.$$

There are no missing arcs from the core to these vertices. Permit every directed nonloop arc between added vertices as an additional optional missing arc.

First consider only old sources and pursuers. The multiples-of-five sets in (T) contain neither adjacent indices nor indices of difference two. In the four compulsory-pair rows of (T), a compulsory cop row can be contained only when it is the source's own row. For $P_0 \rightarrow P_d$, the line set $\{0, -1, d - 4, d - 6\}$ acquires an additional adjacent pair only at $d = 2, 4, 5, 7$; values 3 and 6 merely duplicate the source row. Only 5 is a multiple of five, and it is excluded from D . For $L_0 \rightarrow L_e$, the apparent

extra difference-two pair at $e = 5$ is just the existing pair $\{4, 6\}$. Thus old legal pursuers always have a witness. These modular statements hold because $5 \mid k$ and $k \geq 15$.

An added cop has point/line index difference three. For an optional $P \rightarrow P$ pair, the point danger has the source's residue, while its line residues differ by 0, 1, 4; difference three is absent. For an optional $L \rightarrow L$ pair, the line danger has one residue and the point residues differ from it by 0, 1, 4, again excluding the row. The four compulsory cases in (T) give the same exclusion except possibly (P_{i+1}, L_{i-1}) or (P_{i+4}, L_{i+2}) . Their actual line-minus-point difference is -2 , which cannot equal three modulo $k \geq 15$. Hence added cops also retain witnesses.

For a new source x_a targeting P_a , the old danger parts are

$$\{P_a\} \cup \{P_{a-d} : d \in D\}, \quad \{L_{a+3}, L_{a-4}, L_{a-6}\}.$$

The first lies in one residue class. The second has pairwise differences 7, 9, 2, none congruent to ± 1 modulo $k \geq 15$. Neither old row type is blocked. An added row from x_b would require $b - a$ to be a multiple of five and one of 0, -7 , -9 , forcing $b = a$, the excluded source. For recipient L_{a+3} , the old point danger is $\{P_a, P_{a+3}, P_{a+4}\}$, whose differences 1, 3, 4 cannot be ± 2 ; the old line danger has one residue class. An added cop would require $b - a \in \{0, 3, 4\}$ to be a multiple of five, again forcing $b = a$.

Finally, an optional pair $x_a \rightarrow x_b$ has old danger exactly $\{P_a, L_{a+3}\}$. It contains neither an old row nor a distinct new row. Incoming arrows from added vertices only add new vertices to other danger sets, whereas every compulsory row lies in the old core. They cannot spoil any witness. This proves (R).

There are $4k + 2s = 2n$ compulsory missing arcs, $k((q-2) + (q-1))$ optional core arcs, and $s(s-1)$ optional added arcs. Their total is exactly $H_5(n) = 10q^2 + 5q + s^2 + s$.

5.2.3.2 A parity family with quarter-density cap Write $n = 4q + s$, $q \geq 11$, $0 \leq s \leq 3$, and put $k = 2q$. Take $(\alpha, \beta) = (4, 7)$. Let D consist of all nonzero even residues except $\pm 2, \pm 4, \pm 6, \pm 8$, and let E consist of all nonzero even residues except $\pm 2, \pm 4, 8, 10, 12$. For each $0 \leq a < s$ add a vertex with compulsory row

$$N_B^+(x_a) = \{P_a, P_{a+5}\},$$

and again permit all nonloop missing arcs between added vertices, with none from old vertices to new ones.

Both forbidden row differences in (T) are odd: one for a point cop, three for a line cop. Since D, E contain only even indices, testing the exceptional even/odd pairs in each row of (T) gives the following possible offsets for a blocked old cop other than the source:

- $P_0 \rightarrow L_0$: 4, -2 in D , or 2 in E ;
- $P_0 \rightarrow L_{-1}$: 2, -4 in D , or -2 in E ;
- $P_0 \rightarrow P_d$: $d = 2, 8$;
- $L_0 \rightarrow P_4$: -6 in D , or $-2, -4$ in E ;
- $L_0 \rightarrow P_7$: 6 in D , or 2, 4 in E ;
- $L_0 \rightarrow L_e$: $e = 0, 10$.

All these offsets are excluded. Coincidences that merely reproduce the source row do not block another cop: adjacent pairs and pairs of difference three have unique source indices when $k \geq 22$.

An added cop has a point pair of difference five. In the same six point-danger sets, such a pair would require respectively offsets 6, -4 in D ; 4, -6 in D ; no possibility in the all-even third case; 2, -8 in D ; $-2, 8$ in D ; or $-2, 8, 2, 12$ in E . Every exception is excluded, so added cops are safe against old pairs.

For a new source x_a targeting P_a , its old point danger is

$$\{P_a, P_{a+5}\} \cup \{P_{a-d} : d \in D\},$$

and its old line danger is $\{L_{a-4}, L_{a-7}\}$. The latter is not adjacent. A difference-three point pair would require $d = -2$ or -8 , excluded from D . For recipient P_{a+5} , the point danger is

$$\{P_a, P_{a+5}\} \cup \{P_{a+5-d} : d \in D\};$$

the same test requires the excluded offsets 2, 8. Its line danger again has difference three. A distinct added row could be contained only if $b = a$ or $b = a \pm 5$ modulo k : one parity occurs just once in the point danger. The first is the source and the others are impossible for $a, b \in \{0, 1, 2\}$ and $k \geq 22$.

For a new-to-new pair, the old danger is the source's difference-five point pair, containing no old row and no distinct added row. As before, new danger vertices belong to no compulsory row. This proves (R).

The eight exclusions in D and seven in E are distinct because $k \geq 22$, so $|D| = q - 9$ and $|E| = q - 8$. There are $2n$ compulsory missing arcs and $k(2q - 17) + s(s - 1)$ optional ones, totaling $H_4(n) = 4q^2 - 26q + s^2 + s$. Part 4 follows by selecting any desired number of optional arcs.

5.2.3.3 Comparison and proof boundary The improvement retains the old small-order domains. For $n \geq 30$,

$$H_5(n) - 3n = 5q(2q - 5) + (s - 1)^2 - 1 \geq 14.$$

Also

$$\begin{aligned} 13H_5(n) - n(n + 11) &= 30q^2 - 45q - 20qs + 12s^2 + 2s \\ &\geq \frac{260(q - 1)^2 - 261}{12} > 0. \end{aligned}$$

Since $\lceil (n - 1)/13 \rceil \leq (n + 11)/13$, this strictly exceeds the cap $n \max\{3, \lceil (n - 1)/13 \rceil\}$ of Theorem 48. For H_4 , the difference from $3n$ is at least 65 and

$$13H_4(n) - n(n + 11) = 36q^2 - 382q - 8qs + 12s^2 + 2s.$$

At $q = 11$ its values are 154, 80, 30, 4, and each increases with q . Thus this family also strictly exceeds the old cap at every $n \geq 44$.

The twenty residue classes modulo 20 show $H_4(n) > H_5(n)$ for every $n \geq 52$. Indeed it holds at 52, $\dots$, 71, and if $n = 20t + u$, the difference increases on replacing n by $n + 20$ by

$$120t - 80 + 40(\lfloor u/4 \rfloor - \lfloor u/10 \rfloor) > 0.$$

For the old orders $n = 19, \dots, 29$, that cap is exactly $3n$, already covered by part 2. Thus the present envelopes subsume its entire numerical statement, preserving the time, algorithm, and arbitrary-subset conclusions at every old order. No earlier domain is lost.

The two cap formulas have ordinary proofs above. The independent review `research/dense-phase2-uniform-independent-audit.md` checks their modular exceptions and counts. The primary receipts `research/dense-phase2-uniform-envelope.json` and `research/dense-phase2-quarter-envelope.json` additionally check bounded full-envelope games; those samples are not used to infer the formulas.

5.2.4 The first forced loss at every order from 16

Use an even core of order $2k \geq 12$, or its odd modification (O) of order $2k + 1 \geq 13$. Add four vertices c, x, y, z and missing arcs

$$c \rightarrow x, \quad x \rightarrow y, z, \quad y \rightarrow a, b, \quad z \rightarrow d, e. \quad (\text{A})$$

In the even case choose $(a, b, d, e) = (P_0, P_3, P_1, P_4)$. In the odd case choose $(a, b, d, e) = (P_1, P_4, P_2, P_5)$. The four attachments are distinct; each branch pair has index difference three, hence neither one nor two modulo $k \geq 6$.

For an old missing pair, old pursuers retain their old safe neighbors, since the danger set only gains y or z . A pursuer at c uses x . A pursuer at x can use y or z : no old recipient receives both branches. The pursuers at y, z have point pairs of difference three, which cannot fit into the old point danger pairs of difference one or two. In the odd core there is also the point danger set $\{P_0, P_3\}$ for a pair ending at w ; neither chosen branch is that pair. The other new old-core pairs have at most one point in their danger set.

For a new pair with source y or z and recipient an attachment P_t , ordinary point pursuers have adjacent line neighbors while the old incoming line pair of P_t has difference two. Ordinary line pursuers have point-neighbor difference two while the branch pair has difference three. In the odd core, the modified point pursuers use w , and the pursuer at w has line neighbors of difference three, which cannot both lie in the incoming pair of difference two. The opposite branch is disjoint, x can use that opposite branch, and c can use x . Thus these four pairs win. The pair $c \rightarrow x$ also wins: no missing neighbor enters c , and every other pursuer has two missing neighbors, at least one outside $\{c, x\}$.

Exactly the pairs $(x, y), (x, z)$ fail. A pursuer at c has closed move set $V \setminus \{x\}$ in the delivery graph and blocks every nonterminal departure from x . All other initial pursuers have safe neighbors for those pairs. The construction has $2n - 1$ missing arcs, so $2n - 3$ pairs count. This proves the first-step assertion for all $n \geq 16$.

5.2.5 Functional roots and the rest of the staircase

Use the spare functional vertex when the order is odd. On $r \geq 2$ vertices put $j = \lfloor (r - 2)/2 \rfloor$, retain the missing chains $s_i \rightarrow t_i \rightarrow a$ and the cycle $a \leftrightarrow b$, and, for odd r , put the remaining vertex $z \rightarrow a$. The set of vertices with no missing incoming arc is

$$R = \{s_1, \dots, s_j\} \quad \text{together with } z \text{ when } r \text{ is odd.}$$

Consequently $|R| = \lfloor (r - 1)/2 \rfloor$.

Add k vertices, each with a distinct two-element missing-neighbor set in R . This is possible whenever

$$k \leq \binom{\lfloor (r - 1)/2 \rfloor}{2}. \quad (\text{F})$$

It guarantees exactly $j+2k$ counted pairs. Against a functional pursuer, its unique missing neighbor is outside R and outside the added vertices, so it is safe for every added source and its two recipients. Against another added pursuer, choose a member of its two-element set outside the source's set. Any two distinct roots are joined in the delivery graph. The original pairs (s_i, t_i) also survive: an added pursuer has a root neighbor different from s_i , and that root has a direct delivery arc to t_i . This argument includes the extra root z . Theorem 27 gives the matching upper bound.

Now fix $n \geq 20$ and $1 \leq r \leq n$, and put $k = n - r$. The first step is already proved. For $r \geq 2$ and $k \geq 10$, join the functional r -vertex graph to the direct k -vertex core; Lemma 28 preserves $j + 2k$ guarantees. If $1 \leq k \leq 9$ remains, then $r \geq 11$, so (F) holds because

$$\binom{\lfloor (r-1)/2 \rfloor}{2} \geq \binom{5}{2} = 10 \geq k.$$

Finally $k = 0$ is the functional endpoint in Theorem 13. Since

$$j + 2k = 2n - r - \lceil r/2 \rceil - 1,$$

this proves part 3. More precisely, the same formula holds whenever $r \geq 2$ and either the core condition or (F) applies, including smaller orders than 20; the endpoint $r = n$ holds at every $n \geq 2$.

All recipes consist of modular arc lists and elementary subset choices, and therefore take polynomial time. The receipt `research/dense-unified-core-checks.json` records bounded complete two-move and robustness checks from `src/dense_unified_core.py`. The parameterized proofs above do not depend on those finite tests. The explicitly isolated eleven-vertex core and small optional envelopes have the finite proofs and certificates described above. The infinite families and their general existence proofs remain ordinary arguments. The finite eleven-vertex core and the six small envelopes have Lean coverage in `ElevenVertexDense.lean` and `SmallDenseEnvelopes.lean`, as detailed in the formal verification section.

5.3 Missing-graph criteria and the dense endpoint

The following criterion applies to arbitrary missing graphs. The numerical constructions are now supplied by the stronger direct Theorem 58.

Lemma 18 (high-girth complement). Let F be an orientation of a finite simple undirected graph of girth at least six. Suppose every vertex has outdegree at least two in F . Let G contain exactly the nonloop arcs absent from F . Then every source–recipient pair in G is guaranteed within two messenger moves.

Proof. Diagonal and direct pairs need no argument. For a missing pair (s, t) of G , the arc $s \rightarrow t$ belongs to F . Set

$$B = N_F^+(s) \cup N_F^-(t).$$

Both s and t belong to B . In the underlying undirected graph, the vertices of B form an induced double star: its centers are s, t , the other outneighbors of s are leaves at s , and the other predecessors of t are leaves at t . The two sets of leaves are disjoint, since an overlap would make a triangle. An additional edge inside B would make a triangle or quadrilateral. Thus the distance between any two vertices in this induced tree is at most three.

Fix an initial pursuer vertex $c \neq s$. If $c \notin B$, two neighbors of c in B would form an undirected cycle of length at most five, so c has at most one outgoing F -neighbor in B . The same bound holds

when $c \in B \setminus \{s\}$: a leaf at s has its only incident edge in the tree directed toward it; a leaf at t has only its edge toward t ; and all tree edges incident with t point into t .

Since c has at least two outgoing F -neighbors, choose

$$u \in N_F^+(c) \setminus B.$$

Then $u \neq s, t$, and $s \rightarrow u$ and $u \rightarrow t$ are arcs of G . Also $u \neq c$ and $c \rightarrow u$ is absent from G . The messenger moves to u , survives every pursuer response, and delivers to t on its next turn. $\square$

The lemma only requires a lower bound on outgoing degrees. In particular, edges can later be added to the underlying graph and oriented arbitrarily, provided its girth remains at least six.

Theorem 19 (the densest noncomplete all-delivery networks). At every $n \geq 10$,

$$M(n, n(n-3)) = 2n.$$

An attaining graph guarantees every pair within two messenger moves.

Proof. Use the direct two-type missing graph, with its odd-order subdivision when needed, and the explicit eleven-vertex core, in Theorem 58. It has exactly $2n$ missing arcs and a two-move guarantee for every pair. The missing-pair upper bound proves equality. $\square$

The density $n(n-3)$ is maximal for a noncomplete all-delivery graph by Theorem 14. The earlier high-girth construction at orders $n \geq 52$ is no longer needed for this numerical theorem. Lemma 18 remains useful as a criterion for arbitrary missing graphs; its structural hypothesis is stronger than the condition needed by the direct construction.

Theorem 20 (a retained dense parameterization). Fix integers $n \geq 52$ and $5 \leq D \leq n-1$, and put

$$B_D = D^4 - 2D^3 + 2D^2 + 1, \quad L_D(n) = \left\lfloor \frac{Dn - (D-4)B_D}{2} \right\rfloor.$$

Every integer $2n \leq h \leq L_D(n)$ satisfies

$$M(n, n(n-1) - h) = h,$$

with a deterministic two-move attaining construction. An empty interval makes no assertion.

Proof. A nonempty interval forces $n \geq B_D \geq B_5 = 426$. For $D \geq 5$, the displayed polynomial gives $B_D - 1 \geq 13D/2$. Hence

$$L_D(n) \leq \frac{Dn}{2} \leq n \left\lceil \frac{n-1}{13} \right\rceil.$$

The robust bridge of Theorem 58 and the protected envelope of Theorem 48 cover every missing budget from $2n$ to the latter endpoint. Their constructions have two-move delivery. $\square$

This parameterization is a consequence of the stronger dense envelope, retained for references to its explicit D -dependent bound. It introduces no additional graph search, girth-growth certificate, or separate construction family.

5.4 Attainment across densities and losses near completeness

Angel Raychev proposed that, for every sufficiently large n , the missing-pair bound should be attained at **every** integer budget

$$2n \leq m \leq n(n-3).$$

Theorem 44 proves this conjecture at every $n \geq 115$ using the explicit constructions of Theorems 38, 52, 58, and 59. This section gives the separate exact-budget probability argument and the dense counting and composition principles. Raychev also proposed separating existence from explicit constructions and studying how domination destroys missing pairs as the graph approaches completeness. These distinctions motivate the statements below; their proofs were developed in the September 2026 follow-up investigation.

Write $L = n(n-1)$ and $h = L - m$. Thus h is the number of missing nonloop arcs, and always $I(G) \leq h$.

5.4.1 An exact-budget existence theorem

Theorem 25. Let $n \geq 4$, let $0 < m < L$ be an integer, and put $p = m/L$. If

$$(n-3)p^2(1-p) > 5 \log n,$$

then some graph with exactly n vertices and m arcs guarantees every source–recipient pair within two messenger moves. In particular,

$$M(n, m) = L - m.$$

Proof. Apply Theorem 54 with the routing set equal to the entire vertex set: $H = V$, $b = n$, and $a = m$. Since $n \geq 4$,

$$5 \log n > 4 \log n + \log(8/3),$$

the displayed hypothesis implies its condition. Theorem 54 supplies an exactly m -arc graph with a safe two-move intermediary for every legal source–recipient–pursuer triple, proving the assertion. $\square$

This is an exact integer-budget existence theorem, rather than a statement about an expected number of arcs. It also gives a finite randomized search: sample graphs with exactly m arcs and check the safe-intermediate condition. No practical running-time bound near the threshold is asserted.

Corollary 26. There is an absolute N such that, for every $n \geq N$ and every integer budget

$$\left\lceil 10n^{3/2}\sqrt{\log n} \right\rceil \leq m \leq n(n-3),$$

we have $M(n, m) = n(n-1) - m$. Attainment can always be achieved with delivery in at most two moves.

Proof. Let $p = m/L$ and $h = L - m$. For $n \geq 6$, we have $n-3 \geq n/2$. If $p \leq 1/2$, the lower bound on m gives $p \geq 10\sqrt{\log n}/n$ and therefore

$$(n-3)p^2(1-p) \geq 25 \log n > 5 \log n.$$

If $p \geq 1/2$ and $h \geq 100n \log n$, then $1 - p = h/L \geq 100 \log n/n$, giving

$$(n-3)p^2(1-p) \geq \frac{100}{8} \log n > 5 \log n.$$

Theorem 25 covers both cases.

For the remaining dense budgets use Theorem 20 with

$$D = \lfloor (n/2)^{1/4} \rfloor.$$

For all sufficiently large n its hypotheses hold, and its upper endpoint satisfies

$$L_D(n) \geq n(D/4 + 1) - 1.$$

Since D grows as a positive constant times $n^{1/4}$, this endpoint eventually exceeds $100n \log n$. Theorem 20 supplies every integer missing-arc count from $2n$ to $L_D(n)$, and the probabilistic range supplies the rest. The intervals overlap, so no admissible integer budget is omitted. Both proofs give two-move delivery. $\square$

The absolute threshold is not optimized or asserted to be moderate. In particular, this corollary does not establish the full interval at $n = 52$. The general sparse interval between $2n$ and order $n^{3/2}\sqrt{\log n}$ remains outside its conclusion.

5.4.2 A growing loss in the final dense band

Let F be the loopless directed complement of G , and write $q(v) = d_F^+(v)$. The number

$$\Delta = h - I(G)$$

counts missing pairs that fail to guarantee delivery.

Theorem 27. For $n \geq 2$ and every admissible integer $n \leq h < 2n$,

$$I(G) \leq h - \left\lceil \frac{2n - h}{2} \right\rceil - 1.$$

Equivalently, writing $r = 2n - h$, we have

$$M(n, n(n-3) + r) \leq 2n - r - \lceil r/2 \rceil - 1 \quad (1 \leq r \leq n).$$

Proof. If $I(G) = 0$, the conclusion holds: the right-hand side is increasing with h and at $h = n$ equals $\lfloor (n-2)/2 \rfloor \geq 0$. Assume henceforth that $I(G) > 0$.

Every $q(v)$ is positive. Otherwise a vertex c with $q(c) = 0$ can intercept every nonterminal departure from any other source, while its own source row has no missing pair. Put

$$S = \{c : q(c) = 1\}, \quad \ell = |S|.$$

Since $h \geq \ell + 2(n - \ell)$, we have $\ell \geq 2n - h \geq 1$. For $c \in S$ let $f(c)$ be its unique missing neighbor; thus $f(c) \neq c$. Set

$$U = f(S), \quad a = |U \cap S|, \quad b = |U \setminus S|.$$

Every source $x \in U$ loses all of its $q(x)$ missing pairs. To see this, choose $c \in S$ with $f(c) = x$. The pursuer's closed outgoing neighborhood in G is $V \setminus \{x\}$. It can wait at c while the messenger

waits at x and intercept every nonterminal departure. An indirect recipient cannot be reached by a direct departure.

A second obstruction applies when two distinct vertices $c, c' \in S$ have the same image x . The missing pair (c, x) loses against a pursuer starting at c' , whose closed outgoing neighborhood contains every possible nonterminal intermediate vertex.

Let R be the set of $c \in S \setminus U$ for which $f(c)$ has exactly one preimage in S , and put $\rho = |R|$. Every missing edge with tail in $S \setminus R$ loses by one of the two obstructions. The killed rows in $U \setminus S$ have at least two missing edges each, with tails outside S . These losses are disjoint, so

$$\Delta \geq \ell - \rho + 2b.$$

If $b \geq 1$, then $\rho \leq \ell - a$ because $R \subseteq S \setminus U$, and $\rho \leq a + b$ because its members have distinct images in U . Consequently

$$\Delta \geq \max\{a + 2b, \ell - a + b\} \geq \left\lceil \frac{\ell + 3b}{2} \right\rceil \geq \lceil \ell/2 \rceil + 1.$$

If $b = 0$, the function f maps S into itself without fixed points. Its finite functional graph has a directed cycle of length at least two. Every cycle vertex belongs to U and already has a preimage in U , so it cannot also be the image of a member of R : that member is outside U and would give a second preimage. Thus $\rho \leq a - 2$, as well as $\rho \leq \ell - a$. It follows that

$$\Delta \geq \max\{a, \ell - a + 2\} \geq \lceil \ell/2 \rceil + 1.$$

Using $\ell \geq 2n - h$ proves the bound. $\square$

Thus the forced loss increases as $2, 2, 3, 3, 4, 4, \dots$ as the arc budget moves one, two, three, four, five, six, and further steps above $n(n - 3)$. At the other endpoint, $h = n$, the bound agrees with Theorem 13.

5.4.3 Combining two-move constructions

Lemma 28. Suppose a missing-edge graph F is a disjoint union of graphs F_i , each with positive outgoing degree at every vertex. Let G_i be the complement on its own block. If an indirect pair in G_i can be guaranteed within two messenger moves, then it remains guaranteed in the complement G of the entire F .

Proof. Let the source and recipient lie in block i . If the initial pursuer also lies there, the original two-move strategy chooses an intermediate vertex outside its closed outgoing neighborhood. The same is true in G . The pursuer may respond by moving outside the block, but this cannot prevent receipt on the messenger's next move.

If the pursuer starts at c in another block, choose any outgoing F -neighbor u of c in that block. The arcs $s \rightarrow u$ and $u \rightarrow t$ are present in G because they cross blocks, whereas $c \rightarrow u$ is absent. Also $u \neq c, s, t$. This gives the required safe two-move delivery. $\square$

The lemma concerns two-move strategies. No preservation theorem for arbitrary longer strategies under this join is assumed.

Theorem 29. Let $r \geq 2$ and $k = n - r$. If $k \geq 10$, then

$$M(n, n(n - 3) + r) = 2n - r - \lceil r/2 \rceil - 1.$$

An explicit polynomial construction attains this value with every counted pair delivered within two messenger moves.

Proof. Use the functional missing graph on r vertices, with $j = \lfloor (r-2)/2 \rfloor$ guaranteed pairs. On the other k vertices use the direct missing core in Theorem 58, which has $2k$ guaranteed missing pairs. Both blocks have positive missing outdegree, so Lemma 28 preserves all $j + 2k$ guarantees in the complement of their disjoint union. There are $r + 2k = 2n - r$ missing arcs, and

$$j + 2k = 2n - r - \lceil r/2 \rceil - 1.$$

Theorem 27 gives optimality. $\square$

This includes the former $k \geq 52$ range and the separate 26-vertex block, without any finite girth-growth dependency. In missing-budget notation the range is $n + 10 \leq h \leq 2n - 2$.

5.4.4 A restriction on permutation constructions

Proposition 30. Suppose the two outgoing neighbors of each vertex are $\alpha(v)$ and $\beta(v)$, where α, β are commuting permutations, both images differ from v , and $\alpha(v) \neq \beta(v)$. Then $I(G) \leq n$. In fact all indirect guarantees are decided by a two-move criterion.

Proof. Fix a recipient t and put $s_t = \alpha^{-1}\beta^{-1}(t)$. Commutation gives

$$N^+(s_t) = \{\beta^{-1}(t), \alpha^{-1}(t)\} = N^-(t).$$

The predecessor camping lemma excludes every indirect source except possibly s_t : a pursuer starting at s_t can intercept the messenger when it first reaches a predecessor of t . Thus each recipient contributes at most one pair.

When $s_t \neq t$, the pair (s_t, t) is missing because neither predecessor of a loopless recipient equals that recipient. This pair is guaranteed exactly when no $c \neq s_t$ has $N^-(t) \subseteq C(c)$. Necessity is the same camping lemma. For sufficiency, from each legal initial pursuer position choose $u \in N^-(t) \setminus C(c)$. The messenger moves from s_t to this safe predecessor and then delivers. $\square$

The bound is sharp: on $\mathbb{Z}/7\mathbb{Z}$, take arcs $x \rightarrow x+1$ and $x \rightarrow x+3$. For recipient zero the predecessors are 6, 4, and the only closed outgoing neighborhood containing both is that of 3. The criterion gives $(3, 0)$; translation gives exactly the seven pairs $(s, s+4)$. Since $n(n-3) > n$ for $n \geq 5$, commuting permutations cannot supply a universal vertex-extremal graph in that range. Noncommutation is necessary, not asserted sufficient.

5.4.5 An extension that fills the small dense offsets

Proposition 31. Let $r \geq 2$, $k \geq 1$, and $n = r + k$. If

$$k \leq \binom{\lfloor (r-1)/2 \rfloor}{2},$$

then a deterministic polynomial construction attains

$$M(n, n(n-3) + r) = 2n - r - \lceil r/2 \rceil - 1,$$

with a two-move strategy for every counted pair.

Proof. The functional construction has $j = \lfloor (r-2)/2 \rfloor$ guaranteed pairs. Its vertices with no missing incoming arc are the j branch sources, together with the spare vertex when r is odd. There are exactly $q = \lfloor (r-1)/2 \rfloor$ such roots. Give each added vertex a different two-element missing-neighbor set among these roots. The functional-root argument in Theorem 58 proves all $2k$ new missing pairs and preserves the original j pairs in two moves. The identity $j+2k = 2n-r - \lceil r/2 \rceil - 1$ and Theorem 27 prove equality. $\square$

This strengthens the former bound $k \leq \binom{\lfloor (r-2)/2 \rfloor}{2}$ when r is odd and preserves it when r is even. In particular, all the earlier $n \geq 75$, $1 \leq n-r \leq 51$ cases remain covered.

Theorem 58 supplies the common direct construction and complete proof. The ordinary counting bound, the arbitrary-block composition lemma, and the commuting-permutation restriction remain independent structural results. No new Lean formalization is asserted here.

5.5 Exact final dense band

Theorem 32. For every $n \geq 16$,

$$M(n, n(n-3) + 1) = 2n - 3.$$

A deterministic attaining graph guarantees all but two of its missing pairs within two messenger moves.

Proof. Use the four-vertex attachment to a direct two-type core in Theorem 58. The graph has $2n-1$ missing arcs. Exactly the two missing pairs leaving the attachment's vertex x are blocked by a pursuer at c ; every other missing pair has a safe two-move intermediary. Theorem 27 gives the matching upper bound. $\square$

Corollary 33 (the complete final dense band). For every $n \geq 20$ and every integer $1 \leq r \leq n$,

$$M(n, n(n-3) + r) = 2n - r - \left\lceil \frac{r}{2} \right\rceil - 1.$$

Equivalently, for every integer $n \leq h \leq 2n-1$,

$$M(n, n(n-1) - h) = h - \left\lceil \frac{2n-h}{2} \right\rceil - 1.$$

There is a deterministic polynomial attaining construction at every such budget, with a two-move strategy for every counted pair.

Proof. Theorem 27 gives the upper bound. The first step is Theorem 32. For $r \geq 2$, put $k = n-r$. If $k \geq 10$, Theorem 29 applies. Every remaining positive k is at most nine, so $r \geq 11$ and

$$\binom{\lfloor (r-1)/2 \rfloor}{2} \geq \binom{5}{2} = 10 \geq k.$$

Proposition 31 applies. Finally $k = 0$ is the functional endpoint of Theorem 13. $\square$

Together with Theorem 19 at $m = n(n-3)$ and Theorem 13 above $m = n(n-2)$, this determines the joint maximum for every $n \geq 20$ and $m \geq n(n-3)$. At the boundary the value is $2n$; the next step has value $2n-3$; the staircase then reaches $\lfloor (n-2)/2 \rfloor$ before becoming zero. The proof does not assert that 20 is the smallest possible whole-band threshold.

The core and attachment formulas and their complete ordinary proofs are given together in Theorem 58. The eleven-vertex core is listed explicitly and independently checked; its two-move guarantee is also formalized in Lean. The general composition and attaining-existence statements remain ordinary proofs.

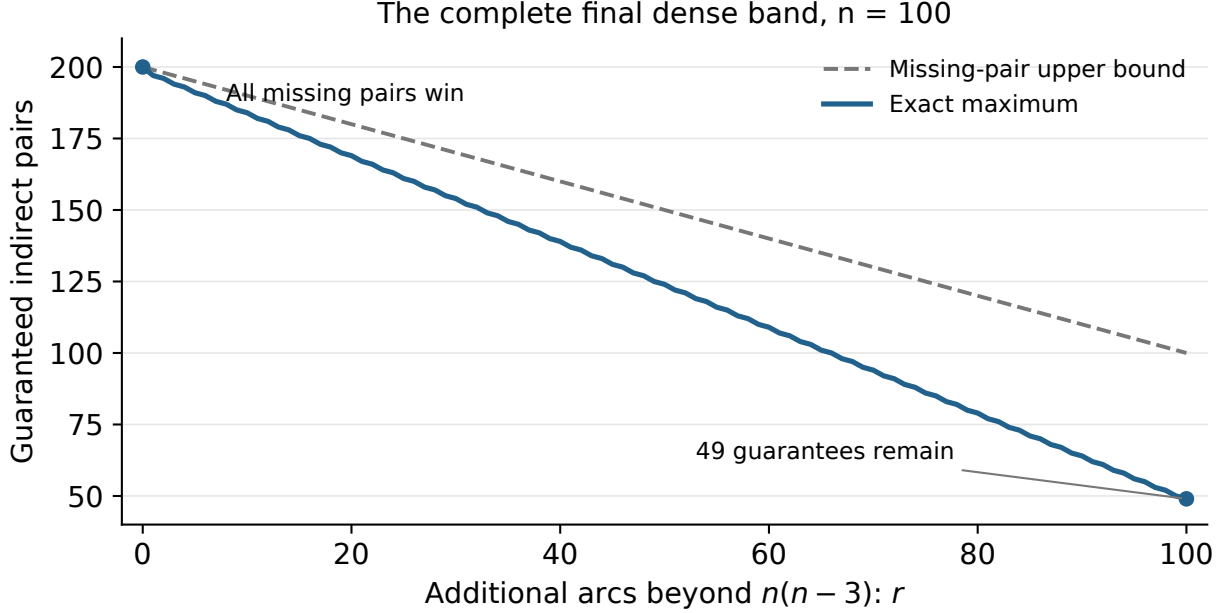


Figure 8: The complete final dense band at $n=100$. The solid curve is the proved exact maximum at every integer budget; the dashed line counts all missing pairs and does not account for forced failures.

5.6 A quadratic dense interval from three safe intermediaries

The sparse missing graph need not retain large girth while its density increases. Three designated outgoing missing arcs per vertex suffice, provided certain differences between their endpoints stay protected. The resulting construction covers a quadratic interval of missing-arc counts by a direct modular rule.

Theorem 48. For every integer $n \geq 19$, put

$$K(n) = \max \left\{ 3, \left\lceil \frac{n-1}{13} \right\rceil \right\}.$$

At every integer missing-arc count $3n \leq h \leq nK(n)$, an explicit graph attains

$$M(n, n(n-1) - h) = h.$$

Every source–recipient pair in the construction is guaranteed within two messenger moves. Arbitrary intermediate subsets of the specified optional missing arcs are permitted.

Proof. Work in $\mathbb{Z}/n\mathbb{Z}$, and put

$$S = \{1, 3, 7\}, \quad D = \{\pm 2, \pm 4, \pm 6\}, \quad E = \{\pm 3, \pm 5, \pm 9\}.$$

We shall construct a set A containing S , not containing zero, with the following properties:

1. Each difference in D has exactly its original ordered representation between two members of S , and no other ordered representation between members of A .
2. No difference between two members of A belongs to E .

For $n \geq 19$, these properties hold for $A = S$: its six ordered nonzero differences are precisely the six distinct members of D , and none belongs to E .

Whenever $|A| = k$, adjoin any residue outside

$$\{0\} \cup A \cup (A + D) \cup (A + E).$$

The exclusion of $A + D$ preserves all protected representations, and the exclusion of $A + E$ preserves the second property; both difference sets are symmetric. At most $13k + 1$ residues are excluded. If $k < \lceil (n - 1)/13 \rceil$, this quantity is strictly less than n , so the process continues. Starting with three members, it produces $|A| = K(n)$; choosing the smallest permissible representative makes the rule deterministic.

Let F be any loopless digraph satisfying

$$\{(x, x + d) : x \in \mathbb{Z}/n\mathbb{Z}, d \in S\} \subseteq E(F) \subseteq \{(x, x + d) : x \in \mathbb{Z}/n\mathbb{Z}, d \in A\}.$$

Let G be its directed complement. We prove all missing pairs of G have safe two-move intermediaries, using only the three designated missing neighbors of each initial pursuer position.

Fix a missing pair (s, t) and initial pursuer position $c \neq s$. Translate so $s = 0$, and write $a = t - s \in A$ and $z = c - s \neq 0$. The three candidates are the members of $z + S$. A candidate is suitable if it lies outside

$$A \cup (a - A).$$

Indeed, this ensures both messenger arcs are present in G , while the candidate is a missing outgoing neighbor of c and therefore cannot be reached on the pursuer's response.

At most one candidate belongs to A . Otherwise, for distinct $u, v \in S$, the difference of $z + u, z + v \in A$ is the protected difference $u - v$. Its unique representation forces $z + u = u$ and $z + v = v$, contrary to $z \neq 0$.

At most two candidates belong to $a - A$. If the candidates with steps u, v belong there, the two elements $a - z - u, a - z - v$ of A have protected difference $v - u$. Its unique representation is (v, u) , so

$$a - z = u + v.$$

If the third candidate, with step w , also belonged to $a - A$, the same argument applied to u, w would give $a - z = u + w$, a contradiction.

If all three candidates were blocked, the remaining candidate $z + w$ would have to belong to A , while the other two are in $a - A$. Setting $b = z + w \in A$, the displayed equality gives

$$b - a = w - u - v.$$

As w runs over $1, 3, 7$ and u, v are the other two steps, this difference is respectively $-9, -5, 3$. Each belongs to E , contradicting the second property of A . Hence a suitable candidate exists.

The suitable candidate differs from c because $0 \notin S$. It differs from both endpoints: $a \in A$ and $0 \in a - A$. Thus the messenger first moves to it, survives every response, and then reaches t ; receipt

precedes any subsequent interception. This argument uses only the envelope containments, so no translation symmetry of the chosen intermediate F is needed.

The compulsory missing graph has $3n$ arcs and the full envelope has $nK(n)$ arcs. Adding any chosen number of optional missing arcs gives each integer h in this interval, with all h missing pairs guaranteed. The missing-pair upper bound proves equality. $\square$

Theorem 58 now subsumes this entire numerical range, including the orders 19 through 21, while preserving two-move delivery and every optional subset. The proof above is retained as a distinct sufficient condition on arbitrary protected difference sets. The main numerical catalogue therefore no longer counts this as an additional dense interval.

The executable construction and bounded checks are in `src/dense_protected_envelope.py`; their exact finite scope is recorded in `research/dense-protected-envelope-checks.json`. The ordinary theorem is independent of those finite pursuit-game checks. No new Lean coverage is asserted.

6 Finite extrema

6.1 Exact extrema with at most seven arcs

Proposition 23. Allowing arbitrarily many vertices, including isolated vertices,

$$(M_E(0), M_E(1), \dots, M_E(6)) = (0, 0, 0, 0, 1, 1, 2).$$

Proof. Write $A = \{v : d^+(v) \geq 2\}$, $B = \{v : d^-(v) \geq 2\}$, $q = |A|$, $r = |B|$, and $b = |A \cap B|$. Every counted pair belongs to $A \times B$, excludes the diagonal, and excludes arcs. Consequently

$$I(G) \leq qr - b - e(A, B), \quad e(A, B) \geq \max\{0, 2q + 2r - m\}, \quad q, r \leq \lfloor m/2 \rfloor. \quad (23.1)$$

For completeness, the second inequality follows by counting the arcs leaving A and those entering B : their respective totals are at least $2q$ and $2r$, and their union has at most m arcs.

A counted pair (s, t) requires two arcs leaving s and two entering t . These four arcs are distinct because $s \rightarrow t$ is absent. Thus $I(G) = 0$ when $m \leq 3$.

We will use two elementary camping observations. If $d^-(z) = 0$, $z \rightarrow t$, and $d^-(t) = 2$, then no indirect pair ending at t is guaranteed. Indeed, let p be its other predecessor. An indirect source is neither z nor p . A pursuer starting at p can wait there: the messenger cannot reach z , and cannot enter p without being captured before reaching t . Dually, if $d^+(z) = 0$, $s \rightarrow z$, and $d^+(s) = 2$, then no indirect pair starting at s is guaranteed. A pursuer waits at the other successor of s ; the messenger's only alternative departure reaches a sink different from the indirect recipient.

Suppose first that $m = 4$ or $m = 5$. Then $q, r \leq 2$. If either is zero there is nothing to count, and if both are one the bound is immediate. When $q = r = 2$, (23.1) gives $I(G) \leq m - 4 - b \leq 1$. When $q = 1$ and $r = 2$, two counted pairs would have the same source and distinct recipients. The source's two outgoing arcs are then disjoint from the at least four arcs entering those recipients, which would require $m \geq 6$. The case $q = 2, r = 1$ follows by the same count with sources and recipients interchanged: at least four arcs leave the two counted sources, and the recipient's two incoming arcs are disjoint from them. Hence $I(G) \leq 1$.

Now let $m = 6$, so that $q, r \leq 3$. If $q = 1$, three counted pairs would require two outgoing arcs from their common source and six disjoint arcs entering their three recipients; this is impossible.

Thus $I(G) \leq 2$. The same argument applies when $r = 1$; cases with a zero coordinate are again immediate. If $q = r = 2$, (23.1) gives $I(G) \leq 2 - b \leq 2$.

Consider $q = 2, r = 3$. Every arc ends in B , and every vertex of B has indegree exactly two. If $A \subseteq B$, then $b = 2$ and (23.1) gives $I(G) \leq 6 - 2 - 4 = 0$. Otherwise choose $z \in A \setminus B$. It has indegree zero and at least two distinct successors in B . The first camping observation eliminates both corresponding recipient columns. At most one column remains, containing at most the two sources in A . Thus $I(G) \leq 2$.

For $q = 3, r = 2$, every arc starts in A and every vertex of A has outdegree exactly two. If $B \subseteq A$, the same count gives zero. Otherwise a vertex $z \in B \setminus A$ has outdegree zero and at least two predecessors in A . The second camping observation eliminates their two source rows, leaving at most one row and two recipients. Again $I(G) \leq 2$.

Finally, if $q = r = 3$, all six arcs go from A to B , and all relevant indegrees and outdegrees equal two. If $A = B$, (23.1) gives $I(G) \leq 9 - 3 - 6 = 0$. Otherwise a vertex of $A \setminus B$ has indegree zero and eliminates two recipient columns by the first camping observation. The remaining column has two distinct predecessors in A , so at most one of the three sources can contribute an indirect pair. This completes the upper bounds in all cases.

For the lower bounds, the four-arc diamond

$$E = \{(0, 1), (0, 2), (1, 3), (2, 3)\}$$

guarantees the indirect pair $(0, 3)$: the messenger chooses the branch not occupied by the pursuer and delivers on the next move. Adding a disjoint arc gives the five-arc example. The six-arc graph

$$E = \{(0, 1), (0, 2), (1, 3), (1, 4), (2, 3), (2, 4)\}$$

guarantees both $(0, 3)$ and $(0, 4)$. If the pursuer starts on one branch, choose the other; if it starts at either sink, choose either branch. In each case it cannot reach the selected branch on its next move, and receipt then occurs on the messenger's second move. Hence $I(G) = 2$, as required. $\square$

These arguments establish the arc-only maximum over every finite vertex set; no restriction on the number of nonisolated vertices is used.

Proposition 24 (with a finite exhaustive computation). We have

$$M_E(7) = 2.$$

Proof. First consider a weakly connected digraph with at most seven arcs and at least seven vertices. Forget the directions but retain one undirected edge for each arc, so that opposite arcs become parallel edges. The resulting connected multigraph has at most as many edges as vertices. It is therefore a tree or has exactly one cycle; a parallel pair is allowed as a cycle of length two.

In a tree, any reachable indirect source–recipient pair has an internal vertex separating its endpoints. A pursuer can start and wait there, preventing delivery. Thus no indirect pair is guaranteed. In the unicyclic case, a pair with either endpoint outside the unique cycle likewise has an internal separating vertex, unless its endpoints are adjacent in the underlying graph. For adjacent endpoints off the cycle, a directed route exists only if their connecting arc has the required direction, making the pair direct. Thus every counted pair must have both endpoints on the cycle.

There are exactly two simple undirected paths along that cycle between two distinct cycle vertices. If only one is a directed path from the source to the recipient, it is either a direct arc or has an internal vertex where the pursuer can wait. If neither is directed, the recipient is unreachable. Excursions into attached trees cannot bypass an internal cycle vertex. Consequently a counted pair requires both cycle paths to be directed from its source to its recipient. This forces the cycle's unique divergence at that source and unique convergence at that recipient; every other cycle vertex has one entering and one leaving cycle edge. A fixed orientation can therefore support at most one such ordered pair. A cycle of length two consists of opposite arcs and supports no indirect pair. We have proved that every weakly connected digraph with at most seven arcs and at least seven vertices has $I(G) \leq 1$.

For a general digraph with seven arcs, distinct weak components contribute additively to $I(G)$. Each component with a positive contribution uses at least four arcs by Proposition 23, so at most one component contributes. If that component uses at most six arcs, Proposition 23 bounds its contribution by two. If it uses seven arcs and has at least seven vertices, the preceding argument bounds its contribution by one. In the only remaining case it has at most six vertices. Adding isolated vertices to make six does not change its indirect guarantee count, and the complete six-vertex enumeration gives a maximum of two at seven arcs.

This last finite assertion is the computational dependency of the proposition. It is reproduced by `src/enumerate_small_n6.cpp`; the result is recorded with `complete: true` in `research/solver-enumeration-n6.json`. The enumeration covers every six-vertex digraph up to the degree-ordering reduction described in the computational section; the witness is also checked by the separate explicit-turn solver. The structural reduction above is what extends that finite calculation to unrestricted vertex counts.

Finally, add one disjoint arc to the six-arc, two-recipient construction in Proposition 23. This preserves its two guarantees and supplies seven arcs, proving the matching lower bound. $\square$

Theorem 49 below settles the next four budgets, eight through eleven.

6.2 Exact arc maxima from eight through eleven

The six-vertex table supplies useful lower bounds, but does not itself settle an arc budget when the number of vertices is unrestricted. At eleven arcs the distinction is essential: seven vertices attain eight guarantees, exceeding the six-vertex maximum of seven.

Theorem 49 (with finite exhaustive certificates). Over all finite vertex sets,

$$(M_E(8), M_E(9), M_E(10), M_E(11)) = (4, 5, 6, 8).$$

Proof of the support reduction. Indirect guarantees add over weak components. Every component with a positive contribution has at least four arcs. At these budgets there can be at most two positive components. If there are two, each has at most seven arcs; Propositions 23 and 24 give a combined maximum of at most three. This cannot improve any of the displayed lower bounds.

There is therefore only one relevant positive component. We argue in increasing order of the budget $m = 8, 9, 10, 11$. If this component has fewer than m arcs, the already established smaller-budget maximum bounds its contribution by the displayed value at m . If it uses all m arcs and has at least m vertices, the tree-or-unicyclic argument in Proposition 24 gives at most one indirect guarantee.

That argument only needs a connected underlying multigraph with at most as many edges as vertices, so applies unchanged at the present budgets.

In the remaining case the component has at most $m - 1$ vertices. Pad it with isolates to exactly $m - 1$ vertices; its arc count and indirect count do not change. Hence each upper bound reduces to the following one finite counterexample test:

Arc budget m	Host order	Excluded threshold for I
8	7	5
9	8	6
10	9	7
11	10	9

Complete finite upper checks. Both enumerators list all prescribed outdegree profiles with total m in sorted order and then all loopless outgoing rows. As in the canonical reduction of Theorem 46, future vertices with equal prescribed degree and equal incoming pattern from completed rows can be permuted while leaving every completed row unchanged. Choosing the new row's neighbors as a prefix of each such contiguous cell retains a representative of every graph. The second implementation instead uses descending degrees and suffixes. Both reduction arguments fix the current vertex and all completed vertices.

The source and predecessor camping conditions of Theorem 46 supply necessary local rejections. A completed degree-two source that points to a prescribed sink also contributes zero, by the earlier sink obstruction. Otherwise a source of degree $d \geq 2$ contributes at most $n - 1 - d$. A target of current indegree d contributes at most $n - 1 - \max(2, d)$, or zero if its indegree cannot reach two.

One additional numerical bound makes these sparse checks short. Let Q contain all sources not yet excluded, and put $q = |Q|$. Every member of Q has prescribed outdegree at least two. If the eventual number of indegree-at-least-two targets is r , the same degree-incidence count as (1) gives

$$I \leq qr - \max(0, q + r - n) - \max(0, 2q + 2r - m).$$

To bound r at a partial assignment, let R be the number of arcs still to be assigned. A target currently of indegree d needs at least $\max(0, 2 - d)$ of these arcs to reach indegree two. Discard targets with too few remaining possible predecessors, and select as many of the others as possible using the smallest such costs first, with total cost at most R . The resulting $r_{\max}$ is an upper bound: each remaining arc enters only one target. Maximizing the displayed bound over $0 \leq r \leq r_{\max}$ is therefore a safe partial rejection. No sufficiency is assumed for these degree or camping tests.

Every remaining complete graph is evaluated by the original finite reachability game. The primary implementation uses synchronous messenger-position masks; the independent implementation uses an explicit alternating-state queue attractor. Both start with terminal receipt states and compute the least winning region, including waiting and all pursuer responses. They reject every candidate at every one of the four thresholds. The exact enumeration counts are supplied in the reproduction receipt; both runs report complete enumeration rather than termination at their time limit. This proves the four upper bounds.

Lower witnesses. The witnesses at eight, nine and ten arcs are the graphs in the six-vertex table. The eleven-arc witness has seven vertices and the following outgoing neighborhoods:

Vertex	Outgoing neighbors
0	1
1	0
2	3
3	4, 5
4	0, 2
5	3, 6
6	1, 2

Its guaranteed indirect pairs are

$$(3, 0), (3, 1), (3, 2), (4, 1), (5, 0), (5, 1), (5, 2), (6, 0).$$

Both independent game solvers check each of the four lower witnesses, and the saved explicit ranks satisfy every legal-move obligation. They attain the upper bounds and complete the proof. $\square$

The complete reproduction command is `python3 src/small_arcs_resume.py`. Its two exhaustive sources are `src/small_arcs_sparse_primary.cpp` and `src/small_arcs_sparse_independent.cpp`; the receipt, source hashes, complete enumeration summaries and all four lower rank certificates are in `research/small-arcs-resume-verification.json`. The final full reproduction took about eighteen seconds. These are finite computer-assisted proofs with an ordinary all-order support reduction; they have not yet been formalized in Lean.

6.3 Small vertex orders

The smallest order attaining the vertex bound is twelve. This statement is about attaining the bound, not about the first noncomplete network in which all pairs are guaranteed: the latter already occurs at seven vertices.

Theorem 46 (with finite exhaustive certificates). The seven- and eight-vertex maxima are

$$M_V(7) = 21, \quad M_V(8) = 32.$$

At the next three orders the following bounds hold:

n	Lower bound for $M_V(n)$	Upper bound for $M_V(n)$
9	48	53
10	62	69
11	84	87

There is no universally winning two-in/two-out regular digraph on any of 7, 8, 9, 10, 11 vertices. Together with the previous exact results through six vertices and Theorem 40, this implies that twelve is the smallest order $n \geq 4$ for which $M_V(n) = n(n - 3)$.

Proof of the lower bounds. The finite witnesses at orders seven through eleven have respectively 21, 24, 24, 28, 26 arcs and guarantee every pair. Their complete graph and winning-strategy certificates are provided in `research/small-exceptional-general-certificates.json`, with the improved nine-vertex witness in `research/root-small-structured-9-certificate.json` and the

eleven-vertex witness in `research/small-exceptional-structured-11-certificate.json`. Two independent attractor algorithms agree; each stored alternating-state rank satisfies all messenger-choice and universal pursuer-response obligations. Thus $I = n(n - 1) - m$ gives 21, 32, 48, 62, 84 respectively. For concreteness, the seven-vertex witness is the following graph, whose guarantees require at most four messenger moves:

Vertex	Outgoing neighbors
0	1, 3, 6
1	0, 4, 6
2	0, 5, 6
3	1, 2, 5
4	0, 2, 3
5	1, 3, 4
6	2, 4, 5

Seven-vertex upper bound. If a graph has at most thirteen arcs, the arc-only bounds give $I \leq 20$; it suffices here to use monotonicity of M_E (pad by disjoint one-way arcs) and $M_E(13) \leq 20$. A graph with at least twenty-one arcs has at most $42 - 21 = 21$ missing pairs. It remains to exclude $I \geq 22$ for $14 \leq m \leq 20$.

We describe the finite reduction and every rejection rule. Write $O_s = N^+(s)$, $P_t = N^-(t)$ and $C_c = \{c\} \cup N^+(c)$. A counted pair (s, t) necessarily satisfies

1. $s \neq t$, $t \notin O_s$, $|O_s| \geq 2$ and $|P_t| \geq 2$;
2. $O_s \not\subseteq C_c$ for every $c \neq s$;
3. $P_t \not\subseteq C_c$ for every $c \neq s$.

The degree conditions are the one-neighbor camping obstructions. For condition 2, a pursuer waiting at c captures the messenger after its first departure from s . For condition 3, it waits at c and captures the messenger upon the first entry into a predecessor of t , before the next messenger move can deliver. Indefinite waiting by the messenger also fails to deliver. These arguments respect receipt priority because an indirect source is not itself a predecessor of its recipient.

Let $U(G)$ be the number of pairs satisfying these three conditions. Then $I(G) \leq U(G)$. A source contributes at most four pairs, so $I \geq 22$ requires at least six sources of outdegree at least two. Every graph can be relabeled in nondecreasing outdegree order, allowing only the first row to have degree below two. We enumerate every loopless outgoing row mask in that order, with total degree between fourteen and twenty. This includes every possible graph, with harmless repetitions. At a partial assignment, the following bounds allow safe pruning:

- A completed row of degree $d \geq 2$ contributes at most $6 - d$, or zero if a known completed row already violates condition 2. Each remaining row contributes at most $6 - \max(2, d_0)$, where d_0 is the smallest outdegree still allowed by the ordering.
- A target currently of indegree d contributes at most $6 - \max(2, d)$. Its contribution is zero if the remaining rows cannot raise its indegree to two.
- The total arc count must remain extendible to the interval $[14, 20]$.

No omitted edge can repair a rejected completed source: the covering closed neighborhood only grows. The incoming-degree bounds likewise use upper estimates on all possible completions. At every surviving complete graph we count $U(G)$ directly. The exhaustive enumeration has 109, 941, 961

recursion nodes and 4, 299, 624 complete leaves before an optional indegree tie normalization. None has $U(G) \geq 22$. A separate implementation uses boolean adjacency, descending outdegree order, and no indegree tie normalization; it completes 123, 280, 819 nodes and the same 4, 299, 624 leaves, again with no survivor. This proves $I \leq 21$ in the remaining arc range. The lower witness proves equality.

Eight-vertex upper bound. At most fifteen arcs give $I \leq M_E(15) \leq 30$; at least twenty-four arcs leave at most thirty-two missing pairs. To exclude $I \geq 33$, it therefore suffices to examine $16 \leq m \leq 23$. Each source contributes at most five pairs, so at least seven sources must have outdegree at least two. The same camping conditions and partial bounds used above apply with eight vertices.

For this slightly larger search we also remove labeling redundancy. Fix a nondecreasing outdegree sequence. After the outgoing rows numbered less than r have been chosen, partition the unassigned vertices into cells with equal prescribed outdegree and identical adjacency from all completed rows. Vertices in such a cell may be permuted without changing any completed row or degree requirement. Therefore, when choosing row r , its neighbors among the future vertices of each cell may be required to form a prefix of that cell. This loses no graph: permute within each cell to put the neighbors first, fixing row r and all previous vertices, and continue recursively. The new cells are obtained by splitting the old cells, so their contiguity is preserved. The analogous suffix rule and reversed degree order are equally complete.

The primary search completes all 178 degree sequences, with 946, 676, 955 recursion nodes and 22, 465, 453 complete leaves. Only 373 leaves satisfy the necessary pair bound $U(G) \geq 33$. Exact least-attractor evaluation rejects all of them; the largest actual I among these survivors is seventeen. Thus no graph has $I \geq 33$, proving $M_V(8) = 32$. The ordinary canonicalization and every pruning rule were independently audited; as an additional small positive check, every labeled loopless four-vertex graph was canonicalized in both prefix and suffix conventions and its relabeling and prefix conditions checked. Independent complete suffix enumeration and the explicit alternating-state game provide a second finite verification: its completed intervals $[0, 160)$ and $[160, 178)$ cover all degree profiles. They visit 1, 179, 975, 770 nodes and 63, 717, 034 complete leaves, with 1, 115 static survivors, all rejected. The differing counts reflect the opposite degree order and suffix convention, which choose a different redundant collection of representatives. Both implementations give the same zero-survivor conclusion.

Nonattainment at orders eight through eleven. By the vertex equality criterion, an attaining graph would be universally winning and have indegree and outdegree exactly two everywhere. Any such directed graph is the union of two arc-disjoint permutation factors. Indeed, its bipartite tail/head incidence graph is two-regular, so alternating the edges around each even cycle gives the two perfect matchings. Both permutations are fixed-point-free. Up to simultaneous relabeling, the first permutation may be put in canonical cycle form for each partition of n having no part one; enumerate every second permutation avoiding both the diagonal and the first permutation's arcs.

The source and predecessor camping conditions above reject a branch as soon as a completed two-element neighborhood is covered by a known closed outgoing neighborhood. For the predecessor condition, at least $n - 3 \geq 2$ sources are indirect, so a covering vertex can be avoided as the initial source. Every remaining complete graph is evaluated by the finite least-attractor algorithm. The exhaustive counts are:

n	Cycle partitions	Recursion nodes	Locally admissible complete graphs	Universally winning graphs
7	4	142	0	0
8	7	1,446	0	0
9	8	10,996	535	0
10	12	231,676	35,610	0
11	14	2,177,946	120,302	0

An independent implementation reverses the permutation assignment order. Its local tests forbid directed transitive triangles, two distinct length-two walks with the same ordered endpoints, and duplicate completed outgoing pairs; these are precisely the same camping obstructions in adjacency-matrix form. It then uses an explicit alternating-state AND/OR queue attractor instead of the primary synchronous bitmask iteration. Both complete enumerations give exactly the displayed counts. As a positive control, a known universal twelve-vertex graph passes every incremental pruning test and the game check of the primary enumeration. Hence equality $M_V(n) = n(n-3)$ is impossible at these orders, which lowers the integer upper bound by one. $\square$

The seven-vertex example also gives the first noncomplete universally winning graph. At orders at most three, the previously established zero maximum excludes any missing guaranteed pair. At orders four through six, the density bound for a noncomplete universal graph would require $m \leq n(n-3)$ and hence $I \geq 2n$, contradicting the exact maxima 2, 4, 7.

Verification boundary. These are finite computer-assisted proofs, with ordinary completeness and pruning arguments; they are not presently Lean formalizations. The negative seven-vertex result needs no game solver after the static camping reduction. The lower witnesses have saved rank certificates checked directly. Every remaining degree-two graph and every eight-vertex static survivor is evaluated by the exact game algorithms in the independent exhaustive runs; full per-graph ranks for those negative enumerations are not stored. The exact values at orders nine through eleven remain open; absence of a two-regular universal graph does not determine their extrema.

Primary sources are `src/small_exceptional_n7_exact.cpp` and `src/small_exceptional_regular.cpp`; independent sources are `src/small_exceptional_n7_independent.cpp` and `src/small_exceptional_independent.cpp`. The eight-vertex sources are `src/small_exceptional_canonical.cpp` and `src/small_exceptional_canonical_independent.cpp`; the independent ordinary audit is `research/small-eight-canonical-independent-audit.md`. The reproduction receipt is `research/small-exceptional-verification.json` and the research record is `research/small-exceptional-progress.md`.

6.4 The exact 23-arc endpoint

The general sparse support reduction and the eleven-vertex core bound reduce the 23-arc endpoint to three finite families.

Theorem 53 (with finite exhaustive checks).

$$M_E(23) = 83.$$

Moreover, $M(n, 23) = 83$ for every $n \geq 12$.

Proof. The previously verified 12-vertex witness has 83 guaranteed indirect pairs. Suppose a graph with 23 arcs had $I \geq 84$. Apply Theorem 57 with $k = 11$, whose sparse support cutoff is

$$k^2 - 4k + 6 = 83.$$

Its induced-core alternative would give an eleven-vertex graph with 22 or 23 arcs and at least 84 guarantees, contradicting the proved eleven-vertex core bound $I \leq 81$.

Consequently, after removing isolates, every possible counterexample has twelve vertices and belongs to one of three families: distinct degree-one source and recipient exceptions with their connecting arc absent; distinct exceptions with that arc present; or one common indegree-one/outdegree-one exception. All other corresponding degrees are two. Their numbers of eligible missing pairs are respectively 90, 89, and 89, by the count in Corollary 57.1. This reduction applies to arbitrary original support sizes and uses the actual-game induced-core implication in Theorem 57.

We exhaust these families using their two-permutation descriptions. For the first, add the missing exceptional arc $u \rightarrow v$; for the second add another copy of the existing arc; for the third add a dummy loop at the common exceptional vertex. The resulting bipartite incidence multigraph is two-regular and decomposes into two perfect matchings. This follows directly by alternating the edges of each even cycle, including a doubled-edge cycle. Regard the matchings as permutations P, Q , placing the added arc in P .

Label the exceptional source 0 and, when distinct, the exceptional recipient 1. Up to relabeling, enumerate the cycle of P containing the added arc and the unordered cycle lengths of its remaining vertices. In the common-exception case, P fixes 0. Every other cycle has length at least two. Then enumerate every admissible permutation Q , allowing only the prescribed doubled exceptional arc in the second family, and delete the added P arc before any game calculation. This covers every graph in the three families.

Partial assignments may be discarded when a stationary pursuer covers the completed outgoing neighbors of a contributing source, or the completed predecessors of a contributing recipient. A source obstruction destroys at least eight eligible pairs. A recipient obstruction destroys at least seven, after omitting a possible source coinciding with the pursuer. Since there are at most 90 eligible pairs, all discarded graphs satisfy $I \leq 83$. Testing coverage with a partially assigned pursuer neighborhood is also safe: that neighborhood can only grow.

Every completed graph surviving these obstructions is evaluated by the exact least reachability fixed point. Evaluation may stop at a leaf once its remaining possible score falls below 84. The three finished enumerations give the following counts; none has $I \geq 84$.

Exceptional configuration	Permutation cases	Partial nodes	Completed graphs surviving camping
Distinct, arc absent	42	164,276,910	16,902,091
Distinct, arc present	42	13,448,216	1,125,588
Common vertex	14	49,814,092	3,558,672

This excludes every proposed counterexample and proves $I \leq 83$. The verified witness gives equality. Adding isolates preserves its guaranteed indirect pairs, proving the fixed-order assertion for every $n \geq 12$. $\square$

Reproducibility and proof boundary. The new source `src/odd_endpoint_near23_threshold.cpp` is a separately saved, parameterized copy of the earlier enumerator; it accepts cutoffs at least 84. The coverage, game calculation, and pruning are unchanged. The complete receipts are `research/odd-endpoint-threshold84-absent.json`, `research/odd-endpoint-threshold84-present.json`, and `research/odd-endpoint-threshold84-same.json`. Their cutoff, completion flags, exact counts, source hash, and dependencies are consolidated in `research/odd-endpoint-threshold84-summary.json`. The unchanged dependency on the eleven-vertex enumeration is audited in `research/odd-endpoint-eleven-enumeration-audit.md`. The current general support reduction is independently reviewed in `research/sparse-stability-independent-audit.md`. The earlier specialized 23-arc reduction and the unchanged cutoff-84 pruning are recorded in `research/odd-endpoint-threshold84-ordinary-audit.md`. A separate software audit of the coverage, pruning, and evaluation is `research/odd-endpoint-threshold84-independent-audit.md`. It also compares 37 leaf-game and early-exit cases across all three families with an explicit alternating-state solver, including full rank and losing-closure checks; the receipt is `research/odd-endpoint-threshold84-independent-tests.json`. The lower-bound certificate is `research/odd-endpoint-m23-lower-certificate.json.gz`, checked by two independently implemented game solvers and every saved rank obligation. The exhaustive upper-bound computation is reviewed at the source and mathematical level; it is not a separately implemented second enumeration or a Lean proof.

7 Verification and open questions

7.1 Exact computation and formal verification

For a fixed recipient t , let $X_j(r)$ be the set of initial pursuer positions from which delivery can be forced in at most j messenger moves. Start with $X_0(t) = V$ and $X_0(r) = \emptyset$ otherwise. Set

$$S_j(v) = \{c : C(c) \subseteq X_j(v)\} \quad (v \neq t), \quad S_j(t) = V,$$

and update

$$X_{j+1}(r) = \left(\bigcup_{v \in C(r)} S_j(v) \right) \setminus \{r\} \quad (r \neq t), \quad X_{j+1}(t) = V.$$

The recurrence includes waits and excludes all nonterminal collisions. Its least fixed point enforces eventual delivery. A separate implementation builds the explicit alternating graph of states (r, c, turn) and computes its reachability attractor. This is standard finite reachability-game machinery (Berwanger 2009), rather than a new general algorithm for graph games.

The certificate checker independently verifies decreasing ranks on the selected messenger moves and on every pursuer response. Outside the winning region it verifies closure under every messenger move and at least one pursuer response. Thus a negative answer comes with an avoidance strategy, not simply failure of a search. Standard adjacency-list attractor analysis gives an all-recipient time bound $O(n^2m + n^3)$; that bound describes the standard implementation, not a claimed running-time bound for the simpler synchronous reference solver.

7.1.1 Certified finite extrema

Complete enumeration gives

n	1	2	3	4	5	6
$M_V(n)$	0	0	0	2	4	7

For $n \leq 5$ every labeled digraph was enumerated. For $n = 6$, sorting the vertex labels by degree reduces the search while retaining at least one representative of every isomorphism class; valid degree bounds prune only candidates unable to improve the current maximum. This certifies extrema, not labeled distributions. Every attaining graph is independently checked by the explicit alternating solver. The complete joint tables, enumeration code, and receipts accompany the paper.

Universal two-regular witnesses attain Theorem 5 for every $12 \leq n \leq 24$. Consequently Theorem 6 is attained at $m = 24, 26, \dots, 48$. Deletion witnesses attain Theorem 7 at $m = 37, 39, \dots, 47$. Theorem 46 proves $M_V(7) = 21$, $M_V(8) = 32$, and bounds the still-open orders 9 through 11.

One clean 12-vertex example is a Cayley digraph of A_4 , with right multiplication by two suitable 3-cycles. Its actual graph and all-recipient finite-rank certificate are included. The Lean theorem checks the explicit 12-vertex graph directly, so correctness does not depend on trusting the group-label interpretation or the search that discovered it.

7.1.2 Lean coverage

The formal artifact uses Lean 4.33.1 and its standard library. The game implementation checks terminal receipt before collision, permits waiting, and quantifies over arbitrary legal history-dependent pursuer choices. Finite winning trees are proved to generate actual delivering plays; rank certificates generate those winning trees. Safe branching and predecessor camping are connected to the same semantics. A fully kernel-checked 12-vertex certificate proves universal delivery and computes $I = 108$. A second kernel-checked graph has 11 vertices, 88 arrows, and 22 guaranteed indirect pairs, each delivering within two moves. The new robust-envelope module proves actual delivery for every graph between supplied base and envelope graphs, assuming explicit safe witnesses. A further module checks the six published base/envelope pairs at orders sixteen through twenty-one, their arrow counts, and their safe witnesses, so the actual delivery theorem applies to every intermediate graph in each of these six envelopes.

The dense-counting module additionally proves the staircase arithmetic from its displayed cardinality inequalities, the attaining count identity, and complementary-block gluing as actual two-move delivery against every legal history-dependent pursuer. The counting inequalities themselves are established by the ordinary graph argument, rather than silently asserted as formal theorems. Existence of the blocks is also a separate premise of the formal gluing result.

`ProtectedRouting.lean` additionally proves that changing arrows wholly outside a routing set preserves its safe two-arrow witnesses and actual delivery within two moves against arbitrary legal history-dependent pursuers. Witness existence is an explicit premise; the probability estimates, exact-budget conditioning and construction algorithm remain ordinary proofs.

These are meaningful statements about the physical game. The general graph-to-counting arguments, arithmetic existence lemma, infinite constructions, probabilistic existence results, and exhaustive-search completeness arguments remain ordinary proofs. The downloadable coverage map lists exact theorem names and source hashes; it does not equate a numerical certificate or assumed interface with a full general formalization.

7.2 Further consequences and open problems

The random statement in Theorem 54, specialized to the full vertex set, implies that for every fixed density $0 < p < 1$ the random loopless digraph guarantees all pairs with probability tending to one. Its failure probability is at most

$$n(n-1)^2(1-p^2(1-p))^{n-3}.$$

Consequently

$$\mathbb{E}[I(G)] = (1-p)n(n-1) - o(1).$$

Indeed, the loss from the missing-pair count is at most $n(n-1)$ times the indicator of failure; the probability decays exponentially. This expectation conclusion is separate from the theorem's exact-budget conditioning and deterministic construction.

Angel Raychev's conjecture that every sufficiently large order attains the missing-pair bound at every $2n \leq m \leq n(n-3)$ is now proved by Theorems 38–44. The sparse endpoint constructions also settle the vertex maximum at every $n \geq 12$, and the even and odd arc maxima at all sufficiently large budgets. Corollary 47 assembles the eventual joint classification.

The remaining central questions are the exact vertex maxima at orders nine through eleven, the arc budgets twelve through twenty-two, the joint maximum at moderate orders outside the stated intervals, and classification of all maximizing graphs. Improving the sufficient 115 threshold would clarify how early the full interval begins. A finite construction that works at one circumference still does not prove a uniform family: the local games and projection arguments are essential to the results above.

A correct future proof must retain enough information about both players. Simple distance-to-recipient potentials fail: some states at a fixed small distance from the recipient require detours whose length grows with the graph. Nor does safe indefinite motion imply delivery. The finite witnesses and failed conjectures are preserved to make those distinctions reproducible.

8 Appendix: provenance and retained formulations

8.1 Correcting the original layered construction

The original graph remains a useful construction, but its printed count is not correct. We give its exact count and separate safe intermediate movement from terminal delivery.

Let $q = 2x$, $a = \binom{2x}{x}/2$, and choose a family $\mathcal{F}$ containing one of each complementary pair of x -subsets of $[2x]$, with every coordinate occurring in exactly $a/2$ members. Such a family exists precisely when positive x is not a power of two; a proof follows below.

There are row vertices A_X, A'_X for $X \in \mathcal{F}$, and coordinate vertices b_i, b'_i, c_i, c'_i for $i \in [q]$. The only arcs are

- $A_X \rightarrow b_i, b'_i$ when $i \in X$;
- $b_i \rightarrow A'_Y$ when $i \in Y$, and $b'_i \rightarrow A'_Y$ when $i \notin Y$;
- $A'_Y \rightarrow c_i, c'_i$ when $i \in Y$;
- $c_i \rightarrow A_X$ when $i \in X$, and $c'_i \rightarrow A_X$ when $i \notin X$.

The order is $N = 2a + 4q$. The four directed layers are $A, B \cup B', A', C \cup C'$.

Theorem 16. For the balanced original family,

$$I(G) = 2a(N - 1 - q) + 4q.$$

At $x = 3$, the 44-vertex graph has exactly 764 indirect guaranteed pairs, rather than the draft's claimed lower bound of 1252.

Proof. The rows form an antichain. The $2q$ literal columns, indicating membership or nonmembership of each coordinate, also form an antichain, with size $a/2 \geq 5$. They cannot coincide: equality on the selected rows would remain true on their complements and hence on every x -subset, which is impossible for distinct literals. Every vertex has at least two successors.

From any noncollision state the messenger can safely advance one layer. If both players are in the same layer, use incomparable outgoing neighborhoods to choose a successor inaccessible to the pursuer. If the pursuer is in the next layer, avoid its occupied vertex; it has no within-layer arc. In the other two cases it cannot reach the next layer in one move. This includes waiting responses.

From A_X to a specified A'_Y there is a two-move guarantee. Against a pursuer A_Z , choose a coordinate in $X \setminus Z$, then choose its primed or unprimed intermediate according to membership in Y . Against a pursuer in the intermediate layer, choose another coordinate of X ; the other layers cannot interfere. The reverse direction is identical. Safely advancing to the appropriate row layer first therefore gives delivery to any row recipient in at most five moves. There are $2a$ such recipients, each with $N - 1 - q$ indirect sources.

For target b_i or b'_i , its predecessor set is exactly $N^+(c_i)$. Lemma 2 excludes every indirect source except c_i . That exceptional source wins in two moves: literal-column incomparability handles a pursuer in its own layer, and the other layer cases are as above. Thus (c_i, b_i) and (c_i, b'_i) both count. Symmetrically (b_i, c_i) and (b_i, c'_i) count, and no other indirect sources do. These give exactly $4q$ further pairs. $\square$

The correction preserves the original $N^2 - O(N \log N)$ leading estimate along its constructed orders. The new circulant family supplies the stronger linear-deficit result at every sufficiently large order. The original construction's regular incidence design and the later extremal construction are distinct contributions.

8.1.1 Balanced complementary halves

Lemma 17. For positive x , the x -subsets of $[2x]$ can be partitioned into two coordinate-balanced families with complementary subsets in opposite families if and only if x is not a power of two.

Proof. Put $C = \binom{2x}{x}$. Each coordinate occurs in $C/2$ subsets, so balance requires $4 \mid C$. The factorial valuation formula gives

$$v_2 \binom{2x}{x} = s_2(x),$$

where $s_2(x)$ is the number of ones in the binary expansion. Thus divisibility by four is equivalent to x not being a power of two.

For sufficiency, distinguish coordinate $2x$ and cyclically rotate the other $L = 2x - 1$ coordinates. Subsets containing $2x$ correspond to binary circular words with $x - 1$ ones and x zeros. Every orbit has length L : a proper repetition would have a repetition count dividing both consecutive integers. The number $C/(2L)$ of orbits is even, since L is odd and $4 \mid C$.

Choose half of those orbits. In the first family put their subsets, and put the complements of the subsets in the unchosen orbits. Exactly one of each complementary pair is selected. The distinguished coordinate occurs $C/4$ times. Rotation invariance gives a common incidence count r for every other coordinate; counting all incidences yields

$$(2x - 1)r + C/4 = xC/2,$$

so $r = C/4$. The other family is balanced as well. $\square$

This proof replaces the draft's incorrect stronger divisibility assertion $4x \mid \binom{2x}{x}$, which already fails at $x = 3$.

8.2 Construction appendix: choosing the circulant parameter

Proof of Lemma 12. If $n \equiv 0 \pmod{4}$, choose $h = n/2 - 1$; any common divisor with n divides two and is odd. If $n \equiv 2 \pmod{4}$, choose $h = n/2 - 2$; the same argument uses four. If $n \equiv 3 \pmod{4}$, use $h = (n - 1)/2$.

It remains to handle $n \equiv 1 \pmod{4}$. Write $n - 1 = 2^e d$ with d odd and $e \geq 2$. If $d \geq 7$, take $h = d$: it divides $n - 1$ and is at most $(n - 1)/4$. If $d \in \{1, 3, 5\}$ and $3 \nmid n$, use $h = (n - 3)/2$.

If $d = 1$ and $3 \mid n$, take $h = (n - 7)/2$. The number $2^e + 1$ is not divisible by seven, since powers of two modulo seven are 1, 2, 4. The case $d = 3$ and $3 \mid n$ is impossible. Finally, if $d = 5$ and $3 \mid n$, then e is even. Use $h = (n - 11)/2$. Divisibility of $5 \cdot 2^e + 1$ by eleven would imply $2^e \equiv 2 \pmod{11}$, whereas even powers have residues 1, 3, 4, 5, 9.

All choices are odd and smaller than $n/2$. Their smallest possible lower estimate is $(n - 11)/2 \geq 7$, valid for $n \geq 25$. $\square$

8.3 The sparse interval as a consequence of the stronger envelope

Theorem 41 (retained reference). For every integer $n \geq 2419$ and every $2n \leq m \leq 3n$, an explicit universally winning graph exists, with delivery within $9\lceil n/6 \rceil + 26$ messenger moves and arbitrary subsets of its prescribed optional arrows.

Proof. Theorem 52 gives the stronger assertion from $n \geq 115$, through a larger exact budget endpoint, with the better time bound $9\lceil n/6 \rceil + 21$. $\square$

The earlier construction separated deletions by seventeen columns and forbade optional arrows within sixteen columns of a deletion. Its separate finite strategy certificates remain available. The new theorem replaces that numerical result and its overlap role; the old proof is retained in `research/sparse-budget-earlier-proof.md` for reproducibility.

Corollary 42 (retained reference). There is an absolute N such that for every $n \geq N$ and $2n \leq m \leq n(n - 3)$,

$$M(n, m) = n(n - 1) - m.$$

Proof. Theorem 44 now permits the explicit choice $N = 115$, with combinatorial attaining constructions throughout. $\square$

8.4 A robust dense bridge

Theorem 50. For every $n \geq 16$ and every integer $2n \leq h \leq 3n$, there is a deterministic n -vertex graph with exactly h missing arcs in which every pair is guaranteed within two messenger moves. Consequently

$$M(n, n(n-1) - h) = h.$$

At each allowed order, one compulsory set of $2n$ missing arcs and a list of n optional missing arcs suffice: **every subset** of the optional list preserves two-move delivery. The construction takes polynomial time.

Proof. This is the robust envelope in Theorem 58. Its parameterized modular families and four odd repairs cover every order from 22; six further explicit optional lists cover 16, ..., 21. The ordinary danger-set arguments and the finite witnesses retain a compulsory safe intermediary against every pursuer, so any optional subset is permitted. Selecting $h - 2n$ optional arcs gives the exact prescribed count. $\square$

Theorem 58 further enlarges this same robust dense interval to

$$2n \leq h \leq H(n)$$

at every $n \geq 30$, where $H(n)$ is its explicit cap and $H(n) = n^2/4 - O(n)$ from the parity construction. The statement here is its uniform initial interval, with its earlier label retained for references. Theorem 58 gives the common constructions and proofs. The earlier projective-incidence decomposition is no longer a separate premise or a separate numerical result. The generic high-girth criterion of Lemma 18 remains available for other graphs.

8.5 The full-vertex specialization

Proposition 51. Put $L = n(n-1)$ and $p = m/L$, where $0 < m < L$. If

$$(n-3)p^2(1-p) > 4 \log n + \log(8/3),$$

there is a deterministic algorithm, polynomial in n , constructing an n -vertex graph with exactly m arcs in which every pair is guaranteed within two messenger moves.

Proof. Take $H = V$, $b = n$, and $a = m$ in Theorem 54. The condition forces $n \geq 4$. That theorem proves the stronger statement with a specified routing set, exact incident budget, arbitrary exterior arrows, and a random-sampling probability bound. $\square$

The former full-vertex conditioning argument and its 1800 overlap are retained in the research history. The counting proof now appears once in Theorem 54, and Theorem 44 gives the improved 115 overlap.

8.6 Provenance and assistance

Alexandra Ignatova's original problem and layered construction, developed under Angel Raychev's mentorship in 2023 and written in the retained February 2024 manuscript, are credited throughout. The upper bound $I(G) \leq n(n-3)$ and the investigation of two-in/two-out-regular graphs also belong to Ignatova and Raychev's original work; the present account does not claim that bound or research direction as a new contribution. The September 2026 work repairs the balancing and strategy arguments, corrects the original count, and develops the new extremal bounds, circulant

constructions, dense families, computational certificates, and formalization. Astra 6, operating through the Codex harness, assisted with these deductions, counterexample searches, proof writing, programming, and Lean development. The current document is a research checkpoint; it has not been submitted to arXiv and does not assign an unresolved final human author line.

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